

Annex 9

Good regulatory practices for market surveillance and control of medical products

For any further information or request, please send an email to the Norms and Standards for Pharmaceuticals (NSP) Team at WHO, at email nsp@who.int.

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Abbreviations

API	active pharmaceutical ingredient
GP	good practice
GSMS	Global Surveillance and Monitoring System
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
IDEAS	insight, data, engagement, action, solutions
INN	international non-proprietary name
INTERPOL	International Criminal Police Organization
NRA	national regulatory authority
TRS	Technical Report Series
WHO	World Health Organization

1. Introduction

1.1 Background

Access to affordable, efficacious, safe and quality-assured medical products,¹¹ in tandem with acceptability and appropriate use, is essential to achieve universal health coverage, particularly for the diagnosis, treatment and prevention of diseases. However, this access is threatened by weak or limited technical and regulatory capacity, insufficient financing, poor procurement practices, supply chain vulnerabilities, criminal activity, and lack of regulatory controls and enforcement when needed. When the quality of medical products that enter the market is unreliable, either through unintended failures in quality assurance or quality control (resulting in product defects) or deliberate actions such as criminal activities, this results in treatment failure, inaccurate diagnoses, patient harm and economic losses ([1](#)). In order to standardize the description of such incidents, the World Health Organization (WHO) replaced the term “substandard/spurious/falsely-labelled/falsified/counterfeit medical products” with “substandard” and “falsified” in 2017 ([Box A9.1](#)) ([2](#)).

¹¹ The term “medical products” refers to medicines, vaccines and medical devices, including in vitro diagnostics.

Box A9.1

Classification of medical products to be used by the WHO Global Surveillance and Monitoring System and the Member State Mechanism

Substandard. Substandard medical products (also termed “out of specification”) are authorized by national regulatory authorities but fail to meet either national or international quality standards or specifications – or, in some cases, both.

Falsified. Descriptive of a product that has been deliberately or fraudulently misrepresented as to its identity, composition or source. Such deliberate or fraudulent misrepresentation refers to any substitution, adulteration or reproduction of an authorized product, or the manufacture of a product that is not an authorized product.

- “Identity” shall refer to the name, labelling or packaging or to documents that support the authenticity of an authorized product.
- “Composition” shall refer to any ingredient or component of the product in accordance with applicable specifications authorized or recognized by the national regulatory authority.
- “Source” shall refer to the identification, including name and address, of the marketing authorization holder, manufacturer, importer, exporter, distributor or retailer, as applicable.

Unregistered or unlicensed. Descriptive of medical products that have not undergone evaluation or approval by the national or regional regulatory authority for the market in which they are marketed, distributed or used, subject to permitted conditions under national or regional regulation or legislation.

Source: World Health Organization (2).

The entry of substandard and falsified medical products into global supply chains and their consumption threatens public health and results in negative socioeconomic impacts. Although quantifying the full extent of their impact remains challenging, a systematic review by WHO of field surveys assessing the quality of medicine in low- and middle-income countries reported a failure rate of 10.5% among 48 218 sampled products. The review also linked these substandard and falsified medicines to serious outcomes such as treatment failure, preventable adverse reactions and deaths, with an estimated economic burden of US\$ 30.5 billion (3). Similarly, a meta-analysis by Ozawa et al. reported an average prevalence of 13.6% of essential medicines as substandard and falsified, and an estimated economic impact of US\$ 31.25 billion based on studies in low- and middle-income countries (4). While both studies acknowledge limitations, particularly the exclusion of costs related to poor-quality treatment, leading to an underestimation, they nonetheless highlight significant consequences of substandard and falsified medical products. These include adverse health effects,

treatment failure, antimicrobial resistance, increased out-of-pocket and system-level health expenditure, diminished public trust in health systems, increased pressure on health care providers, and losses in productivity and income.

Various factors and behaviours contribute to the existence of substandard and falsified medical products in the market, including:

- poor manufacturing practices;
- weak regulatory oversight, including inadequate dossier reviews and lack of regulatory inspections;
- inadequate quality control and quality assurance due to lack of access to quality infrastructure, expertise or services, such as laboratory testing;
- globalization and the increasing complexity of supply chains;
- unregulated alternative supply pathways (informal markets), such as internet sales and online pharmacies;
- criminal activities or networks involved in falsifying medical products that take advantage of weak regulatory oversight;
- poor governance resulting in negligence, corruption or bribery;
- patient or user demand;
- lack of public awareness or education, including among health professionals.

[Fig. A9.1](#) groups these within three converging forces that enable the availability of substandard and falsified medical products.

Fig. A9.1

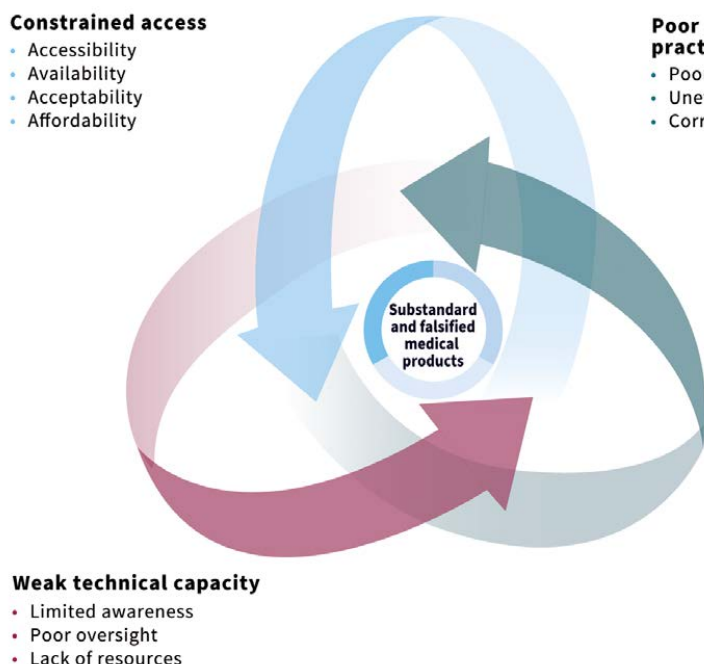
Converging forces that enable the availability of substandard and falsified medical products

Constrained access

- Accessibility
- Availability
- Acceptability
- Affordability

Poor governance practices

- Poor procurement practices
- Unethical practices
- Corruption



Weak technical capacity

- Limited awareness
- Poor oversight
- Lack of resources

Source: World Health Organization (3).

Addressing these forces requires robust regulatory oversight and decisive action by national regulatory authorities (NRAs) and stakeholders to eliminate vulnerabilities and to prevent, detect and respond to substandard and falsified medical products. Although substantial regulatory oversight and resource investments may be prominent in the early stages of a medical product's life cycle, these efforts often diminish over time, even as the number of transactions and stakeholders involved continues to grow. This imbalance contributes to supply chain vulnerabilities as the product gets closer to the patient or end user. While the implementation of controls at the beginning of the life cycle of medical products is critical to ensure that they are safe, efficacious and of high quality, the maintenance of quality controls throughout the supply chain is equally crucial to preserve these attributes across the entire distribution process. However, regulatory oversight becomes more challenging with highly complex globalized supply chains involving numerous intermediaries.

1.2 Importance of market surveillance and control

Market surveillance and control are complementary regulatory functions that together safeguard the safety, quality and efficacy of medical products throughout their life cycle. “Control” refers to “the state wherein correct procedures are being followed, and criteria are being met”, while a “control measure” is defined as “any action and activity that can be used to prevent or eliminate a (product) quality hazard or reduce it to an acceptable level” (5). For example, the implementation of a requirement that delivery vehicles that transport cold storage medicines should maintain refrigerator temperatures in the range 2–8 °C is a control measure.

“Market surveillance” in the context of this guideline refers to “the activities carried out and measures taken by public authorities to ensure that products comply with the requirements set out in the relevant legislation and do not endanger health, safety, or any other aspect of public interest protection” (6). In the context of medical devices, this includes ensuring that the performance of the medical device remains adequate (7). Following the earlier example, periodic checks of temperatures of refrigerated delivery vehicles and tests of the quality of medicines transported would be a surveillance activity.

Therefore, market controls aim to ensure that products circulating in the given market’s supply chain remain within specifications (that is, they do not become substandard) and falsified versions do not enter or penetrate the regulated supply chain, whereas market surveillance undertakes checks to detect the presence of substandard and falsified versions of products to ensure that they are not used or they are prevented from entering the supply chain.

Medical product vigilance, commonly known as pharmacovigilance (in the context of pharmaceuticals), encompasses the scientific methods and regulatory activities aimed at preventing, detecting, assessing and understanding adverse effects – such as adverse events following immunization in the case of vaccines – and other problems related to medical products (8). This core regulatory function supports the ongoing assurance of product safety and efficacy once products are on the market and is closely linked to market surveillance and quality control, as it involves the systematic collection and analysis of data on adverse reactions or therapeutic failures. Such data can reveal the presence of substandard and falsified products; for example, a spike in treatment failure reports may point to a defective batch or counterfeit product, prompting further investigation and regulatory action.

Operational themes of market surveillance and control

Considering the complementary nature of market surveillance and control, the primary goal of this function is to ensure that medical products comply with predetermined criteria for quality, safety and efficacy (or performance) following market approval. In well resourced settings, this function generally involves four

operational themes: (a) control of import activities; (b) prevention of, detection of, and response to substandard and falsified medical products; (c) market surveillance for the monitoring of quality of medical products throughout the supply chain; and (d) control of promotional, marketing and advertising activities (9). However, successful market surveillance and control require robust regulatory systems with sufficient capacity for their implementation and effective operationalization of other related regulatory functions, such as regulatory inspections, licensing of establishments, vigilance and laboratory testing. Given the complexity of manufacturing and distribution of medical products, the work to prevent, detect and respond to substandard and falsified medical products requires a robust regulatory framework with oversight throughout the product life cycle, coordination among local stakeholders, and an active and effective global network of regulators and international stakeholders from various sectors.

1.3 WHO Member State Mechanism and priorities

The WHO Member State Mechanism on Substandard and Falsified Medical Products was established in 2012 by the World Health Assembly with the mandate “to protect public health and promote access to affordable, safe, efficacious and quality medical products; promote through effective collaboration among Member States and the Secretariat, the prevention, control of substandard/spurious/falsely labelled/falsified/counterfeit medical products and associated activities”, whereby intellectual property considerations were excluded (10). The Member State Mechanism includes representatives of Member States who are guided by its terms and conditions (World Health Assembly resolution WHA65.19) (11). Since its inception, the Member State Mechanism has developed a biennial list of priorities for its workplan, including prioritized activity A, which includes the development of a technical guideline to strengthen the capacity of national and regional regulatory authorities to plan, perform and assess risk-based market surveillance (12). Although the priorities of the Member State Mechanism may change periodically, this guideline is intended to support market surveillance and control functions of NRAs over a longer duration.

1.4 Prevent–detect–respond strategy

In November 2017, as part of the report of the Global Surveillance and Monitoring System (GSMS), WHO presented the prevent–detect–respond strategy as a policy framework to guide Member States in formulating national action plans to address substandard and falsified medical products (13). The strategy, which has been adopted by the Member State Mechanism, presents 12 actions for NRAs based on six objectives within three pillars. It may be used to develop strategic goals, actions and activities within national action plans.

Although presented under different objectives, the actions and stakeholders are interrelated and interdependent, requiring careful planning by Member States (13). The implementation of the prevent–detect–respond actions will also support regulatory system strengthening and global benchmarking for market surveillance and control. Table A9.1 presents the prevent–detect–respond objectives and actions, and the related regulatory functions outlined in the WHO Global Benchmarking Tool for the evaluation of regulatory authorities (9, 13).

Table A9.1
Elements of the prevent–detect–respond framework and interrelated regulatory functions

Objectives	Actions	Regulatory functions
Prevention 1. Demand quality 2. Secure supply	Comprehensive legal framework	market surveillance and control, regulatory systems
	Education and awareness	market surveillance and control
	Supply chain integrity	licensing of establishments, lot release, market authorization/registration, market surveillance and control
	Multistakeholder engagement	market surveillance and control, regulatory systems, vigilance
Detection 3. Improve detection 4. Increase reporting	Border control	clinical trials, market surveillance and control
	Risk-based inspections and surveillance	market surveillance and control, regulatory inspections
	Access to laboratories and screening technologies	laboratory testing
	Reporting systems	market surveillance and control, vigilance

Table A9.1 *continued*

Objectives	Actions	Regulatory functions
Response 5. Protect public health 6. Prevent recurrence	Alerts and recalls	market authorization/ registration, market surveillance and control, regulatory systems, vigilance
	Regulatory strengthening	regulatory systems
	Transparent legal process	market surveillance and control, regulatory systems
	Evidence-based policy and procedure	market surveillance and control

1.5 How to use this guide

1.5.1 Target audience

This guide is developed for NRAs that are seeking to strengthen market surveillance and control of medical products to protect public health. It will enable NRAs to assess their current function and plans and implement strategies to conduct market surveillance and control. It is intended for administrators, including division leaders and technical officers of NRAs who are responsible for activities that directly or indirectly contribute to market surveillance and control. Other stakeholders involved in or supporting market surveillance and control – such as health professionals, academia, students, manufacturers, importers, procurement agencies and donors – may consult this guide to better understand the roles, responsibilities and recommended activities of NRAs, and their own respective roles.

1.5.2 Context

This consolidated guidance aims to provide NRAs with a structured framework and practical recommendations for undertaking the market surveillance and control function. It draws from published guidance and expertise of contributing NRAs to support the adoption or adaptation of WHO standards and other relevant scientific best practices for efficient and effective oversight of medical products. The guideline is designed to support a risk-based approach by NRAs for market surveillance and control of medical products used for humans. It focuses on the identification of priority products and facilities, proactive monitoring, information sharing and optimal resource allocation to mitigate risks associated with substandard and falsified medical products.

The good practices for NRAs are presented within each section as numbered action statements preceded by the acronym “GP” in bold font according to the section number, followed by statements with additional practices, considerations and recommendations (for example, “GP2.1 Support the development of competence of health professionals”). The sequence restarts with each section, and a list of the key overarching recommendations is provided at the end of the document. Although the guideline refers to responsibilities and actions of regulated entities and stakeholders, the recommendations are framed using a regulatory perspective. For instance, the statements focus primarily on what the NRA should implement, or what the NRA should ensure is implemented, by regulated entities or stakeholders.

1.5.3 NRA application

There is no single universal model for the regulation of medical products that can be directly applied to all country contexts. Therefore, each NRA should adopt or adapt the good practices provided according to the context of national health policies and priorities, national socioeconomic status, the availability of resources (including financial, human and digital resources) and infrastructure, the health system’s capacity, the national legal framework, research and development capacity, and local production capacity.

NRAs should utilize the guide to:

- assess current capacity and execution of market surveillance and control;
- develop national action plans for risk-based market surveillance and control based on identified gaps and local needs;
- adapt or adopt the practices, approaches, tools and resources for implementation, and according to the national context, regulatory scope and capacity.

2. Glossary

active pharmaceutical ingredient. Any substance or mixture of substances that is intended to be used in the manufacture of a pharmaceutical dosage form and that, when so used, becomes an active ingredient of that pharmaceutical dosage form. Such substances are intended to furnish pharmacological activity or other direct effect in the diagnosis, cure, mitigation, treatment or prevention of disease or to affect the structure and function of the body.

adverse drug reaction. A response to a drug that is noxious and unintended, which occurs at doses normally used in humans for the prophylaxis, diagnosis or therapy of disease or the modification of physiological function.

adverse event following immunization. Any untoward medical occurrence that follows immunization. It does not necessarily have a causal relationship with the usage of the vaccine. The adverse event may be any unfavourable or unintended sign, abnormal laboratory finding, symptom or disease. It is also referred to as “an event supposedly attributable to vaccination or immunization”.

adverse event (medical devices). Any event associated with a medical device that led to death of or serious injury to a patient, user or other person, or that might lead to death of or serious injury to a patient, user or other person if the event recurs.

adverse event (pharmaceuticals). Any untoward medical occurrence in a subject administered a pharmaceutical product; it does not necessarily have a causal relationship with the treatment.

aide-memoire. A practical tool that summarizes key points and recommendations for managing incidents of substandard or falsified medical products.

assessment. This term encompasses any evaluation conducted for a regulatory function (for example, evaluation of a clinical trial application or an initial marketing authorization for a medical product, any subsequent post-authorization changes, evaluation of safety data, or evaluation as part of an inspection).

authorized product. A product in compliance with national and regional regulations and legislation. National or regional regulatory authorities can, according to national or regional regulations and legislation, permit the marketing and distribution of medical products with or without registration or licence.

batch. A defined quantity of starting material, packaging material or product processed in a single process or series of processes so that it is expected to be homogeneous.

competence. Knowledge, skills and attitude required for successful work performance.

competent authority. Any person or organization that has the legally delegated or invested authority, capacity or power to perform a designated function. This term is often used in legal and regulatory contexts to refer to entities that have the official power to make decisions and enforce laws within a specific area of public policy.

consignment (or delivery). The quantity of medical products made by one manufacturer and supplied at one time in response to a particular request or order. A consignment may comprise one or more packages or containers and may include material belonging to more than one batch.

corrective and preventive actions. A system for implementing corrective and preventive actions arising from investigations into complaints, product rejections, non-conformances, recalls, deviations, audits, regulatory inspections, and findings and trends from process performance and product quality monitoring.

decommissioning. Removal of medical devices from their originally intended uses in a health care facility to an alternative use or disposal.

distribution. The procuring, purchasing, holding, storing, selling, supplying, importing, exporting or movement of medical products, with the exception of dispensing or providing pharmaceutical products directly to a patient or their agent.

distribution chain. A collective term for local manufacturers' authorized representatives, importers, and distributors established within the jurisdiction.

excipient. A substance or compound, other than the active pharmaceutical ingredient and packaging materials, that is intended or designated to be used in the manufacture of a pharmaceutical product.

falsified product. A product that has been deliberately or fraudulently misrepresented as to its identity, composition or source. Such deliberate or fraudulent misrepresentation refers to any substitution, adulteration or reproduction of an authorized product or the manufacture of a product that is not an authorized product. This definition excludes breaches of intellectual property rights.

“Identity” shall refer to the name, labelling, packaging or documents that support the authenticity of an authorized product.

“Composition” shall refer to any ingredient or component of the product in accordance with applicable specifications authorized or recognized by the national regulatory authority.

“Source” shall refer to the identification, including name and address, of the marketing authorization holder, manufacturer, importer, exporter, distributor or retailer, as applicable.

field safety corrective action. An action taken by a manufacturer to reduce or remove a risk of death or serious deterioration in the state of health associated with the use of a medical device that is already placed on the market.

field safety notice. A communication sent out by a manufacturer or its representative to the device users in relation to a field safety corrective action. *Note:* A field safety notice may also be of a non-safety nature, for example, quality-related customer product information.

Global Benchmarking Tool. A WHO technical instrument used to assess the maturity and performance of national regulatory systems for medical products. Its purpose includes supporting regulatory system strengthening as a means to mitigate the global threat of substandard and falsified medical products. The Global Benchmarking Tool helps WHO Member States and partners to strengthen capacities to prevent, detect and respond to such products through structured evaluation, institutional development planning and monitoring of progress.

Global Surveillance and Monitoring System (GSMS). The system aims to work with WHO Member States to improve the quality of reporting of substandard and falsified medical products and to ensure that the data collected are analysed and used to influence policy, procedure and processes to protect public health at the national, regional and global levels.

Global Surveillance and Monitoring System focal point. Global Surveillance and Monitoring System focal points are national points of contact who provide, receive and share information about potentially substandard or falsified medical products with WHO, regional networks and other partners. They are nominated to exchange information and report substandard and falsified incidents to the WHO GSMS portal.

good distribution practice. That part of quality assurance that ensures that the quality of a medical product is maintained by means of adequate control of the numerous activities that occur during the distribution process as well as providing a tool to secure the distribution system from unapproved, illegally imported, stolen, falsified, substandard, adulterated, or misbranded medical products.

good manufacturing practice. That part of quality assurance that ensures that pharmaceutical products are consistently produced and controlled to the quality standards appropriate to their intended use and as required by the marketing authorization.

good practice. An umbrella term describing a collection of regulatory quality standards intended to ensure that medical products are consistently safe, effective, and of required quality; meet their intended use; and adhere to quality processes during production, control, storage and distribution. Note: the various types of good practice are often referred to generically by the term “good x practices”, or GxP”.

good storage practice. That part of quality assurance that ensures that the quality of medical products is maintained by means of adequate control throughout the storage thereof.

governance. Refers to the different ways that organizations, institutions, businesses and governments manage their affairs. Governance is the act of governing and

thus involves the application of laws and regulations, but also of customs, ethical standards and norms. Good governance means that affairs are managed well, not that the laws, regulations or norms are themselves necessarily “good”.

guidelines or guidance documents. Non-statutory advisory publications intended to assist those parties affected by legislation to interpret requirements.

health professional. Any individual who studies, advises on or provides preventive, curative, rehabilitative and promotional health services based on an extensive body of theoretical and factual knowledge in the diagnosis and treatment of disease and other health problems. Health professionals may conduct research on human disorders and illnesses and ways of treating them and supervise other workers. The knowledge and skills required are usually obtained as the result of study at a higher educational institution in a health-related field for three to six years, leading to the award of a first degree or higher qualification.

importation. The act of bringing or causing any goods to be brought into a customs territory (national territory, excluding any free zone).

importer. An individual or company or similar legal entity importing or seeking to import a medical product. A “licensed” or “registered” importer is one who has been granted a licence for the purpose.

incident. The discovery of substandard or falsified medical products at one time and place. A “product” refers to a particular medicine, vaccine or medical device. An incident can refer to a single dose of a medicine or a container filled with millions of doses.

indicator. The criterion used to measure changes, directly or indirectly, and to assess the extent to which the targets or objectives of a programme or project are being attained. Indicators should meet the criteria of clarity, usefulness, measurability, reliability, validity (see below) and acceptance by key stakeholders.

informal market. A sector of the national or local economy where:

- the manufacture, import or export, distribution, sale, supply or purchase of medical products takes place outside the legal, regulatory or administrative oversight of relevant public health or regulatory authorities;
- the medical products have or have not been assessed for safety, efficacy or quality by public health and regulatory authorities;
- the aforementioned activities may be conducted by persons or entities with or without appropriate qualifications and may take place in a physical, virtual or hybrid environment.

International Criminal Police Organization (INTERPOL). An intergovernmental organization consisting of member countries that helps police in member countries to work together, enabling access to and sharing of data on crimes and criminals. INTERPOL also offers a range of technical and operational support.

International Medical Device Regulators Forum. The International Medical Device Regulators Forum strategically accelerates international medical device regulatory convergence to promote an efficient and effective regulatory model for medical devices that is responsive to emerging challenges while protecting and maximizing public health and safety.

in vitro diagnostic medical device. A medical device, whether used alone or in combination, intended by the manufacturer for the in vitro examination of specimens derived from the human body solely or principally to provide information for diagnostic, monitoring or compatibility purposes.

Note: in vitro diagnostic medical devices include reagents, calibrators, control materials, specimen receptacles, software, and related instruments or apparatus or other articles and are used, for example, for the following test purposes: diagnosis, aid to diagnosis, screening, monitoring, predisposition, prognosis, prediction and determination of physiological status.

licence holder. An individual or a corporate entity possessing marketing authorization for a medical product.

manufacturer (medical devices). Any natural or legal person with responsibility for the design or manufacture of a medical device with the intention of making the medical device available for use under its name, whether or not such a medical device is designed or manufactured by that person themselves or on their behalf by another person or persons.

Note: this “natural or legal person” has ultimate legal responsibility for ensuring compliance with all applicable regulatory requirements for the medical device in the countries or jurisdictions where it is intended to be made available or sold, unless this responsibility is specifically imposed on another person by the regulatory authority within that jurisdiction.

manufacturer (pharmaceuticals). A company that carries out operations such as production, packaging, repackaging, labelling and relabelling of pharmaceuticals.

marketing authorization (product licence, registration certificate). A legal document issued by the competent national regulatory authority that authorizes the marketing or free distribution of a medical product in the respective country

after evaluation for safety, efficacy and quality. In terms of quality, it establishes, inter alia, the detailed composition and formulation of the medical product and the quality requirements for the product and its ingredients or components. It also includes details of packaging, labelling, storage conditions, shelf-life and approved conditions of use.

marketing authorization holder. The entity (company or organization) that is granted the regulatory approval to commercialize the finished product in a country.

market surveillance and control. The market surveillance and control function plays a crucial role in assuring the consumer safety of medical products. Its objective is to ensure compliance of the products placed on the market with preset criteria for quality, safety and efficacy (that is, verification of compliance with marketing authorization and good practice guidelines). Market surveillance and control function activities are primarily concerned with four themes: (a) control of import activities; (b) prevention and detection of and response to substandard and falsified medical products; (c) market surveillance programme for monitoring the quality of medical products throughout the supply chain; and (d) control of promotional, marketing and advertising activities. The aforementioned activities may or may not be undertaken by a single entity (for example, organization, division or department).

medical device. Any instrument, apparatus, implement, machine, appliance, implant, reagent for in vitro use, software, material or other similar or related article intended by the manufacturer to be used, alone or in combination, for human beings for one or more of the specific medical purposes of:

- diagnosis, prevention, monitoring, treatment or alleviation of disease;
- diagnosis, monitoring, treatment, alleviation of, or compensation for an injury;
- investigation, replacement, modification, or support of the anatomy or physiological process;
- supporting or sustaining life;
- control of conception;
- cleaning, disinfection or sterilization of medical devices;
- providing information by means of in vitro examination of specimens derived from the human body.

The scope includes any medical device that does not achieve its primary intended action by pharmacological, immunological or metabolic means in or on the human body but that may be assisted in its intended function by such means.

medical product. Any finished pharmaceutical product or medical device, including in vitro diagnostic medical devices and vaccines.

national regulatory authority (NRA). The national agency responsible for the marketing authorization of and other regulatory activities concerning medical products.

pharmaceutical product. Any product intended for human use or veterinary product intended for administration to food-producing animals, presented in its finished dosage form, which is subject to control by pharmaceutical legislation in either the exporting or the importing state. It includes products for which a prescription is required, products that may be sold to patients without a prescription, biologicals and vaccines. It does not, however, include medical devices.

prevent–detect–respond. A three-pronged strategy developed by WHO that encompasses mutually reinforcing actions and systems to tackle substandard and falsified medical products.

product recall. A process for removing a medical product (or its batches) from the distribution chain because of defects in the product, complaints of serious adverse reactions to the product, or concerns that the product is or may be falsified. The recall might be initiated by the manufacturer, importer, wholesaler, distributor or a responsible agency.

quality control. All measures taken, including the setting of specifications, sampling, testing, and analytical clearance, to ensure that raw materials, intermediates, packaging materials, and finished pharmaceutical products conform with established specifications for identity, strength, purity and other characteristics.

quality control (medical devices). An internal procedure that monitors the testing process to verify the system is working correctly and gives confidence that the results are reliable enough to be released.

quality management system. An effective infrastructure encompassing the organizational structure, procedures, processes, resources and systematic actions necessary to ensure adequate confidence that a product or service will meet the given requirements for quality.

quality manual. A document that includes the quality policy and objectives and describes the various elements of the quality management system.

quarantine. The status of medical products isolated physically or by other effective means while a decision is awaited on their release, rejection or reprocessing.

rapid alert. An urgent notification submitted by a national regulatory authority concerning measures taken against a product placed on the market that poses a risk to consumers' health or safety.

regulation. A written instrument containing rules having the force of law.

regulations. The second stage of the legislative process (the first stage being legislation). Regulations are specifically designed to provide the legal framework for achieving the administrative and technical objectives of the legislation.

regulatory system. The combination of institutions, processes and the regulatory framework with which a government controls particular aspects of an activity.

reliance. The act whereby a regulatory authority in one jurisdiction takes into account and gives significant weight to assessments by another regulatory authority or trusted institution or to any other authoritative information in reaching its own decision. The relying authority remains independent and is responsible and accountable for the decisions taken, even when it relies on the decisions, assessments and information of others.

risk assessment. A systematic process of organizing information to support a risk decision to be made within a risk management process. It involves identifying hazards and evaluating the risks associated with exposure to those hazards.

risk evaluation. The comparison of the estimated risk to given risk criteria using a quantitative or qualitative scale to determine the significance of the risk.

risk management. The systematic application of quality management policies, procedures and practices to the tasks of assessing, controlling, communicating and reviewing risk.

screening technologies. The qualitative or semiquantitative technologies that can rapidly acquire information or analytical data for preliminary identification of suspect medical products in the field.

special authorization. A form of market authorization granted to products that have been deemed urgently needed or necessary for unique situations but have not been assessed via the official market authorization procedure. This includes products granted authorization for compassionate use, investigational use, or temporary or conditional authorization, where there may be limits on quantities permitted and conditions of use.

specification. A list of detailed requirements with which the products or materials used or obtained during manufacture must conform. They serve as a basis for quality evaluation.

substandard and falsified (medical products). A term used to collectively refer to medical products considered “substandard” or “falsified” (see definitions in this glossary).

substandard product. “Substandard” medical products (also called “out of specification”) are authorized by a national regulatory authority but fail to meet either national or international quality standards or specifications – or, in some cases, both.

unregistered or unlicensed (medical products). Medical products that have not undergone evaluation or approval by the national or regional regulatory authority for the market in which they are marketed, distributed or used, subject to permitted conditions under national or regional regulation and legislation.

vigilance (medical products). Medical product vigilance, defined as the science and activities related to the prevention, detection, assessment and understanding of adverse effects or any other medical product-related problems, is crucial for ensuring that medical products continue to meet requirements for quality, safety, efficacy and performance throughout the product life cycle.

3. National regulatory systems

The NRAs are ultimately responsible for overseeing and regulating medical products in local markets. They require the authority, enforcement power and capacity to plan, implement and assess implemented strategies to prevent, detect and respond to activities, actions and behaviours that result in substandard or falsified medical products. In countries without an existing NRA, there is a need for a competent authority that enforces a national regulatory system that includes, at a minimum, pre-market assessment, market control measures, vigilance and market surveillance. Countries with existing regulatory authorities should continuously seek to strengthen their regulatory systems to ensure effective oversight.

For effective market surveillance and control of medical products, NRAs should be able to perform the following activities (7):

- implement strategies and controls to prevent the entry and use of substandard and falsified medical products, including standards, policies and guidelines;
- detect substandard and falsified medical products using appropriate resources and mechanisms (for example, field testing or laboratory testing);
- receive reports from stakeholders (including the public) to enable detection of and response to substandard and falsified medical products;

- facilitate reporting of substandard and falsified medical products to the appropriate system and stakeholders;
- monitor and assess the risks posed by substandard medical products;
- conduct routine and targeted inspections of facilities, including manufacturing sites and distribution channels, especially for higher-risk products;
- enforce penalties and sanctions against non-compliant entities or unauthorized companies that undermine supply chain integrity or facilitate the entry or use of substandard and falsified medical products;
- recall products or facilitate the recall of products (where applicable) from the market and apply other corrective actions to remove or correct substandard and falsified products;
- implement risk mitigation strategies, including risk communication, to protect the public from substandard and falsified medical products;
- revise or improve national standards, guidelines, procedures and other regulatory tools for market surveillance and control;
- collaborate with other NRAs, international organizations, and other stakeholders to strengthen cross-border control measures and global systems.

Therefore, effective regulatory systems require comprehensive legal frameworks, good governance, adoption of international norms and standards, competent staff, functional quality management systems, collaborative agreements with other NRAs and stakeholders, and the implementation of good regulatory practices appropriate to the national context and developmental goals. In settings where the financial and technical resources may be limited, and regulation of medical products is minimal, a stepwise approach to regulation is recommended, with development from basic controls (for example, publication of law, import controls, resourcing the NRA) to expanded controls, such as inspection of establishments (6).

3.1 Comprehensive legal framework

A robust legal framework for the regulation of medical products, including promulgated and enforced laws, is crucial for effective regulation of medical products. The elements of the framework include guidelines with the most detail and flexibility (easy to change), regulations that describe conditions for compliance, and laws with the least detail and flexibility (most difficult to change) (Table A9.2) (14).

Table A9.2
Architecture of regulatory frameworks

Component	Features
Laws	Define the mandate of the regulatory authority Define the authorities for making regulations State what behaviours are authorized or prohibited (products, persons, actions) Enacted by the legislative branch of the government
Regulations	State, at a high level, conditions to be met (for example, the responsible authority may issue market registration if sufficient evidence) Enacted by the executive branch of the government
Guidelines	Provide detail on how the conditions may be met (for example, what is considered sufficient evidence) Provide flexibility and adaptability Issued by regulatory authority

3.1.1 Regulatory frameworks for market surveillance and control

GP1.1 Ensure that the regulatory framework clearly defines or describes the NRA's mandate

The regulatory framework should enable the NRA to effectively conduct market surveillance and control based on a clear mandate, which empowers the NRA with the necessary oversight. The appropriate development and communication of the legal framework will prevent duplication of regulatory efforts, facilitate collaboration among enforcement agencies, and ensure a focused remit for regulatory oversight of medical products.

The regulatory framework (14) should:

- define the role, structure, functions and scope of authority of the NRA to enable appropriate regulatory actions to prevent, detect and respond to substandard and falsified medical products;
- identify and describe the roles of regulated entities, health workers, customs and law enforcement agencies, and other stakeholders;
- delineate standards, guidelines, and procedures for the regulation of all medical product categories;
- enable the necessary flexibility and responsiveness, particularly for public health emergencies and the introduction of new technologies or practices;

- promote regulatory cooperation and regulatory efficiencies with local, regional and international partners;
- facilitate alignment of provisions with international standards to promote consistency for regulatory harmonization and convergence, and facilitate international cooperation, global health security and trade.

The regulatory framework should empower the NRA to implement the following activities:

- develop regulations supported by guidelines that:
 - define regulatory conditions for market surveillance and control, such as licensing requirements, good manufacturing practices, traceability standards, and good storage and distribution practices;
 - define specific procedures or pathways for regulatory processes for routine activities and public health emergencies;
- collect and analyse samples and information relevant to potential risks, incidents and trends, enabling timely interventions;
- conduct inspection of facilities and reporting of quality complaints, product quality issues, and relevant safety concerns for timely response, assurance of public safety and development of trust;
- manage risks associated with medical products via guidelines, standards, education of consumers and health professionals, risk assessment, and risk communication to ensure that products on the market meet quality and safety requirements;
- collaborate with other regulatory agencies for information sharing, regulatory harmonization, and coordination of actions in a timely manner;
- implement regulatory actions, including restrictions of use, market suspension, withdrawal, seizure, product recalls, warnings, and destruction, in a timely manner in response to substandard and falsified medical products that pose an immediate or imminent threat to public health.

GP1.2 Ensure that the legal framework clearly describes the scope of medical products and facilities to be regulated

The legal provisions within the framework should:

- identify and define the types of medical products covered (for example, pharmaceuticals, biologics including vaccines, medical devices including in vitro diagnostics), and the populations for use (for example, humans, animals);
- specify the relevant aspects of the medical product life cycle that are subject to regulatory oversight by the NRA;
- specify the elements that constitute the legal supply chain, including the types of facilities or sites that are subject to regulation (for example, manufacturing facilities, importers, distributors, warehouses, dispensers, retail pharmacies, hospitals);
- describe the scope and extent of authority, roles and responsibilities of the NRA and partner agencies involved in law enforcement for the prevention of, detection of, and response to substandard and falsified medical products.

GP1.3 Ensure that legal provisions facilitate activities to prevent substandard and falsified medical products

The legal framework should empower the NRA to proactively prevent the entry of substandard and falsified medical products into the market by setting standards and penalties, assessing medical products in the market, licensure of establishments, assessment of risks, and provision of education.

The legal provisions should enable the NRA to set standards, requirements and penalties, specifically:

- quality standards, safety requirements and efficacy criteria for medical products, active ingredients or constituents to be imported and used in local markets;
- criteria and conditions for online sellers;
- guidelines for the production, handling, storage and distribution of medical products using good manufacturing practices and good storage and distribution practices;
- regulatory and legal actions that would be applied in the event of violations, such as illicit online sales.

Legal provisions should enable the NRA to assess or review authorized medical products in active distribution by:

- enabling the NRA to require product information relevant to the quality, safety and efficacy of medical products to be used in local markets according to defined standards as a requirement for market authorization;

- authorizing the NRA to periodically review conditions of market authorization and post-authorization changes of medical products and, where necessary, suspend or cancel the market authorization;
- enabling the use of regulatory efficiencies and tools, such as reliance, recognition, certificates of pharmaceutical product, national databases, traceability systems, artificial intelligence, data mining, research, and quality control testing for regulatory decisions to prevent substandard and falsified medical products.

The legal provisions should empower the NRA to implement licensing of establishments and approval for manufacturing, importation, distribution and disposal of medical products by:

- issuing licences, permits or registrations to manufacturers, importers, distributors, dispensers and disposal facilities by the NRA;
- defining standards, requirements for compliance, responsibilities and permitted activities of licensed establishments that constitute the legal supply chain, including online sellers, and the respective conditions for maintaining licensure;
- establishing systems for the approval of medical products imports, including starting materials, in collaboration with customs and other relevant authorities;
- facilitating periodic review, modification, suspension or cancellation of licences or import approvals, as necessary, based on risk or non-compliance.

Legal provisions should allow the NRA to assess possible risks and provide education and awareness by:

- developing and implementing policies and procedures to assess risks based on incidents and alerts from other NRAs, regional networks and international organizations;
- developing policies and procedures for education and awareness of stakeholders (for example, health professionals, patients, consumers) to prevent market entry and use of substandard and falsified medical products.

GP1.4 Ensure that legal provisions enable the detection of substandard and falsified medical products

The legal framework should empower the NRA to detect substandard and falsified medical products through systematic surveillance using inspections and

audits, information systems, sampling and testing, reporting, assessment of risks and sharing of information.

Legal provisions should support inspections and audits by the NRA, specifically:

- empowering the NRA to conduct inspections at manufacturing and distribution facilities using global standards;
- authorizing inspectors to enter premises, review records (administrative, technical and financial), and collect samples;
- enabling the use of regulatory efficiencies and tools, such as reliance, recognition, digital intelligence, and national databases of other NRAs that publicize inspection findings, good manufacturing practice status, and quality testing results.

Legal provisions should facilitate the use of information systems for product traceability, data analysis, vigilance, and monitoring of promotions and sales by (15):

- allowing the use of appropriate traceability systems with clearly defined governance structures, standards, databases and requirements;
- facilitating the use of information systems, digital tools, research and alerts from regional or global systems for the detection of falsified medical products, interoperability with vigilance systems and information exchange;
- monitoring media and the internet for violations related to unauthorized online sales, unethical or misleading promotions, quality complaints, and reports of suspected falsification of medical products, and enabling a regulatory response to such violations.

Legal provisions should facilitate sampling and testing by:

- allowing the NRA to sample and test medical products throughout the supply chain to verify compliance with standards and nature of quality issue;
- enabling the implementation of policies and procedures for risk-based sampling and testing.

Legal provisions should support reporting, risk assessment and information sharing by:

- empowering the NRA in the establishment of reporting systems to encourage timely reporting of incidents of therapeutic failure,

quality defects, adverse events and illicit sales where there is suspicion of substandard and falsified medical products;

- allowing the NRA to collect and utilize reports and other relevant information from the public, industry, testing laboratories, WHO GSMS and other stakeholders;
- empowering the NRA to conduct a risk assessment and analysis of inspection findings and incident reports received from stakeholders;
- allowing the NRA to implement policies and procedures for information sharing with WHO GSMS, other regulatory agencies and international bodies.

GP1.5 Ensure that legal provisions empower the NRA to respond to detected substandard and falsified medical products

The legal framework should equip the NRA with the necessary authority and provisions to take swift and effective action through risk communications, regulatory actions, and coordination with law enforcement, health professionals, manufacturers and other stakeholders.

Legal provisions should enable risk communications, product recalls and withdrawals by:

- authorizing the NRA to recall, suspend, withdraw, seize and dispose of substandard and falsified medical products from the market by:
 - enabling the NRA to implement procedures for appropriate risk communications for stakeholders, including the issuance of rapid alerts;
 - empowering the NRA to establish regulations, requirements and procedures for appropriate safe storage and disposal of substandard and falsified medical products.

Legal provisions should support the application of sanctions, penalties and criminal prosecution, specifically by:

- empowering the NRA to define conditions and penalties for non-compliance (for example, fines, licence revocation, seizure, safe disposal);
- empowering the NRA to develop and enforce regulatory actions proportionate to intention, risk and harm, where applicable.

Legal provisions should support coordination and collaboration with law enforcement, environmental protection authorities, health professionals, industry, regional or international networks, and civil society by:

- defining roles, terms and conditions for collaboration with customs, law enforcement agencies and environmental authorities for regulatory actions (for example, seizure, safe disposal) and criminal investigations;
- defining roles, terms and conditions for collaboration with customs, law enforcement and environmental protection agencies for identification and action against informal markets, non-compliant or unlicensed establishments, sellers of falsified products, and agents of illicit or unauthorized trade of medical products;
- delineating roles, terms and conditions for collaboration, cooperation, and information sharing with industry, distributors, health professionals, the public, other NRAs, and regional and international networks or reporting systems, where appropriate.

GP1.6 Ensure that legal provisions empower the NRA to effectively manage, monitor and evaluate the market surveillance and control function

The legal framework should enable the implementation of a quality management system and the collection of data for regular monitoring, evaluation and improvement of the national regulatory system, including the market surveillance and control function. When developing or strengthening the quality management system, the NRA should:

- clearly describe the scope, functions and processes relevant to market surveillance and control delineated by the legal framework;
- clearly describe the organization of the NRA;
- describe the technical and administrative functions and processes that constitute the routine market surveillance and control operations of the NRA and any interrelated functions (for example, regulatory inspections);
- clearly describe the scope for each institution involved in regulation of medical products of the country, where applicable;
- support consistency, effectiveness, efficiency and systematic collaboration to improve coordination among institutions and the divisions of the NRA;
- develop performance indicators for the evaluation of the market surveillance and control function;
- describe the methods or procedures and tools for performance monitoring, analysis and evaluation of operational indicators, quality management system effectiveness, resources and customer satisfaction;

- describe the procedures for continuous improvement of market surveillance and control based on the data collected.

Additional information to support the development or strengthening of the quality management system is provided in the *WHO guideline on the implementation of quality management systems for national regulatory authorities* (16) and the *Quality management system requirements for national inspectorates* (17).

The evaluation of the national regulatory system and market surveillance and control function should be informed by the respective indicators and sub-indicators provided by the Global Benchmarking Tool, according to the scope of medical products to be regulated and the national context (9).

3.1.2 Resources for developing or strengthening legal frameworks

In the WHO Technical Report Series (TRS) 885, Annex 8 (1999), WHO provides guidance on the essential aspects of medicines regulatory legislation that may assist NRAs with limited capacity to formulate laws and regulations for authorization of medical products and their control while on the market. It identifies critical objectives, including the responsibility of NRAs, and provides recommendations for the required systems for the NRA to be effective, the scope of the legislation, the structure of the NRA, authorization of products and facilities, enforcement, monitoring and evaluation, and technical competence (18).

The WHO Global Model Regulatory Framework provides additional details and recommendations for basic-level and expanded regulatory controls and enforcement for medical devices within the legal framework, which may also inform controls for the regulation of medical products (6).

3.1.3 Supplemental resources for model legislation

In 2019, the United Nations Office on Drugs and Crime launched *Combating falsified medical product-related crime: a guide to good legislative practices* to support countries in implementing or strengthening domestic legislation and in protecting public health. The guide includes sample provisions and guidance that may be adapted to local conditions, as well as principles, structures and information on enforcement arrangements (19).

The *Reference manual on legislation on substandard and falsified medical products: an African perspective*, published in late 2023 via the European Union MEDISAFE project, provides guidance on the necessary legal framework for national and regional responses to substandard and falsified medical products. It addresses regulatory systems based on the African Union Model Law on Medical Products Regulation and related guidance, following the context of the nine functions delineated by the Global Benchmarking Tool for the evaluation of national regulatory systems. It also covers the criminal justice response, describing

the offences to be established for effective enforcement, tools, international cooperation, and measures to effectively prevent and respond to falsified medical products (20).

These, along with the WHO guidance mentioned above, may be used by Member States to develop or strengthen legislation for the local context to address substandard and falsified medical products and the crimes that facilitate their entry.

3.2 Good regulatory practices

With the growing complexity of medical products and supply chains, it is increasingly difficult for NRAs to regulate medical products. As a result, NRAs need robust regulatory systems that are efficient, effective and responsive to change. However, a sound system requires not only an effective legal framework supported by sufficient resources, capacity and competence, but also good regulatory practices for the effective implementation of functions, setting clear requirements for regulated entities, and ensuring accountability to the public and stakeholders. In 2021, WHO published guidance for good regulatory practices presenting system components, enablers, and nine high-level principles applicable to all NRAs irrespective of resources, maturity or regulatory approach, in response to the need for guidance to address common gaps in regulatory practices and for the development of regulatory instruments (Table A9.3) (14). These principles and enablers may inform policies and practices of national regulatory systems according to resources, goals, health policies and the medical product environment.

Table A9.3
Principles and enablers of good regulatory practices

Main principles	Description
Principles	
Legality	Regulatory systems and the decisions that flow from them must have a sound legal basis
Consistency	Regulatory oversight of medical products should be consistent with existing government policies and legislation and be applied in a consistent and predictable manner
Independence	Institutions that execute regulation of medical products should be independent
Impartiality	All regulated parties should be treated equitably, fairly and without bias

Table A9.3 *continued*

Main principles	Description
Principles <i>continued</i>	
Proportionality	Regulation and regulatory decisions should be proportional to risk and to the regulator's capacity to implement and enforce them
Flexibility	Regulatory oversight should not be prescriptive but rather be flexible in responding to a changing environment and unforeseen circumstances
Clarity	Regulatory requirements should be accessible to and understood by users
Efficiency	Regulatory systems should achieve their goals within the required time and at reasonable effort and cost. International collaboration promotes efficiency by ensuring the best use of resources
Transparency	Regulatory systems should be transparent; requirements and decisions should be made known, and input should be sought on regulatory proposals
Enablers	
• Political and governmental support • Good organization, governance and leadership • Effective communication, collaboration and coordination • Robust and well functioning quality management system • Sufficient and sustainable financial resources • Competent human resources • Preset organizational ethics and values • Science- and data-driven regulatory decision-making process	

The impact of implementing good regulatory practices has been demonstrated through lessons from regulatory system strengthening shared by national and regional regulatory authorities, inclusive of strategies for post-market surveillance, considerations for emergency situations, and surveillance of online sales by reference NRAs ([21](#), [22](#)). All nine principles are important, but the level of priority may vary based on the principles that are already implemented in the national system, particularly within the market surveillance and control function. The following highlights considerations for the implementation of some of the principles of good regulatory practice, but additional practices and examples are further elaborated in WHO TRS 1033, Annex 11 ([14](#)).

GP1.7 Implement good regulatory practices

The NRAs should assess the extent to which the nine principles and related elements of good regulatory practice are implemented within its conduct of market surveillance and control and complementary regulatory functions.

Based on the gaps identified, each NRA should prioritize the implementation or strengthening of the relevant principles by importance or urgency, for instance, where there are conflicting legislative frameworks between the NRA and other oversight bodies (for example, law enforcement agencies) or an absence of regulations enabling international cooperation.

GP1.8 Practice good governance

For good governance in the context of market surveillance and control ([22](#)), the NRA should:

- build trust with stakeholders by engagement and collaboration in anticorruption, integrity and good governance initiatives;
- collaborate with anticorruption agencies for the integration of transparency principles in policies and procedures, international cooperation and timely information sharing;
- develop and implement codes of conduct to highlight norms and expectations of inspectors and other regulatory personnel, regulated entities, advisory committees for regulatory units involved in or complementary to market surveillance and control, and other stakeholders involved in regulatory decision-making to promote ethical behaviour;
- manage conflicts of interest of regulatory personnel, advisory committees and focal points of enforcement agencies;
- communicate core values through norms, guidelines, procedures, training and advocacy (for example, media, awards);
- build capacity in good governance among staff, leaders, experts and health professionals in the medical product sectors.

GP1.9 In the absence of a dedicated system for substandard and falsified medical products, utilize existing systems for vigilance

Given that the quality of medical products is a prerequisite for their safety, and some adverse events may arise from the use of substandard and falsified medical products, national systems for vigilance may support the detection of such products. For instance, adverse events, lack of efficacy, product defects and misdiagnosis due to poor performance of an in vitro diagnostic device may be reported to national vigilance systems. Ideally, a dedicated system for receipt and analysis of incidents

involving substandard and falsified medical products should be established. However, where this is not feasible, the existing vigilance system should be utilized.

To optimize vigilance and market surveillance (21), NRAs should:

- increase stability of and allocate appropriate resources to vigilance and market surveillance systems;
- strengthen collaboration and information sharing with national programmes, regional networks and international reporting systems for vigilance for active support and engagement;
- collaborate with national vigilance systems for information sharing, capacity-building, case management, risk assessment and regulatory response to substandard and falsified medical products, including global reporting.

GP1.10 Utilize evidence in decision-making

The NRA should utilize evidence-informed decision-making to develop and revise policies and procedures for market surveillance and control.

Evidence-based activities conducted by NRAs may include reviews of:

- data and metrics collected from the NRA's surveillance programmes, such as findings from regulatory inspections, user complaints, digital intelligence, information from investigations, quality testing results, and reviews of clusters of adverse events or incidents involving medical products (for example, adverse drug reactions, adverse events following immunization or medical device user feedback);
- guidelines and formal publications from other NRAs, regulatory networks, country cooperations (for example, African Union, Asia-Pacific Economic Cooperation, Pharmaceutical Inspection Co-operation Scheme) and international organizations;
- evidence-based systematic reviews and research relevant to market surveillance and control, particularly for prevention of, detection of, or response to substandard and falsified medical products;
- reports and alerts from international agencies or other NRAs (for example, alerts related to substandard and falsified medical products, news releases, information regarding law enforcement actions against high-risk medicinal products, WHO GSMS alerts), including where regulatory reliance has been applied;
- lessons learned and best practices derived from the NRA's real-world experience with significant incidents, such as recalls and confirmed substandard and falsified medical products.

The NRA should have established policies on the type of evidence and information that will be reviewed and how different types of evidence will be used to identify procedural weaknesses and to inform decision-making. Each NRA should critically evaluate evidence and its sources, including from artificial intelligence, considering limitations inherent in the source and the level of evidence.

The NRA should consider relevant quality research and published reviews, particularly aspects related to context, study design, possible biases, generalizability of conclusions, inclusion of public opinion, equity, feasibility of implementation of recommendations, affordability, sustainability and acceptability to stakeholders (23).

The NRA should exercise caution when using individual case reports and observations as a basis for significant policy and procedure revisions, as there may not always be a causal link and root cause sufficient to support such changes. Further case analyses and causality assessments should be conducted to confirm any association between the suspected product and the adverse event or incident reported. Quantitative studies and systematic reviews may provide sufficient information to support changes to policies, whereas qualitative data, such as case reports and incident investigations, may be more useful to identify emerging issues or potential gaps, which may help prioritize existing surveillance activities and inform assessments of the effectiveness of surveillance measures.

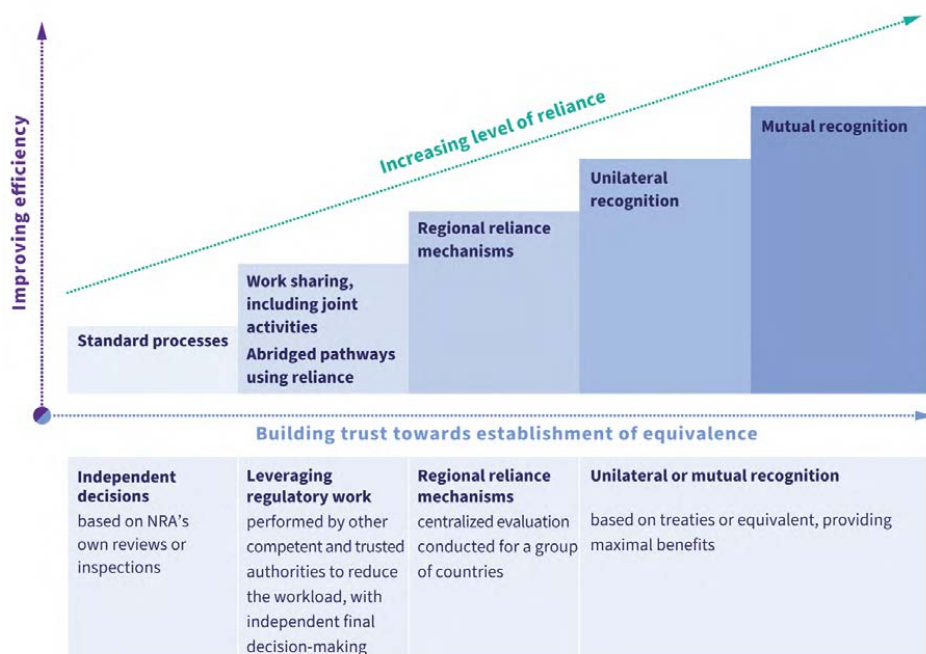
When implementing new policies or procedures or making significant revisions to existing ones, the NRA should conduct preliminary assessments, pilot tests, simulations or feasibility studies to determine the effectiveness of the changes before scaling up (13, 23).

3.2.1 Regulatory reliance for market surveillance and control

Regulatory reliance facilitates timely access to safe, effective, quality-assured medical products and can support regulatory preparedness and response, particularly during public health emergencies. For countries with limited regulatory capacity to undertake all aspects of market surveillance and control, NRAs may apply regulatory reliance to leverage the outputs of other NRAs while maintaining autonomy. The WHO guidance *Good reliance practices in the regulation of medical products: high-level principles and considerations* (WHO TRS 1033, Annex 10) provides recommendations supporting reliance between NRAs in order to make the best use of available resources and expertise (24). This principle allows leveraging the output of others whenever possible while placing a greater focus at a national level on value-added regulatory activities that other authorities cannot undertake. These include vigilance and oversight of local manufacturing and distribution. The guideline presents a range of concepts for regulatory efficiency, including recognition of reliance-based approaches

(for example, work sharing and joint reviews) and provides information on the features of the various approaches (Fig. A9.2). The WHO Global Model Regulatory Framework for medical devices (Annex 3, TRS 1045) provides information on the development of reliance-based pathways that NRAs may use (6).

Fig. A9.2
Key concepts and levels of reliance



As a regulatory strategy, reliance can result in more evidence-based, better-quality decisions, increase efficiency through reallocation of resources of the NRA, and enable learning from assessments and decisions by reference NRAs, where the same product is under consideration.

In the context of market surveillance and control, regulatory reliance may facilitate sharing or exchange of information (21) about:

- regulatory inspections and compliance with good manufacturing practices of common facilities for verification of quality;
- substandard medical products and the conditions of non-compliance with specifications for local monitoring where the same products are on the local market;

- falsified medical products identified in the supply chains of other NRAs, particularly where there are shared channels of distribution;
- unregistered or unlicensed medical products identified in local markets;
- identification of online sellers and informal markets;
- regulatory actions and enforcement in response to substandard and falsified medical products by other NRAs, and global operations against medical product crime.

GP1.11 Determine the use of regulatory reliance for market surveillance and control

The decision to apply reliance (24) should consider:

- the NRA's existing capacity and resources;
- the needs of the national regulatory system and public health priorities;
- the NRA's strategy for reliance, for example, using a risk-based approach;
- the availability of authorities or organizations on which the NRA can rely with confidence;
- opportunities where reliance could complement the capacity to increase efficiency and optimize available resources.

When applying reliance to aspects of market surveillance and control (24), the NRA should determine:

- the type of reliance approach to be used (unilateral versus mutual agreement versus recognition);
- the extent of reliance and how it will inform decision-making;
- the criteria for selection of trusted reference authorities, including maturity level, scope of functions, alignment of requirements with international standards and availability of information for verification;
- the extent of collaboration required with trusted reference authorities, and feasibility;
- the scope of reliance, based on medical products and the related regulatory functions (for example, inspections, vigilance, laboratory testing);
- the procedures, requirements, systems and pathways needed to enable operationalization of reliance for market surveillance and control.

In the context of applying reliance to regulatory inspections ([21](#)), NRAs should:

- examine international inspection strategies to find an optimal combination of risk and efficiency, including relying on trusted authorities;
- leverage trusted good manufacturing practice information, including the exchange of certificates and inspection reports with NRAs of reference and members of regulatory networks or systems (for example, Pharmaceutical Inspection Co-operation Scheme, European Medicines Agency);
- utilize available tools providing good manufacturing practice information, such as public databases (for example, WHO prequalification inspections) to check the good manufacturing practice compliance of sites;
- improve regulatory inspection transparency by making information available on NRA websites and encouraging manufacturers to approve sharing of inspection reports among NRAs;
- intensify active pharmaceutical ingredient (API) and excipient manufacturing oversight, with priority given to facilities where a moderate or high risk of quality issues or incidents of non-compliance leading to substandard products have been identified.

3.2.2 Transparent legal processes

The establishment of a comprehensive legal framework requires activation of multiple tiers, including regulations that frame the legal provisions, guidance for users to comply with regulations, policies to establish operating principles, and procedures to delineate the steps involved. Therefore, both internal and external stakeholders need to be familiar with guidance, requirements and procedures involved in market surveillance and control, according to their respective roles and responsibilities.

GP1.12 Ensure transparency in legal processes

The NRA should clearly describe the criteria, requirements and procedures involved in market surveillance and control, including:

- licensure of establishments or facilities;
- regulatory inspections and investigations;
- product sampling and evidence gathering, where necessary, including APIs and excipients;

- product testing;
- reporting of suspected or confirmed substandard and falsified medical products;
- risk assessment;
- regulatory decision-making and enforcement of regulatory decisions, including seizure, quarantine, product withdrawals and market suspensions;
- submission of queries concerning or appeals against regulatory decisions;
- information sharing with other NRAs;
- communication of medical product information via public interfaces;
- risk communication;
- public consultation, where applicable.

The NRA should (18):

- provide clear, succinct, non-conflicting guidelines, policies and procedures;
- develop feasible procedures to enable implementation by stakeholders and staff;
- communicate regulations, guidelines, policies and procedures to staff, regulated entities, enforcement agencies and other key stakeholders involved in market surveillance and control prior to their implementation and in response to detected non-compliance;
- consult with stakeholders in a transparent manner, including other regulatory agencies (for example, customs or police) and regulated entities that would be affected by or involved in the implementation of the new or revised provisions prior to implementation;
- ensure that current versions of guidelines and policies are accessible to external stakeholders, including health professionals and the public.

3.3 Regulatory competence

A competent workforce, combined with a strong organizational culture and a sound quality management system, contributes to the success of an NRA – leading to predictable, consistent, transparent and high-quality regulatory decisions (25). To conduct market surveillance and control activities, the NRA will need to be adequately resourced with a workforce that is sufficient, qualified, suitably

skilled, trained and appropriately assigned (9). The WHO Global Competency Framework for Regulators of Medicines provides strategies to harmonize the development of the regulatory workforce of NRAs through a set of best practice competencies. It describes competency requirements for regulatory personnel at two levels – organizational requirements and role-specific requirements – which are further categorized into competencies, activities and domains (Fig. A9.3). In addition, role-specific competencies for four regulatory roles are included, namely for reviewers, inspectors, laboratory analysts and vigilance personnel (25).

Fig. A9.3

Overview of WHO Global Competency Framework for Regulators of Medicines organizational and role-specific requirements for NRAs

Organizational requirements



Meta-competencies:

- competency requirements that are essential for every organization's work environment
- **Domains:** collaboration, communication, decision-making, evidence-informed practice

Core activities:

- activities required for the regulation of medicines, that are common to all regulatory functions
- **Domains:** organizational awareness, report preparation, quality management system, regulatory framework, policies and processes, surveillance and enforcement

Core competencies:

- knowledge, skills and attitudes that support the core activities
- **Domains:** computer skills, decision-making, organizational awareness, quality management system, regulatory framework, policies and processes, surveillance and enforcement

Role-specific requirements



Role-specific activities:

- tasks specific to a regulatory role that contribute to the NRA's regulatory functions

Role-specific competencies:

- knowledge, skills and attitudes that underpin the performance of role-specific practice activities

Details on the domains and competencies can be found in the World Health Organization Global Competency Framework for Regulators of Medicines (25). The publication provides information on the competencies that each NRA, depending on its level of operations, needs to be able to carry out effective medicine regulation. The current framework addresses competencies for regulators of medicines and vaccines, but it is expected to expand to address competencies for regulatory personnel for medical devices. In addition, the WHO Global Competency Framework for Regulators of Medicines may be used by other stakeholders, including professionals planning for improvement, training providers, and industry. The recommendations presented in this guideline will be based on organizational requirements and the role-specific competencies described by the WHO Global Competency Framework for Regulators of Medicines for inspectors and vigilance personnel of the NRA in the context of market surveillance and control.

GP1.13 Develop an NRA competency model based on organizational requirements

The NRA should:

- clearly define its organizational requirements for competencies for personnel involved in market surveillance and control, considering its mandated scope of regulatory activities, the maturity level of its system, areas for strengthening, the regulated medical products and facilities, the local legislative environment, and the national and regional contexts;
- develop a regulatory competency model inclusive of role-specific competencies for market surveillance and control activities and the organizational reality;
- utilize the NRA competency model to inform policies and activities for recruitment, performance, identification of learning needs, and development and training of market surveillance and control staff.

GP1.14 Determine required staff competence and recruitment criteria, and assign staff according to functions

The NRA should develop policies and procedures for recruitment, assignment and training of staff for market surveillance and control to meet the needs of the agency.

Market surveillance and control personnel should (25):

- possess meta-competencies and core competencies for the environment (see Fig. A9.3);
- be able to perform core activities at an appropriate level of proficiency for their responsibilities;

- possess role-specific competencies to undertake market surveillance and control;
- be assigned based on qualifications, training and experience, particularly where specialized training or experience is needed for complex situations.

The NRA should develop job profiles for market surveillance and control personnel describing all the necessary competencies, the key responsibilities of the given roles, knowledge requirements and personal suitability (25). The job profiles should be used to inform recruitment, performance appraisals, training needs, training and development plans, and retention strategies.

The criteria and procedures for hiring, evaluating and monitoring staff performance should be clearly described and available for staff as a reference.

3.3.1 Competency requirements for market surveillance and control personnel

The NRA should clearly define the role-specific competencies and activities relevant to market surveillance and control according to its maturity level, and the scope of surveillance and enforcement activities, considering the four groups of regulatory personnel identified in the WHO Global Competency Framework for Regulators of Medicines in the context of medicines and vaccines (Table A9.4). Where the NRA is at maturity level 1 or 2, it should eventually establish competencies for functioning at maturity level 3. The equivalent framework for regulators of medical devices is yet to be developed.

Table A9.4

Examples of role-specific activities for regulatory personnel involved in market surveillance and control and related functions

Role-specific activities for reviewers	Role-specific activities for inspectors	Role-specific activities for laboratory analysts	Role-specific activities for vigilance personnel
Review of bioavailability, bioequivalence	Data management and use of informatics	Application of standards and international best practices	Post-market surveillance: critical appraisal, evaluation of individual case study reports, medication errors, incidents, reports of substandard and falsified medical products; risk management and signal management
Data review on safety, efficacy and quality	Evaluation of submitted documentation for compliance with regulations and requirements	Maintenance of laboratory systems and equipment	Pre-market vigilance
Maintenance of a register of approved products	Management of the inspectorate system	Participation in analytical method verification, validation and method transfer	Pharmacovigilance system strengthening: capacity-building, system development, regulations
Management of the product review system	Performance of regulatory inspections	Participation in investigations of out-of-specification results	Implement regulatory actions
Evaluation of product information and labelling	Assessment of product quality reports	Performance of support activities for routine laboratory operations	Support for product recalls (for example, precautionary approach, risk communication)
Assessment of product quality information	Regulatory decision-making	Performance of measurements, tests, assays	
Regulatory decision-making		Verification of analytical reports	

Appendix 3 presents sample activities and competencies of inspectors, which may be used to develop job descriptions and training.

GP1.15 Implement appropriate staff training and development

The NRA should:

- develop a training programme for market surveillance and control personnel, including continuing professional development according to needs, and using appropriate training approaches;
- establish and maintain a documented system for staff qualifications and training;
- identify training needs of market surveillance and control personnel utilizing the institutional competency framework;
- develop and implement individual training plans to support achievement of job requirements and continued development that are outcome based, considering the individual's competency level, experience and area of interest;
- participate in networks that provide harmonized training of regulatory personnel, based on the Global Competency Framework;
- periodically review and document training needs of staff and training received.

The WHO Training Toolkit on Substandard and Falsified Medical Products has been introduced to provide resources that may be used by NRAs in developing training plans for personnel specific to substandard and falsified medical products (26). In addition, the WHO *Health products regulation and prequalification learning catalogue* may be used to identify additional learning resources that will supplement the toolkit and help maintain competencies in regulatory functions, including market surveillance and control (27).

The NRA should consider the following in its selection of training approaches (25, 28):

- learning curve or development pathway for market surveillance and control staff, which may be informed by the proficiency levels presented in the WHO Global Competency Framework for Regulators of Medicines (from entry level to proficient or expert level) and job requirements;
- training programme that includes certification of training;
- financing of training programmes to ensure sustainability;
- inclusion of tools that support the application of efficiencies, for example, regulatory reliance, standardized incident reporting forms, electronic databases of reports of substandard and falsified medical products;

- training or accreditation status;
- training modalities (approaches), for example, attendance at conferences, on-the-job training, participation in postgraduate programmes, secondments.

A continuing education or continuing development programme should be implemented that should include introductions to established and revised policies, procedures and tools. When new policies are introduced, there should be a training session and an evaluation of learning. Where there are shortcomings noted when inspectors are evaluated periodically, a retraining plan should be developed and implemented (29).

3.4 Multistakeholder engagement and reliance practices

In globalized markets, the problem of substandard and falsified medical products is shared, and multistakeholder engagement serves to involve all key stakeholders in working across different sectors and scales to address the problem through an integrated approach (30). A sustained multistakeholder response depends on informed, engaged and collaborative stakeholders, and the implementation of effective risk-based activities by all stakeholders.

The engagement of various stakeholders for market surveillance and control should follow agreed dynamic processes and time frames to establish responsibilities and effective communication channels, foster collaboration and learning, balance bottom-up and top-down approaches, minimize conflicts, and generate regular feedback for regulatory activities, enforcement, adaptations and institutional changes, as needed.

The roles, responsibilities and enforcement powers of the NRA, enforcement agencies, industry, distributors, laboratories and other stakeholders involved in the prevention of, detection of and response to substandard and falsified medical products should be clearly established, agreed on and communicated to stakeholders.

Examples of how multistakeholder engagement may contribute to the work of NRAs for the prevention and detection of and response to substandard and falsified medical products include the following:

- implementation of widely available communication channels for the receipt of product quality complaints from users, health professionals, manufacturers, economic operators (for example, distributors, importers, retailers), internal offices, and other governmental and law enforcement institutions;
- utilization of harmonized approaches and resources for the prevention of, detection of and response to substandard and

falsified medical products by international cooperation initiatives (for example, Pharmaceutical Inspection Co-operation Scheme, Asia-Pacific Economic Cooperation) and international agencies or networks (for example, WHO);

- sharing of information about detected substandard and falsified medical products with respective companies and other NRAs for further evaluation, investigation and the adoption of preventive actions;
- establishment of inter-institutional and thematic working groups, networks, committees or councils, formalized at the national or regional level, to enable regular discussions, joint activities, rapid communication, sharing of relevant information and mutual support among participating stakeholders for regulatory actions, including investigations and law enforcement;
- implementation of national databases and regulatory dashboards that integrate and synchronize data and activities across interrelated functions for improved coordination, data sharing and regulatory efficiency;
- active participation of the NRA in multilateral international cooperation initiatives, such as the WHO Member State Mechanism on Substandard and Falsified Medical Products, promoting collaboration among NRAs and WHO;
- access to and active engagement in regulatory harmonization or convergence forums for medical products (for example, the Pharmaceutical Inspection Co-operation Scheme and the International Medical Device Regulators Forum), including participation in the technical discussions and implementation of international standards, where applicable and feasible;
- submission of data collected by national or regional vigilance centres to global databases, such as the WHO GSMS for incident reports of substandard and falsified products;
- utilization of data collected and analysed by academic institutions, where relevant, to inform surveillance priorities and regulatory decision-making.

Note: International collaboration may include the exchange of experiences, lessons learned, best practices and information on ongoing activities at national, regional and global levels; training and targeted technical discussions; and consultations and networking with relevant stakeholders in a transparent and coordinated manner, including regional and other global efforts.

GP1.16 Map key stakeholder groups

The NRA should map the various stakeholder groups involved in market surveillance and control, particularly in the context of the prevent–detect–respond framework. Mapping of stakeholders should include:

- identification of key stakeholders;
- analysis of stakeholders' perspectives, interests, roles and engagement in implementation;
- mapping of objectives and potential contributions to achieving desired outcomes;
- prioritization of engagement, considering level of influence, type of influence, interest and phase of engagement.

The *WHO implementation handbook for national action plans on antimicrobial resistance: guidance for the human health sector* describes a sample process for stakeholder mapping and analysis that may be adapted for the development of national action plans (31). In addition, WHO guidance on reproductive health and family planning service delivery provides a stakeholder mapping tool that may be utilized.¹²

The NRA should convene workshops and focus groups with key stakeholder groups to identify relevant activities that are under way, roles, critical issues and concerns, perspectives and opportunities for collaboration (Fig. A9.4). Given that the roles of stakeholders may vary with the type of product, the NRA should plan according to the types of products (medicines, vaccines or medical devices). An example of an approach to delineate stakeholder roles for medical devices is provided in Table A9.5 (7).

¹² <https://cdn.who.int/media/docs/default-source/reproductive-health/contraception-family-planning/stakeholder-mapping-tool.pdf>.

Fig. A9.4

Key stakeholder groups for prevent–detect–respond in relation to substandard and falsified medical products



Table A9.5

Examples of stakeholders and their roles in post-market surveillance and market surveillance of medical devices

Stakeholder	Activity
User and clients, patients	Observe and detect issues
	Document feedback
	Provide feedback
	Follow manufacturer's instructions
Manufacturers and economic operators (for example, marketing authorization holders, distributors)	Implement a system for post-market surveillance
	Classify feedback and escalate if required
	Establish if reporting to NRA is required
	If required, undertake root cause analysis
	If required, make a correction
NRA	If required, implement corrective or preventive actions
	Ensure user feedback is forwarded to manufacturers
	Conduct risk assessment, as appropriate
	Collect reports, review manufacturer investigation and other actions
	Conduct or coordinate testing using a risk-based approach
	Collect other market information
	Take regulatory action in a timely manner, if needed, and ensure its implementation
Regional and international organizations or networks (for example, WHO)	Share information with other NRAs and WHO, if applicable
	Provides timely support to manufacturers for post-market surveillance of medical devices within their scope of work
	Provides support to NRAs for market surveillance of WHO-recommended and other medical devices
	Provides support to users of medical devices to provide feedback

Adapted from WHO *Guidance for post-market surveillance and market surveillance of medical devices, including in vitro diagnostics* [7].

GP1.17 Engage stakeholders according to priority

The NRAs should utilize the findings of stakeholder mapping activities to inform the level of engagement and the type of engagement needed to support prevention of, detection of and response to substandard and falsified medical products. [Fig. A9.5](#) presents a matrix of stakeholder prioritization and recommended level of engagement, based on the WHO implementation handbook for national action plans on antimicrobial resistance: guidance for the human health sector (31). [Fig. A9.6](#) presents the type of engagement over a spectrum from the WHO stakeholder mapping tool.

Fig. A9.5
Stakeholder prioritization

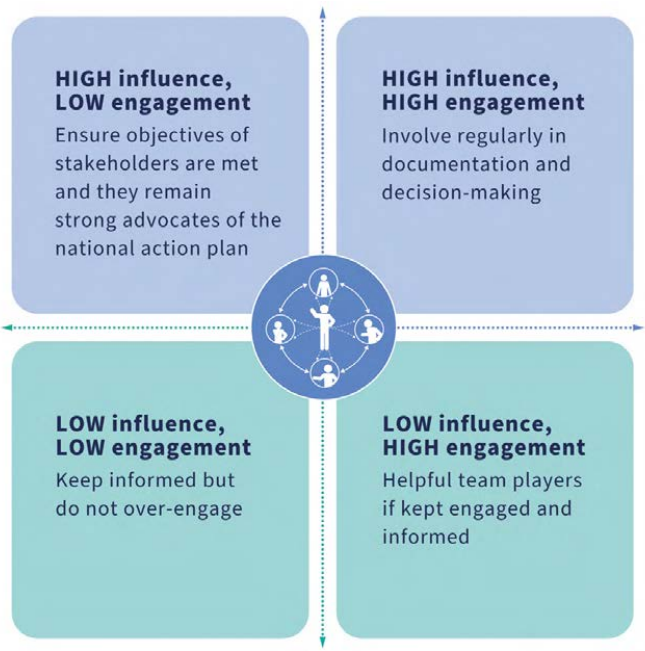


Fig. A9.6
Spectrum of engagement



NRAs should strive to systematically build mutual trust through national, regional and international partnerships and cooperation. The establishment of confidentiality arrangements, data-sharing agreements, and secure online platforms for the exchange of inspection reports and other regulatory data also fosters reliance practices among regulators and, consequently, supports optimal management of the time and the resources (both human and financial) required for risk assessment, management and communication. Considering that the prevention, detection and response pillars for market surveillance and control of substandard and falsified medical products may generate information that can be mutually relevant for policy design, strategic and action planning, and global surveillance and response, NRAs may reach greater success by maintaining regular and efficient interaction between the different stakeholders involved, including industry, the WHO GSMS and other NRAs.

Additional recommendations relevant to stakeholder engagement are included within the various sections of this guideline in the context of the prevent–detect–respond framework.

3.5 Regulatory system strengthening

To determine the readiness and effectiveness of market surveillance and control by the national regulatory system, the NRA would need to evaluate the system's performance and plan for regulatory system strengthening as a result. Based on World Health Assembly resolution WHA67.20 ([32](#)), WHO supports regulatory system strengthening primarily by (a) building capacity of Member States to implement good regulatory practice; and (b) promoting regulatory cooperation, convergence and transparency through networking, work sharing and reliance ([33](#)). Such measures are supported by tools for the evaluation of national systems (Global Benchmarking Tool) and for the development of national institutional development plans, which would inform government investment and technical assistance by WHO and other development partners ([33](#)).

3.5.1 WHO Global Benchmarking Tool

The WHO Global Benchmarking Tool is an internationally accepted method for the objective evaluation of regulatory systems for medical products, as part of regulatory system strengthening. The Global Benchmarking Tool enables WHO, NRAs and other stakeholders interested in supporting regulatory strengthening to:

- identify strengths and areas for improvement;
- facilitate the formulation of an institutional development plan;
- prioritize investments in implementation of the institutional development plan;
- monitor progress and achievements.

The Global Benchmarking Tool is designed to support the evaluation of the respective component functions of regulatory systems, including market surveillance and control, using indicators and sub-indicators that support consistency in evaluation, documentation and rating. The Global Benchmarking Tool also enables WHO and regulatory authorities to determine the overall maturity of different regulatory systems, which would facilitate regulatory strengthening through reliance and harmonization (see Appendix 2, Fig. 1) ([9](#)). For the purposes of this document, reference to the Global Benchmarking Tool will include additional indicators that have been developed to include blood products and medical devices, such as the Global Benchmarking Tool Plus Medical Devices (see Appendix 2, Table 1) ([28](#)).

Towards regulatory system strengthening in the context of market surveillance and control to prevent, detect and respond to substandard and falsified medical products, the following are recommended practices, as appropriate and according to NRA capacity. These are based on recommendations provided in World Health Assembly resolution WHA67.20 and lessons in regulatory system strengthening in the Americas ([21](#), [32](#)).

3.5.2 Benchmarking of market surveillance and control

Benchmarking of the national regulatory system, including its market surveillance and control function, will facilitate strategic planning by the NRA for regulatory system strengthening through institutional development plans. In addition, it will sensitize other NRAs, regulatory partners, civil society, international agencies and other stakeholders about the scope and capacity of the benchmarked NRA for regulatory reliance on decisions arising from market surveillance and control activities.

However, the market surveillance and control function is interdependent with other regulatory functions as there are cross-cutting aspects essential to medical product regulation. These are as follows:

- marketing authorization and registration as the primary gatekeeper to prevent entry of substandard and falsified medical products;
- vigilance, whereby reports of suspected adverse drug reactions, adverse events following immunization, medication errors, quality complaints, and reports of suspected substandard, falsified or both products are collected, investigated and analysed;
- licensing of establishments to ensure the application of good storage and distribution practices throughout the supply chain, to prevent entry and use of substandard, falsified or both medical products;
- regulatory inspections for the detection and investigation of suspected substandard, falsified or both medical products;

- laboratory testing to determine compliance with product specifications, thereby informing investigations and regulatory decisions for substandard, falsified or both products.

Therefore, systemwide benchmarking would be important for effective planning and regulatory strengthening. Further, collaboration throughout the NRA is important, for example, coordination with the legal services and communications departments to establish appropriate messaging and formal administrative arrangements with stakeholders.

GP1.18 Implement approaches for regulatory system strengthening

The NRA should:

- identify the core set of regulatory activities for market surveillance and control, and a core set of regulatory functions that support or complement market surveillance and control;
- utilize the WHO Global Benchmarking Tool indicators to develop procedures for the evaluation of its market surveillance and control function;
- identify and allocate appropriate resources for the evaluation of the market surveillance and control function, for example, data sources, personnel, information systems and tools;
- undertake self-benchmarking of the regulatory system using WHO Global Benchmarking Tool indicators and sub-indicators to identify strengths and opportunities for improvement of market surveillance and control and related functions;
- utilize findings of self-benchmarking to inform the national action plan and the development of the institutional development plan for regulatory system strengthening;
- plan for benchmarking every three to seven years using a risk-based approach;
- allocate appropriate resources (including funding, staff, training and information systems) to vigilance and market surveillance and control to enable timely responses to substandard and falsified medical products and the growing complexity of the products and supply chains.

GP1.19 Utilize the WHO Global Benchmarking Tool for self-assessment and the development of national action plans

The NRA should utilize the relevant WHO Global Benchmarking Tool (for example, WHO Global Benchmarking Tool, WHO Global Benchmarking Tool

Plus Medical Devices) and any supplemental resources according to the scope of medical products regulated in the country.

The sub-indicators of the market surveillance and control function should be used to conduct self-assessment of the national regulatory system, and to prepare for collaborative assessment with support from WHO (see Appendix 2, Table 1) (28), with consideration of other interrelated regulatory functions of market authorization or registration, vigilance, licensing of establishments, regulatory inspections, clinical trials and laboratory testing where applicable.

The NRA should identify sub-indicators that are relevant to its scope of work, medical products, current maturity level (if applicable), and local context for the assessment of market surveillance and control.

NRAs should collaborate with WHO and utilize its resources for benchmarking of the national regulatory system. This will support the development of trust and collaboration with other NRAs.

Findings of the benchmarking activities should be used to inform national action plans for market surveillance and control and the formulation of institutional development plans for regulatory strengthening.

NRAs should utilize published information about maturity levels assigned based on WHO benchmarking, and the regulatory scope of other NRAs that may be considered as reference authorities for regulatory reliance in the context of market surveillance and control.

3.6 National action plans

A combination of constrained access, poor governance and weak technical capacity creates a fertile environment for the emergence of substandard and falsified medical products. There are known strategies to prevent, detect and respond to substandard and falsified medical products but, for most countries, implementing these requires awareness, political commitment, investment of resources, building capacity, multistakeholder collaboration and partnership, and advocacy of the development of dedicated national action plans. The national action plan would provide a framework to address the threat of substandard and falsified medical products and support stakeholder collaboration among parties involved in the manufacture, importation, distribution and supply of medical products.

In 2017, in a report to the Seventieth World Health Assembly, WHO presented guidance for the development of national plans for preventing, detecting and responding to actions, activities and behaviours that result in substandard and falsified medical products. The document describes guiding principles, goals and desired outcomes for the national plan, and a roadmap of activities with examples. A multidisciplinary approach is recommended for the formulation of national action plans, which will include the development or strengthening of legislative frameworks and enforcement (34).

Subsequently, WHO developed a draft handbook for all stakeholders involved in eradicating substandard and falsified medical products and ensuring that medical products entering the national supply chain are quality assured, safe to use and work as intended. The handbook, which will be published in 2026, outlines a process for developing national action plans, including the roles of the multiple stakeholders whose engagement is essential in forging a response to substandard and falsified medical products and suggesting mechanisms for governance and coordination. This resource and the *WHO implementation handbook for national action plans on antimicrobial resistance* ([31](#)) may guide the development of national objectives, goals, actions, inputs, outputs and outcomes for strategic and operational plans, and provide insights into approaches for operational planning, monitoring and evaluation.

GP1.20 Develop a national action plan for substandard and falsified medical products

NRAs should develop a national action plan to address the threat of substandard and falsified medical products with strategic objectives based on the prevent–detect–respond framework. The objectives of the national action plan should include the achievement of a stable, integrated and well functioning regulatory system at maturity level 3, as per the Global Benchmarking Tool.

The national action plan should aim to ensure:

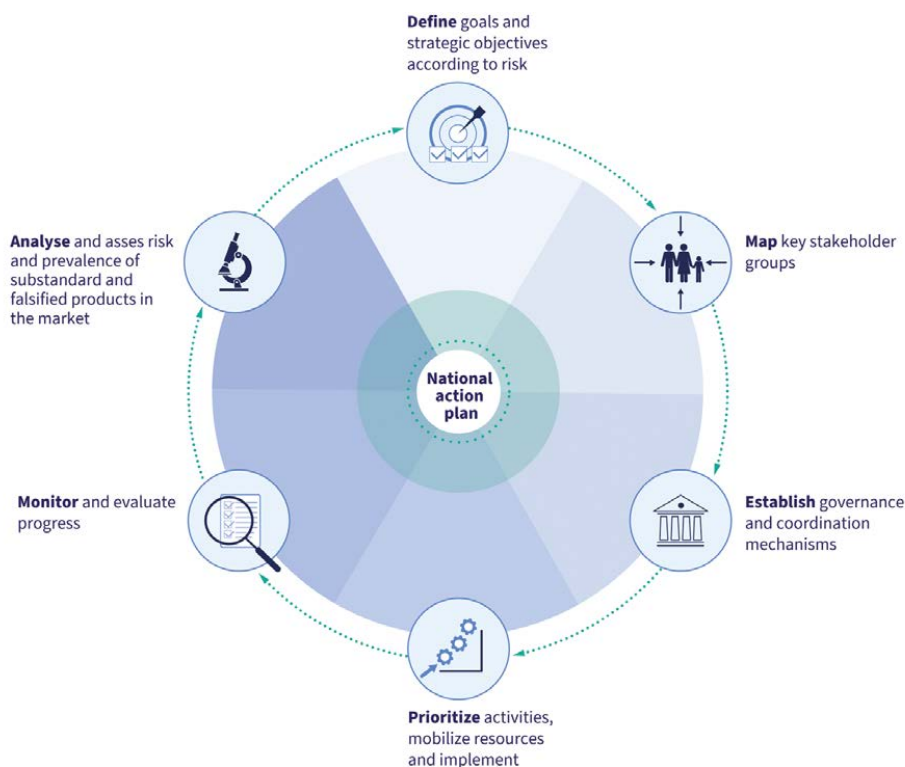
- all stakeholders within the supply chain are committed and equipped with the required knowledge, skills and equipment to identify and report suspicious products to the NRA;
- activities are planned to mitigate the risk of substandard and falsified medical products;
- adherence to international good manufacturing, procurement, distribution and storage practices;
- monitoring and evaluation of outputs;
- the NRA is suitably resourced to respond to reports in a timely and effective manner.

The NRA should implement a continuous process for the development, implementation and evaluation of the national action plan for substandard and falsified medical products ([Fig. A9.7](#)) ([31](#)).

The strategic objectives of the national action plan should be adapted to each setting, taking into account the economic, cultural and contextual drivers, characteristics of the health system, capacity, capability and specific risks.

Activities within the national action plan should be prioritized based on risk to public health, patient safety, available resources and potential for mutual reliance.

Fig. A9.7

Continuous process for national action plan development, implementation and evaluation

4. Prevention

4.1 Education and awareness

Efforts to eliminate substandard and falsified medical products from supply chains should focus both on the demand side and on the supply side (for example, strengthening supply chain integrity through effective regulation). Two major sets of factors contribute to public demand for medical products that are potentially substandard or falsified:

- lack of knowledge and awareness of the existence and prevalence of substandard and falsified medical products, the risks posed, and ways to reduce exposure;
- constraints on access to quality-assured medical products, which may be caused by limited regulatory capacity, lack of manufacturing capacity or costs.

Both factors are important drivers of public demand and must therefore be addressed simultaneously, especially in limited-resource settings (35, 36). Public education and awareness are essential to prevent exposure to and demand for substandard and falsified medical products; such campaigns help to empower consumers, enhance reporting, support regulatory efforts, and thereby protect public health (13). This section provides recommendations for strategies to inform and educate the public, as well as other key actors, including health professionals, procurers and distributors, about substandard and falsified medical products, specifically:

- awareness of substandard and falsified medical products, the associated enablers and risks;
- education on how to reduce risks of exposure (how to avoid informal markets and how to identify and avoid substandard and falsified medical products);
- encouragement to report suspected adverse reactions and provide feedback on performance (medical devices) and products suspected to be substandard or falsified.

However, to be effective, health professional education and public education and awareness campaigns must be paired with parallel investments in the health system and supply chain strengthening to address factors constraining access to quality-assured medical products from authorized sources, especially for the poorest and most vulnerable sectors of the population.

4.1.1 Education of health professionals

Health professionals are key stakeholders in access to, use of and handling of medical products, including in such areas as selection, prescribing, therapeutic monitoring, receiving complaints about product safety or quality, storage, distribution, and product administration. Educating health professionals about the scope of, contributors to and impact of substandard and falsified medical products and the existing solutions will assist the NRA with prevention, detection and response activities. This should be done using a multidisciplinary approach, as all health professionals should be educated about substandard and falsified medical products and their role in prevention, detection and response. In 2021, WHO and the International Pharmaceutical Federation published a curriculum guide and competency framework to improve reporting on and response to substandard and falsified medical products by pharmacists, which may be used by institutions and organizations involved in the education of health professionals (for example, in universities or professional societies) (37). In addition, the recently published WHO Training Toolkit on Substandard and Falsified Medical Products

supports the standardization of training and education of health professionals on the topic of substandard and falsified medical products. It includes a competency framework, a curriculum guide, a trainer's guide, and technical resources that may be used by NRAs, health professionals and educators (26).

GP2.1 Support the development of competence of health professionals

Where possible, the NRA should collaborate with academic institutions and organizations providing continuing education to health professionals to:

- support the development of curricula for undergraduate and postgraduate education, and for professional development;
- provide or support the delivery and training of health care providers via lectures and experiential training at the undergraduate, postgraduate and professional education levels.

NRAs should collaborate with professional societies and academic institutions to facilitate the delivery of lectures, seminars and workshops for health professionals to support the prevention of, detection of and response to substandard and falsified medical products. Where possible, sessions should be multidisciplinary, involving physicians, pharmacists, nurses and other health workers, according to their roles and responsibilities.

Education and training of health professionals should include the scope of the problem of substandard and falsified medical products, contributing factors, local and global impact, the roles and responsibilities of the NRA, the role of health care providers, relevant regulations, applicable guidelines, the prevent–detect–respond framework, methods of detection of substandard and falsified products, the national reporting system and available resources.

4.1.2 Public education and awareness campaigns

GP2.2 Develop and implement appropriate education and awareness campaigns

Investing in a public education and awareness campaign

Before deciding to invest in an education and awareness campaign, the NRA should assess the relative importance of the three main drivers of penetration of and exposure to substandard and falsified medical products, namely:

- low knowledge and awareness;
- constraints on access to quality-assured products (for all or some sectors of the population);
- insecurity of supply chains.

If low knowledge and awareness is the biggest contributor to potential exposure to substandard and falsified products, a well designed education and

awareness campaign may be a worthwhile and effective intervention. If, however, the target population has limited capacity to act on the information they receive, or if supply chain integrity is weak, the value of an awareness campaign may be limited without parallel investments. Education and awareness campaigns can be designed for the public or can specifically target those involved in the distribution of medicines to the public, for example, health care professionals and retailers, according to the likelihood of exposure and the NRA's capacity.

When considering investing in a public education and awareness campaign for substandard and falsified medical products, the NRA should determine:

- the target audience and their respective level of knowledge and awareness of substandard and falsified medical products, including the risks;
- whether the target audience will be able to act meaningfully on the information that it receives;
- which stakeholders, partners and personnel should be involved;
- the effectiveness of current regulatory measures in eliminating substandard, falsified or both products from regulated supply chains and licensed pharmaceutical outlets.

Considerations for designing and implementing an effective public education and awareness campaign for substandard and falsified medical products

In order to be effective in changing awareness and behaviour, a public-facing education and awareness campaign needs to convey messages that:

- reach the intended audience through appropriate messaging;
- can be understood by the target audience;
- are perceived to be important and trustworthy;
- provide recipients with the capacity to act on guidance meaningfully ([36](#)).

The following steps can help achieve each of these targets.

Reach: identifying the target audience

For public education campaigns, the NRA should determine if the intention is to reach the whole population, or specific groups (for example, older adults or parents). According to population needs, the focus may involve proactively targeting sectors of the population that are the most vulnerable to or likely to be exposed to substandard and falsified medical products, for example, caregivers, persons of lower socioeconomic or educational status, rural populations, older persons or persons living with disabilities ([36](#), [38](#)).

Reach: selecting relevant messages

Depending on the context and target audience, it may be effective to focus on particular messages or modify messages according to what is accessible and achievable. For example, if the level of literacy is low among most of the target audience, advice to check for spelling mistakes and grammatical errors may be less helpful than advice to source medicines from trusted outlets or modifying the message to ask someone else to check the packaging on their behalf.

WHO has provided advice to patients and consumers to minimize the purchase and use of substandard and falsified medical products ([Fig. A9.8](#)) ([39](#)).

Fig. A9.8

WHO advice to patients and consumers

The following steps will help minimize the risk:



Source from trusted and licensed outlets, e.g. pharmacies, clinics, doctors, hospitals, health centres.



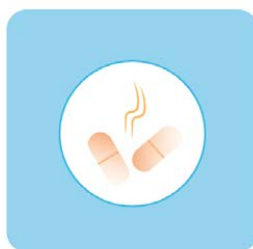
Avoid unregistered websites supplying medical products that conceal their physical address or landline telephone number.



Examine the packaging for condition, spelling mistakes or grammatical errors.



Check the manufacturing and expiry dates.



Ensure the medicine looks correct, is not discoloured, degraded or has an unusual smell.

**If you suspect a product is not working properly or if you suffer an adverse reaction, discuss with your pharmacist or health care professional as soon as possible.
Always buy your medicines from known and reliable sources.**

Reach: determining media

Multiple options are available to communicate messages to the target audience, including:

- broadcast media (for example, radio, television);
- digital media, via mobile phones or tablets (for example, social media, phone adverts);
- physical materials (for example, posters, leaflets, bumper stickers);
- in-person sensitization (for example, community-based organizations, local health professionals).

In selecting the most appropriate form of media for the target audience and messages, the following should be taken into consideration:

- **Reach.** The proportion of the target audience that may be reached. For example, what proportion have radios, televisions or mobile phones?
- **Inclusivity.** The sectors of the population that are more or less likely to have access to different media forms, and channels whereby the most marginalized groups may be reached most effectively.
- **Resource feasibility.** Available resources for effective reach (for example, the number of personnel needed to deliver in-person sensitization and for what geographical coverage, availability of national or local radio or television stations).

Ensuring understandability

To ensure the campaign's messages are understood, two main considerations apply:

- Written media are only accessible to literate individuals, and may thus exclude persons with low literacy levels, who may be among the most vulnerable groups. Alternative forms of media (for example, radio) should also be considered.
- Where multiple languages are spoken, written and audio materials produced in the most widely spoken languages may exclude those in minority ethnolinguistic groups. Materials should be translated into all relevant languages of the target audience.

Although visual aids may be more easily accessible, they may not easily convey complex messages through images alone.

Perceived importance and trustworthiness

In contexts where trust in governments and health systems is low, additional efforts would be needed to convince populations that the messages are important and trustworthy (40). The proliferation of “fake news” and conspiracy theories may undermine public health messaging. Moreover, where people face multiple risks and challenges in their daily lives (for example, conflict, food insecurity, displacement or other health risks), substandard and falsified medical products may not be perceived as a priority concern.

The NRA will need to be transparent about its role and actions, and convey messages about the risk to health and seriousness of the issue without increasing fear, in order to avoid undermining trust in the use of essential medicines and vaccination programmes.

Ability to act on the advice

Recent research has shown that constraints to acting on public health advice are often bigger obstacles to safe medicine practices than lack of awareness and education (35, 36, 38). Parallel investment in overcoming barriers to service access may be necessary to maximize a campaign’s success and avoid exacerbating health inequities.

For individuals, the inability to act may increase feelings of anxiety and powerlessness and may reduce the effectiveness of the education and awareness campaign and lead to diminished trust in health systems and health care providers. Where access to quality-assured health care is limited (for example, due to high prices) or unreliable (for example, in remote rural areas or among displaced populations), campaign messages should be realistic, actionable by the target audience, and appropriately tailored to the local context, taking into account the following factors (40).

- Advice to examine packaging for spelling mistakes or grammatical errors, or to check manufacturing and expiry dates, may not be feasible for persons with limited literacy.
- Advice to source medicines from trusted and licensed outlets and to report adverse events to health care professionals is likely to be difficult for persons who lack access to these services.
- Advice to report suspect products relies on people being able to identify the specific suspect products. For example, polypharmacy and a high prevalence of comorbidities may make it difficult to discern the therapeutic effect (or lack thereof) or adverse effect of a single product, and thus decide whether to report it.

WHO Member State Mechanism risk communication recommendations: the IDEAS campaign

Recommendations for effective risk communication and public awareness campaigns to combat substandard and falsified medical products have been developed by the WHO Member State Mechanism as a campaign based on the themes insight, data, engagement, action, solutions, using the acronym IDEAS as a mnemonic. The recommendations present a communication framework based on the five key elements that provide pillars for the development of communications about substandard and falsified medical products ([41](#)). NRAs may utilize this framework in tandem with the steps listed to inform the design and conduct of public education and awareness campaigns.

Health professional and retailer education and awareness campaigns

Education and awareness campaigns intended for health professionals or retailers might focus on raising awareness of the dangers of substandard and falsified medical products and the importance of verifying sources and suppliers, detection, and reporting ([42–44](#)).

Target audience

It is important to identify and target personnel who are legally authorized to prescribe, administer and dispense medical products to the public via the public, private and voluntary sectors. These should include retailers at pharmacies and shops authorized to sell medical products over the counter, which are often the first avenue for medicines in many communities.

Messaging and media

WHO provides advice to health care professionals and retailers in a pilot campaign, as shown in [Box A9.2](#), which may be modified to include all medical products ([35](#), [36](#), [38](#)).

Box A9.2

WHO advice to health care professionals and medicine retailers

- Purchase medicines only from licensed or registered pharmaceutical outlets or manufacturers.
- The price of medicines should be fair and affordable to patients.
- Beware of suspiciously low-priced medicines.
- Advise patients about medicine quality and encourage reporting of product defects or adverse reactions, including lack of efficacy.
- Report cases of adverse reactions or substandard, falsified or both medicines to the NRA.

Reach and media

Targeted media should focus on maximizing reach among health care professionals or providers, especially those in harder-to-reach areas, making use of existing networks and communications systems (for example, via professional publications, social media groups, email lists, or in-person sensitization by trained NRA representatives).

Ability to act on the advice

Health professionals may also face challenges in acting on information. Constraints may include:

- lack of access to licensed outlets, for example, businesses in rural areas, or affordability of medical products from licensed outlets ([43](#));
- lack of confidence of health professionals in their ability to differentiate substandard, falsified or both medical products from genuine products;
- lack of sufficient resources, system access, or knowledge to report, especially where the system for reporting is opaque or cumbersome;
- lack of trust in the health system, including fear of repercussions as a result of reporting ([42](#), [44](#)).

GP2.3 Monitor and evaluate the impact of education and awareness campaigns

To ensure that resources are being deployed effectively, the NRA should periodically monitor and evaluate the effectiveness of education and awareness campaigns. This may be done in collaboration with research institutes and universities.

Prior to the implementation of public education and awareness campaigns, the NRA should clearly define each campaign's objectives, methods and procedures for monitoring and evaluation, with consideration of the target audience, modes of communication, costs, and systemic challenges or limitations. The frameworks for the education and awareness campaigns should be relevant to the target audience, including the respective performance indicators. [Table A9.6](#) provides suggested frameworks and considerations for monitoring and evaluating public-facing and provider-facing education and awareness campaigns, respectively.

Table A9.6

Frameworks for public-facing and provider-facing education and awareness campaigns

Category	Public-facing	Professional or provider-facing
Objectives	1. To estimate the proportion of the population (or target audience) that have encountered, understood, and acted on campaign messages 2. To identify any sectors of the population (or target audience) that might have been excluded from (any aspects of) the campaign 3. To measure impacts of the campaign: (a) immediate or short-term: increased knowledge and awareness (b) longer-term outcomes: for example, increased reporting rates and improved health outcomes 4. To understand the reasons for the patterns observed and identify mechanisms underpinning any inequities in reach and outcomes 5. To work collaboratively with communities to address shortcomings and adapt campaigns	1. To estimate the proportion of health professionals or medical product providers that have encountered, understood and acted on campaign messages 2. To identify any groups of health professionals or providers that might have been excluded from (any aspects of) the campaign 3. To measure impacts of the campaign: (a) immediate or short-term: increased knowledge and awareness (b) longer-term outcomes: increased detection and reporting rates; compliant purchasing and dispensing practices 4. To understand the reasons for the patterns observed and identify mechanisms underpinning any inequities in reach and outcomes 5. To work collaboratively with providers to address shortcomings and adapt campaigns

Table A9.6 *continued*

Category	Public-facing	Professional or provider-facing
Methods	1. Purposively select sufficient representative sites (for example, two to four) per country with varying geographical and sociocultural characteristics 2. Within each site, conduct focus group discussions with different demographic groups (for example, older women, younger women, older men, younger men) to: (a) obtain quantitative estimates of the education and awareness campaign's reach, understandability and behaviour changes (b) understand reasons and mechanisms through group discussion, focusing particularly on which groups may have been excluded and why (c) elicit local views on how the effectiveness of the campaigns may be improved	1. Purposively select sufficient representative sites (for example, two to four) per country with varying geographical and sociocultural characteristics 2. Within each site, conduct individual interviews with different types of health professionals and providers (for example, health centre staff, private retailers) to: (a) obtain quantitative estimates (such as ratings) of the education and awareness campaign's reach, understandability and behaviour changes (b) understand reasons and mechanisms, what has worked, what has not, and why (c) elicit views on how the effectiveness of the campaigns may be improved

Table A9.6 *continued*

Category	Public-facing	Professional or provider-facing
Interpretation	1. Overall, how effective is the campaign? What proportion of respondents have encountered the campaign, understood it, perceived it as important, and have acted on it? 2. Which (if any) of the four criteria (reach, understandability, perceived importance, ability to act) is the biggest obstacle, and how might this be addressed? 3. Which (if any) sectors of the population are being excluded, how, and how might this be addressed?	1. Overall, how effective is the campaign? What proportion of providers have encountered the campaign, understood it, perceived it as important, and have acted on it? 2. What are the main obstacles to behaviour change? 3. Which (if any) types of providers are being excluded, how, and how might this be addressed?
Key considerations	• Does the campaign represent good value for money (in terms of impact and costs)? • Is the campaign equitable? Are disadvantaged groups more or less likely to benefit? • Is the campaign sustainable? What resources and other inputs are required? • What parallel investments are required to maximize the campaign's effectiveness?	

4.2 Supply chain integrity

Regulating medical product supply chains is essential to ensuring the safety, efficacy and quality of medical products, particularly the monitoring and control of every stage of production, distribution and use, which is critical to prevent the emergence of substandard and falsified products and the diversion of quality products. The maintenance of medical product supply chain integrity aims to (a) ensure the quality of authorized medical products during storage, distribution and use by the relevant stakeholders; (b) prevent diversion or theft of authorized products; and (c) prevent the production and importation of falsified, unauthorized or otherwise unsafe medical products. NRAs play a critical role in upholding supply chain integrity by establishing and enforcing standards to be met, from a product's manufacture to its storage and distribution while the product is on the market.

This includes ensuring that manufacturers, importers, online sellers and distributors are duly authorized and that they implement appropriate systems and practices to prevent entry or emergence of substandard and falsified medical products.

4.2.1 **Market authorization information**

The function of market authorization or market registration is essential to quality assurance of medical products entering the national market and their control once approved. The NRA is responsible for approving medical products for market, ensuring that they are of good quality, safe, and efficacious, and for determining conditions of their use to protect public health. The information to be provided for the NRA's assessment of the product for market authorization will serve as a reference for the product's attributes and specifications for market surveillance and control, including verification of source, composition, manufacturing process and authenticity. In the absence of key information, or where this information is incomplete or outdated, post-authorization activities to verify product quality, sameness or authenticity would be undermined. Hence, the scope and quality of information provided during the marketing authorization procedures remain important for market surveillance and control (45).

It is expected that during the lifespan of the product, the manufacturer may need to implement changes based on emerging quality, safety and efficacy (performance) data, including post-authorization safety studies, changes in production, and clinical studies in novel populations. Post-authorization changes that impact quality, safety and efficacy should be submitted as notifications to the NRA, or evaluated and approved by the NRA, according to national or regional regulations. This enables the NRA to reassess risk and to have updated information as reference for market surveillance and control activities. In situations where an NRA applies regulatory reliance for market authorization decisions, access to product assessments and inspection reports of reference NRAs or WHO, along with registration dossiers, may facilitate and accelerate national decisions by assuring the relying NRA of the positive benefit–risk ratio and the identical quality of the product at the time of approval, which may serve as reference for market surveillance and control (46). The application of reliance to subsequent post-authorization changes will further support the maintenance of market authorization, provided that the product maintains a positive benefit–risk balance and the same quality while on the market.

GP2.4 Ensure access to current critical product information and specifications

The NRA should implement requirements for market authorization that will provide critical reference information for market surveillance and control, inspections and laboratory testing, such as product information, product

specifications, pharmacopoeial methods or in-house testing methods, certificates of analysis, manufacturing sources, labelling, stability studies and risk management.

The NRA should require all applicants to submit sufficient scientific and technical information on the product to enable verification, quality control testing and other relevant investigations (for example, inspections). The required reference information should include (47):

- validated test methodology, including reference standards for use in quality testing of samples;
- stability data, including testing conditions;
- the nature of impurities found in stability studies, and the testing methods for detecting and quantifying them;
- information on the degradation of products or contaminants, and acceptable limits;
- sites of manufacture of the active substance or manufacturing intermediates, where applicable;
- sites of manufacture of the finished product;
- packaging and product labelling.

The NRA should reserve the right to request additional information, particularly about impurities that may pose a risk to the product's quality and public health, such as diethylene glycol, ethylene glycol and nitrosamines (based on pre-existing or emerging evidence). These may be considered based on reports or alerts issued by WHO GSMS, international organizations that monitor illicit goods (for example, INTERPOL and the World Customs Organization), manufacturers, and other NRAs, considering local context and global supply chains.

The minimum required product information listed above should also be applicable to products granted special authorization, for example, compassionate use, investigational use, or emergency use, or to support market surveillance and control activities.

For medical devices, the technical information provided should be according to the NRA's requirements based on device classification and risk.

Where the NRA utilizes regulatory reliance for market authorization decisions, the authority should include the submission of key technical information by the applicant and, where feasible, develop agreements with reference NRAs for information sharing. An example of information, documentary evidence and assessment activity of an NRA applying the reliance model is provided in Appendix 5 of WHO TRS 1019, Annex 6 (46).

Within the NRA, policies and procedures should be implemented to facilitate information access and sharing between personnel or units responsible

for market authorization and market surveillance and control. This should include information relevant to post-authorization changes (including manufacturing sites and product information), product registration status, conditions of market authorization, quality complaints, safety signals, inspections, incidents of suspected and confirmed falsification, and related regulatory decisions.

Resource: The WHO manual on marketing authorization of multisource pharmaceuticals for NRAs (the “Blue Book”) provides detailed guidance for various aspects of this function, inclusive of guidance for approval of post-authorization changes (47). *Note:* The principles outlined in the manual also apply to innovator products.

GP2.5 Ensure access to evidence of manufacturing quality

Evidence of quality manufacturing, and information on quality risk management processes provided by manufacturers to the NRA during market authorization, should be available as reference information for market surveillance and control. This includes the identification of manufacturers, respective roles, certification of compliance with good manufacturing practice or ISO 13485 (for medical devices), or equivalent, and updates to product specifications and testing methods. Where needed, market surveillance and control officers should be able to access this information, as well as information on the quality of manufacturing of authorized products, investigational products, donations and products granted special authorization (for example, compassionate use or emergency use).

The *WHO guidelines on quality risk management* provide information on the quality risk management process, its application to pharmaceuticals, tools, and considerations for NRAs (48).

GP2.6 Apply a risk-based approach and efficiencies for preregistration inspection of manufacturing sites

Where resources do not permit direct inspection, the NRA should use a risk-based approach to determine the need for inspection of manufacturing sites to inform market authorization decisions, prioritizing locations where there are known or suspected risks, or incidents involving the integrity of manufacturing sites or the supply chain.

Based on risk and local context, the NRA should utilize regulatory harmonization systems or efficiencies, such as:

- recognition of WHO inspections of manufacturing sites;
- application of regulatory reliance on the inspection of NRAs of maturity level 3 or 4 with sufficient capacity for inspection of the relevant product types (for example, non-biologics, biotherapeutics, vaccines and medical devices);

- utilization of the WHO certification of pharmaceutical products programme ([49](#));
- inspections recognized by regional regulatory authorities or regulatory networks utilizing international standards, such as the Pharmaceutical Inspection Co-operation Scheme, African Union standards, the European Community database on good manufacturing practice (EudraGMP), the International Medical Device Regulators Forum, and the Medical Device Single Audit Program.

The NRA should develop criteria, conditions and procedures for the application of regulatory efficiencies or harmonization mechanisms, including reliance or recognition of inspections by selected reference NRAs, and the use of the WHO certification scheme on the quality of pharmaceutical products moving in international commerce, where applicable.

In situations where regulatory reliance or recognition is applied, whereby the NRA inspectorate has access to inspection reports of the reference NRA or the WHO Prequalification Team, the NRA should apply a desk review process utilizing key product information (for example, artworks or specifications) instead of on-site inspections ([46](#)).

Resource: The WHO compendium of guidelines for the quality assurance of pharmaceuticals, with a focus on good manufacturing practices and inspections, may be used as a reference by NRAs to strengthen oversight of manufacturing practices for finished pharmaceutical products and starting materials ([50](#)).

GP2.7 Publish a database or lists of authorized medical products

The NRA should establish and publish a list or database of medical products approved for the market with sufficient information to allow health workers and the public to identify the product and verify its registration status. This database should be reviewed and updated regularly to include newly approved products, recent post-approval changes, information on market suspensions or revocation, and changes to product information needed to enable identification and verification.

The lists should include, at a minimum, the following information about each product ([47](#)):

- the international nonproprietary name (INN) or generic name, dosage form and strength;
- its trade (or brand) name (where applicable);
- the marketing authorization holder;
- the product marketing authorization number, or equivalent product identifier;

- the name and address of the finished product manufacturer;
- product information (for example, indications, contraindications, warnings);
- photographs of the finished product and packaging;
- its registration status (for example, active, suspended, special authorization);
- its registration date and validity of registration.

The national product database should provide the current regulatory status and presentation (for example, images and artworks) of authorized or registered medical products, including post-approval changes relevant to the published information. As a tool for product or source verification, the database will support the detection of (a) falsified medical products; (b) authorized products that may be suspected as substandard; and (c) unauthorized products that may be substandard or falsified.

Where possible, the NRA should implement linkages between public databases for authorized medical products and national traceability systems.

4.2.2 Importation of medical products

The control of import activities is integrated as part of the market surveillance and control function by NRAs, with the aim of ensuring appropriate authorization and upholding standards to maintain product quality and the integrity of supply chains (9). Effective border control strategies should incorporate a range of regulatory measures, surveillance technologies and international partnerships.

The following are based on the WHO *Guidelines on import procedures for medical products* (TRS 1019, Annex 5), with a focus on substandard and falsified medical products. Further details are provided in the guidance document, including required documentation, implementation of controls and procedures (51). Additional reference is made to the WHO certification scheme on the quality of finished pharmaceutical products moving in international commerce, which will enable NRAs to provide or receive verification of the production quality according to the issuing NRA, and certification of product registration from participating Member States (49). The WHO Global Model Regulatory Framework for medical devices further supplements the following recommendations for import control in the context of medical devices, including in vitro diagnostics (6).

GP2.8 Collaborate and communicate with customs and border agencies

The NRA should collaborate with customs, police, judiciary and other partners in law enforcement to agree on and implement strategies to prevent the entry and

distribution of substandard, falsified or both medical products. In collaboration with customs and border control authorities, the NRA should:

- facilitate training of border control officers and inspectors on the procedures relevant to their function, where needed;
- establish communication protocols and channels utilizing secured digital platforms, emails, and other forms of official correspondence to ensure timely communication and decision-making as it relates to importers, consignments, and suspected substandard, falsified or both medical products and unregistered medical products;
- identify focal point persons who would facilitate communications for urgent and routine matters;
- develop and implement emergency procedures, considering various scenarios, such as the absence of an inspector or NRA officer.

The NRA should reserve discretionary powers to waive product authorization requirements in respect of consignments imported in response to emergencies, investigational use, or donations, and in response to requests for limited supplies of unregistered products from clinicians for a named patient. In this regard, the NRA should develop, implement and communicate guidelines and procedures for import approvals in the context of special conditions or waiver of market authorization procedures for key stakeholders, including sponsors of clinical trials and importers.

GP2.9 Implement standards and procedures for importer authorization with relevant guidelines

The NRA should implement processes for the authorization and licensure of importers of medical products, with clearly defined criteria for maintaining authorization. Importer qualifications should be clearly described, along with their responsibilities in the importation process.

The NRA should clearly establish requirements for import authorization of medical products, including documentation of the quality of batches prior to shipping (for example, certificate of analysis), marketing authorization or certification of pharmaceutical products from the NRA of the exporting country, and product labelling.

A risk-based approach should inform the need for product quality testing or screening prior to shipping to the country, or upon receipt at ports of entry. The respective conditions, requirements and procedures for pre- or post-shipment testing or screening should be clearly established in regulations or guidelines to be communicated to applicants and importers. There should be clear justification for quality testing prior to importing or release of medical products, including

vaccines, considering the limited utility of pre-market testing as an indicator of post-market quality.

The NRA should register qualified importers and publish lists of authorized importers, inclusive of the name of importer, the importer's address, the authorized medical products for importation, nature of authorization (for example, emergency use), and the status of the importer (active or suspended). The list should be regularly updated to reflect changes in products and authorization, particularly for inquiry by customs and border authorities. Where feasible, the lists should be published on the NRA's website.

Customs officers and NRA inspectors should have access to the registry or database of authorized importers and the national product registry, inclusive of medical products for emergency use and other products granted special authorization for use (for example, investigational medical products or compassionate use).

The NRA should develop and publish complementary guidelines for importers, detailing criteria, process overview, conditions, required documentation (for example, batch release certificate, certificate of pharmaceutical products), and requirements for labelling of consignments and products.

GP2.10 Identify and monitor entry points for medical products

The NRA should clearly define and publicly identify entry points or customs ports authorized for importation of medical products and their storage. The selection of ports should be informed by risk assessments and capacity and strength of controls, and these should include staff of the NRA, where feasible. The NRA should document deviations in good practices, inadequate quality management system, unauthorized medical product imports, and incidents involving substandard and falsified medical products at each entry point to inform risk-based surveillance and to strengthen border controls.

The NRA should ensure that storage facilities at ports and related distribution sites are secured and compliant with good storage practices, with the appropriate environmentally controlled conditions.

The NRA should ensure that cold storage areas for refrigerated and frozen products are fully equipped with appropriate storage facilities, such as cold rooms, cold stores or refrigerators, along with the necessary equipment for environmental control (for example, thermostats with integrated monitoring and alarm systems).

GP2.11 Establish appropriate clearance procedures for imported products

The NRA should collaborate with customs authorities to establish or revise procedures for the clearance of imported medical products, including desk reviews, product verification, and initial screenings using visual inspection methods or field screening technologies.

Procedures, requirements and criteria for conditional clearance of medical products under routine and urgent conditions should be clearly established, inclusive of documentation of relevant justification and decision-making processes. The NRA should ensure that where conditional clearance is under consideration the quality, safety and efficacy of the medical products would not be compromised.

The NRA should establish conditions for the release process in collaboration with customs officials to ensure that handling and storage was appropriate, and to avoid product mix-up or diversion.

Procedures should be established whereby the NRA or importers may inform customs in advance about the anticipated arrival of consignments, particularly those requiring cold chain storage, or for emergency use, necessitating prioritization and prompt transfer to designated storage areas.

The NRA should monitor importation process time frames, ensuring timely submission and review of required documents for clearance by the respective authorities.

GP2.12 Assign inspectors according to risk, turnover and qualifications

The NRA, in collaboration with customs authorities, should assign either full-time or part-time inspectors to designated ports based on risk, with plans for supplemental backup personnel and communication.

The allocation and assignment of inspectors should be determined based on the turnover of medical product imports, the risk level of the port and the availability of qualified personnel. For instance, NRAs should assign full-time inspectors to entry points or ports with a high turnover of medical product imports.

4.2.3 Product traceability

The mapping of the supply chain from manufacture to importation to dispensing and distribution to the public will assist NRAs to identify vulnerable “pinch points” for increased oversight. The traceability of medical products throughout the supply chain serves to assist in the verification of product authenticity and facilities for storage and distribution. The Member State Mechanism recommends that traceability technologies are implemented, along with a national traceability system for medical products. In two reports, the Member State Mechanism identifies available traceability (“track and trace”) technologies, and provides guidance on available authentication technologies for the prevention and detection of substandard and falsified medical products ([52](#), [53](#)). In its policy paper on the traceability of finished medical products, WHO offers guidance to Member States on policy and regulatory approaches, including governance of traceability systems ([15](#)).

Traceability technologies and related systems may support the integrity of supply chains and improve their efficiency by enabling tracing along supply

chains, from the final manufacturing stage to dispensing or patient administration. Access to real-time (or near real-time) information and appropriate data provides visibility of products and enables prompt regulatory responses by (15):

- ensuring only authorized, registered or approved products are circulating in legal supply chains;
- preventing the distribution, dispensing or administration of substandard and falsified medical products;
- enabling prompt and efficient market recalls;
- facilitating efficient inventory management;
- identifying shortages and reasons for stockouts or shortages.

GP2.13 Establish traceability systems, databases and standards for authorized medical products

The NRA should establish a traceability system for approved medical products, with an appropriate system of governance based on national specificities. The decision should consider product features (for example, real-time digital verification, serialization), interoperability, cost, security, regulatory control, and access to safe, quality medical products (Fig. A9.9).

The NRA should apply international standards for product identification, identification of manufacturing, data capture and data exchange for efficient implementation of the traceability system (15).

The NRA should establish a registry or database of authorized medical products with identifiers to enable regulatory personnel and enforcement officers (for example, customs) to track the origins, storage and distribution points, availability, and regulatory status of the products. This should include products authorized under special conditions, such as for compassionate use, investigational use or emergency use.

The NRA should encourage manufacturers and marketing authorization holders to implement traceability technologies for medical products to facilitate tracing and verification of authenticity, for example, e-labelling and use of serialization data from accredited systems. Manufacturers, marketing authorization holders, sponsors, or agents of medical products granted special authorization (for example, investigational use, compassionate use) or conditional authorization should be required to implement traceability technologies.

The NRA should require importers and distributors that have been granted approvals for import or distribution of medical products under special conditions to maintain product databases to enable tracing of the products through the supply chain, as well as their distribution, storage and disposal, with current inventory balances.

Fig. A9.9

Considerations for nine common features of traceability systems



4.2.4 Good storage and distribution practices

The receipt of authorized quality medical products is a successful initial step, but controls are needed to ensure that the conditions of storage and distribution are appropriate to maintain the product's quality. Further medical products may be exposed to various risks along the supply chain, which involves storage, repackaging, relabelling, transportation, distribution and disposal. It is essential to protect the supply chain against the penetration of substandard and falsified medical products in the market to ensure the quality and safety of medical products, prevent the exposure of patients to substandard and falsified products, and ensure that the integrity of the distribution chain is maintained. The WHO guideline *Good storage and distribution practices for medical products* (TRS 1025, Annex 7) may be used to inform standards for supply chain integrity for the prevention of distribution of substandard and falsified products. NRAs may use it to establish or revise standards for good storage and distribution practices and related inspection frameworks (54). The following recommendations are extracted from this guideline, which should be checked for additional details.

GP2.14 Establish and communicate standards for storage and distribution of medical products

The NRA should collaborate with all stakeholders responsible for the supply of medical products, including customs agencies, law enforcement agencies, manufacturers, importers and distributors, to communicate and uphold national standards for good storage and distribution practice.

In addition to the registration of importers noted above, the NRA should implement policies and procedures for the registration of distributors of medical products. The regulated entities should be advised of requirements, guidelines and the respective regulations in order to uphold the national standards.

The principles of good storage and distribution practices should be included in national guidelines for storage, transportation and distribution of medical products, and they should apply to:

- medical products moving forward in the legal supply chain from the manufacturer;
- medical products moving backwards in the legal supply chain, for example, due to returns or recalls;
- donations of medical products;
- medical products granted special authorization (for example, investigational use, compassionate use).

All activities and operations should be conducted in accordance with national legislation, good storage and distribution practices, and the associated guidelines (54).

GP2.15 Conduct inspections according to established standards and guidelines

Risk-based inspections of storage and distribution facilities should be done periodically by the NRA, including at ports of entry, in collaboration with the relevant stakeholders.

Inspectors should assess compliance with good storage and distribution practice and related guidelines, as appropriate.

Inspections should cover premises, equipment, personnel, activities, quality system, qualification and validation, and other related aspects.

Corrective and preventive actions for observations of non-compliance noted in the inspection reports, with the national legislation and guidelines, should be submitted for review by the inspectors within a defined period, as defined by the inspectors (54). Where feasible, the NRA should include the compliance status of distributors and warehouses on a national database for inspections.

GP2.16 Ensure quality systems are implemented

The NRA should evaluate the extent to which entities involved in storage and distribution of medical products have implemented comprehensive quality systems that incorporate good storage and distribution practice, quality risk management and management review (54).

The NRA should evaluate the relevant quality systems to ensure the following:

- good storage and distribution practices are adopted and implemented to ensure that the quality of medical products is maintained throughout their shelf-life, and medical products are appropriately procured, stored, distributed, delivered and disposed of (in compliance with legislation);
- there are written procedures that clearly specify and describe operations;
- responsibilities are clearly specified in job descriptions;
- all risks are identified, and the necessary effective controls are implemented;
- processes are in place to assure appropriate management of outsourced activities;
- there is a procedure for self-inspection and quality audits;
- there is a functional quality risk management system;
- there are appropriate functional systems for managing returns, complaints and recalls;

- procedures for handling returned goods include appropriate quarantine, security of goods, appropriate storage conditions, safe transport, appropriate destruction and comprehensive record keeping;
- there are systems to manage changes, deviations, and corrective and preventive action.

The NRA should require that entities involved in storage and distribution adhere to good storage and distribution practice to ensure that:

- premises and equipment are adequate and suitable for product storage, handling, transportation and distribution;
- personnel, systems and their management are appropriate;
- procedures for the distribution of medical products are appropriate;
- dispatch procedures are appropriate and adequate.

GP2.17 Identify and separate substandard and falsified medical products

The NRA should ensure that distributors and stakeholders involved in storage and distribution of medical products implement procedures to assist in the identification and handling of suspected substandard, falsified or both medical products. The procedures should include:

- steps for storage in a secure, segregated area;
- clear identification of areas and products to prevent further distribution or sale;
- access control;
- maintenance of records reflecting investigations and action taken, such as product disposal.

Additional details on good storage and distribution practice are provided by WHO in its guidance on *Good storage and distribution practices for medical products* (WHO TRS 1025, Annex 7) ([54](#)). Guidelines developed and adopted by mutual agreement among countries may serve as references for the development of national guidelines, such as guidance published by the European Medicines Agency¹³ and the Pharmaceutical Inspection Co-operation Scheme ([55](#)).

¹³ <https://www.ema.europa.eu/en/human-regulatory-overview/post-authorisation/compliance-post-authorisation/good-distribution-practice#:~:text=medicines%20are%20stored%20in%20the,within%20a%20satisfactory%20time%20period.>

4.2.5 Unethical promotion

The control of promotion of medical products remains important to market surveillance and control, as it allows the NRA to set standards for ethical behaviour, establish legal requirements for marketing authorization holders, and provide guidance for health professionals. This control should also extend to promotion via external communication networks, such as television, the internet and social media, that reach countries with no regulatory frameworks to address electronic modes of marketing and promotion. Unregulated, unethical marketing of medical products may include incomplete, inaccurate or misleading information (for example unproven claims), which can create misconceptions of safety and efficacy, increase demand from unregulated sources including informal markets, and create parallel supply channels with weak oversight, particularly in the case of unregulated internet or social media sales.

Ethical criteria and similar guidance developed by WHO (56) and the Pan American Regulatory Network for Drug Regulatory Harmonization of the Pan American Health Organization (57) may be used by NRAs to inform regulatory standards, policies and procedures to control and monitor medical product promotions.

GP2.18 Collaborate with stakeholders to enforce standards for ethical product promotion

The NRA should collaborate with the relevant national authorities, professional associations, universities and marketing authorization holders to uphold ethical standards for medical product promotion, according to local context.

The NRA should establish reporting or communication channels to enable stakeholders to confidentially report unethical product promotion.

The NRA should apply a risk-based approach to the evaluation of promotional materials in order to analyse potential risks and impact on public health, particularly in cases of incomplete product information or unjustified claims, for regulatory intervention.

4.2.6 Informal markets

[Box A9.3](#) presents a definition of an informal market.

Box A9.3

Adopted definition of “informal market” by the WHO Member State Mechanism on Substandard and Falsified Medical Products

An informal market in the context of medical products is considered:

“A sector of national or local economy where:

- the manufacture, import or export, distribution, sale, supply, or purchase of medical products takes place outside of the legal, regulatory, or administrative oversight of relevant public health or regulatory authorities;
- the medical products have or have not been assessed for safety, efficacy, or quality by public health and regulatory authorities*; and
- the aforementioned activities may be conducted by persons or entities with or without appropriate qualifications, and may take place in a physical, virtual, or a hybrid environment.”

* Authorized products found on the informal market are not considered substandard, falsified or both products per se.

Informal markets create vulnerabilities in supply chains, as they may offer avenues whereby medical products that have not been assessed by the NRA for quality, safety and efficacy may enter markets. In addition, informal markets may be physical or virtual pathways that facilitate the re-entry of authorized products that have been diverted and possibly tampered with into the supply chain. Therefore, quality is no longer assured, as quality controls for the composition, storage and distribution of these diverted products could not be guaranteed, further increasing the likelihood of substandard and falsified products entering the market.

The emergence of internet sales of medical products has presented opportunities for additional informal markets, which require oversight. This is particularly the case in countries with limited regulatory capacity and oversight of physical markets, e-commerce and related imports.

However, no formal guidance was found at the time of the development of this guideline. In an update on the list of prioritized activities for 2022–2023, the WHO Member State Mechanism on Substandard and Falsified Medical Products summarized a commissioned literature review linked to the development of strategies for NRAs to mitigate public health risks posed by distribution of substandard and falsified medical products through informal markets. The report identified drivers of informal markets in low-, middle- and high-income countries, along with intervention strategies. Common drivers included high prices, unavailability or delays in the formal market, geographical accessibility, trust in informal sellers, and weak regulatory oversight (58). In addition, WHO is developing guidance to help NRAs protect supply chains with consideration of informal markets.

With e-commerce and international sales of medical products, regulators would need to become familiar with the existing informal markets in their country. Collaboration with law enforcement authorities, other NRAs, regional networks, international organizations, industry and civil society would strengthen the NRA's ability to identify informal markets and to develop strategies to deactivate them. Examples of international security organizations, such as INTERPOL, the United Nations Office on Drugs and Crime, and the World Customs Organization are provided in Appendix 8. Recommendations for the NRA's oversight over various establishments (including online sellers) are provided in earlier sections, for example, section 3 on national regulatory systems. The following are supplementary to the earlier recommendations.

GP2.19 Establish systems to regulate and monitor channels for promotion, marketing and sales of medical products

The NRA should establish guidelines, policies and procedures for the oversight of various modes or channels of promotions, marketing and sales of medical products, including online sellers.

Online sellers

Recommendations for regulatory oversight of online sellers by NRAs are listed by operational categories for ease of reference.

For online seller registration and disclosure, the NRA should:

- implement legal provisions to require the registration and licensure of online sellers with the NRA with clear criteria and standards for operation and continuity;
- review qualifications and compliance of online sellers with established standards;
- assess the online platform's quality and safety assurance capabilities, including data protection and information security;
- require that online sellers implement medical product quality management systems as part of licensure or registration requirements.

As it relates to information management, the NRA should:

- establish standards for online information about medical products, as per national regulations, for example, conditions of sale;
- verify practices of prescription review, product display, product information, product sales and distribution, handling of complaints and incident reports;

- ensure accuracy, scientific integrity and completeness of product information;
- confirm the implementation of appropriate systems for data security, including security of client information.
- As it relates to partnerships and agreements, the NRA should:
- require that online sellers implement appropriate agreements with partners that clearly define responsibilities for product quality and safety, in alignment with national regulations;
- require that online sellers implement procedures for the review of the qualifications and quality assurance capabilities of partners seeking to use the online platform;
- establish profiles for partner enterprises;
- evaluate partner compliance with legal requirements and standards.

To support market surveillance and incident handling, the NRA should:

- require that online sellers implement systems for product traceability and for reporting and monitoring product sales, complaints, incidents and recalls;
- require online sellers to report incidents, complaints of substandard, falsified or both products, and violations to the NRA promptly;
- require that online sellers and their associates cooperate with NRA inspections, investigations and regulatory actions in response to incidents.

The NRA should establish a publicly available database or listing of authorized medical products and establishments (for example, manufacturers, distributors, dispensers, online sellers) with registered locations (physical or virtual) for verification. Education and awareness campaigns should include ways to identify authorized medical products, as well as establishments licensed for manufacture, distribution and sale of medical products, including online sellers.

The NRA should establish a digital regulatory observatory to monitor social media, online sellers and web-based product promotions to identify emerging risks, and to deter unauthorized marketing. The NRA should select appropriate digital tools, including artificial intelligence solutions, to review platforms and websites.

The NRA should map the various media or channels where promotion and marketing of medical products are active, including promotions by health professionals.

The NRA should adopt or adapt the Member State Mechanism's definition of informal markets and apply this to identify the various types, sources and

locations of physical and virtual informal markets where medical products are sold (58). The NRA should plan to monitor and sanction unauthorized advertising or unauthorized sales of medical products across different media, including social media, to identify promotion of unregistered medical products or import and sale of medical products by unauthorized entities. Where applicable, the NRA should collaborate with law enforcement agencies responsible for data security and cybersecurity.

Supplemental resource

Asia-Pacific Economic Cooperation, in collaboration with regulators, industry stakeholders, representatives from nongovernmental organizations, international organizations and academics from across the globe, has published a supply chain security toolkit using the prevent–detect–respond framework. This toolkit provides a quick reference for guidelines that may be used by NRAs seeking to strengthen supply chain security (59).

5. Detection

5.1 Border control

The identification of potential and actual channels for the entry of substandard and falsified medical products into the market is a critical part of detection, which requires collaboration and cooperation among NRAs, local enforcement agencies (for example, police, customs and border agencies), and international border agencies (for example, World Customs Organization). This is particularly important considering activities of transnational criminal networks involved in smuggling or trafficking of illicit goods. Given the complexity of supply chains (including illicit channels) and the need for efficient use of resources, a risk-based approach to the detection of substandard and falsified medical products via border control, including mapping the various entry points, is expected to improve the effectiveness of the controls. Activities for the detection of substandard and falsified medical products may include targeted risk-based surveys, investigation of complaints, follow-up on suspicious observations (for example, inventory discrepancies), and discrepancies during verification of product authorization (51, 60).

GP3.1 Collaborate with border control agencies for detection of substandard and falsified medical products

The NRA should collaborate with border control agencies, such as customs, to define processes and develop procedures to enable detection and reporting of suspected substandard and falsified medical products identified at borders, specifically:

- good storage and distribution practices and methods for internal audits;
- procedures for verification of product admissibility (for example, registration, market authorization, import approval);
- access to and use of national databases for registered medical products for verification;
- methods for visual inspection and authentication, including recognition of overt and covert technologies;
- methods for the use of screening technologies, where applicable;
- policies, procedures and resources for reporting suspected substandard and falsified medical products to the NRA;
- policies, procedures and tools for sampling of medical products for quality testing;
- procedures for handling suspected substandard and falsified medical products, including collection containers, quarantine, labelling and monitoring storage conditions.

The NRA should facilitate training of border control personnel in policies, procedures, guidelines and resources needed for the detection, reporting and handling of substandard and falsified medical products imported through or seized at borders.

Additional details on implementation of controls by customs for the detection, handling of samples, quarantine of consignments, and storage are provided in the WHO *Guidelines on import procedures for medical products* (TRS 1019, Annex 5) ([51](#)). The guidance includes overarching recommendations that may be used to develop policies and procedures for the national context.

5.2 Risk-based inspection

Regulatory inspection is one of the tools for detecting substandard and falsified medical products in the market, which encompasses product manufacturing and distribution throughout the regulated supply chain. Inspections provide an opportunity for on-site evaluation of the physical structures, processes, company documentation or information systems, quality management system, and risk management. Broadly, inspections aim to assess the regulatory compliance of establishments, identify practices that could lead to the manufacture, transportation, distribution, storage, or trade of substandard and falsified medical products, and conduct timely investigations of complaints or suspected cases.

GP3.2 Structure an inspection programme as part of the NRA's quality management system

The inspection section or unit of an NRA should operate within a written, implemented quality management system that includes a description of the inspectorate's scope and context within the regulatory mandate, including all its functions, processes and activities ([17](#)).

Standard operating procedures should be established and implemented for activities, including inspection planning, conduct of inspections, documentation, review of corrective and preventive actions after inspections, and complaint handling and investigation ([48](#)).

Elements of the quality management system should be developed and documented in a quality manual, which should include such aspects as ([17](#)):

- administrative and organizational structure of the inspectorate:
 - o management of data and documentation, including inspection reports, document review, storage, distribution of reports, access control and change control;
 - o resources and support needed, for example, technologies, digital resources, software, databases and personnel;
- operations of the inspectorate and processes for decision-making, for example, conduct of inspections or audits, handling of complaints, receipt and review of corrective and preventive action, issuance of licences and good manufacturing practice, and good storage and distribution practice certificates;
- performance evaluation, including internal audit and evaluation of indicators;
- quality improvement of the inspectorate based on data collected and analysed.

Additional details on quality management system concepts, requirements, and implementation methods for NRAs are provided in WHO TRS 1025, Annex 13 ([16](#)), and guidance on quality management systems for inspectorates is provided in TRS 1025, Annex 5 ([17](#)). These may be supplemented by guidance developed by regional regulatory authorities or regulatory cooperation groups, such as the Pharmaceutical Inspection Co-operation Scheme ([61](#)).

5.3 Quality risk management

Quality risk management is the overall and continuing process of appropriately managing risks to product quality throughout the product's life cycle to

optimize its benefit–risk balance and use of resources. It is a systematic process for the assessment, control, communication and review of risks to the quality of the medicinal product (62). While quality risk management may be applied to activities by multiple stakeholders involved in the supply chain, from production to dispensing, this section will focus on quality risk management considerations for NRAs in the context of risk-based inspections by national inspectorates. Routine activities by the national inspectorate may exert a certain level of control; however, to ensure that resources are properly and efficiently utilized, implementation of quality risk management is required to enable the identification, evaluation, classification and mitigation of risks through appropriate and robust controls. Fig. 1 in Appendix 4 presents an overview of the quality risk management process, following the *WHO guidelines on quality risk management* (48), which incorporates the concepts described in International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) Harmonized Guideline Q9 for quality risk management frameworks (62).

GP3.3 Establish and document quality risk management processes

General principles

Risk management principles should be applied to regulatory inspections, particularly during inspection planning, to establish the best allocation of inspection resources in order to determine, evaluate and monitor risks identified during inspection for appropriate mitigation, and to identify opportunities for process improvement (48).

The concept of risk-based management should apply throughout the product's life cycle at various stages of the inspection process, for example, during inspection planning, the conduct of inspections and vigilance activities.

Decisions regarding review, inspections or audits, and inspection or audit frequency should consider product risk and the extent of compliance with good practices.

The level of effort, formality and documentation of the quality risk management process should be commensurate with the level of risk.

The evaluation of the risk to quality should be based on scientific knowledge and ultimately linked to the protection of the patient.

In the risk assessment the following basic questions should be addressed:

- What might go wrong?
- What is the nature of possible risks?
- What is the probability of their occurrence and how easy is it to detect them?
- What are the consequences (the severity)?

Based on the criticality or level of risk, specific corrective actions should be developed to prevent the recurrence of adverse events, both at the inspected site and at other relevant locations.

Appropriate systems should be in place to ensure periodic monitoring and review to assess new information that may impact the original risk-based decision.

The quality risk management process and methodologies should be verified, communicated and documented (62).

GP3.4 Apply quality risk management approaches to the NRA inspection strategy

The primary aim of risk-based inspection is to identify the sources of hazards and hazardous situations that will compromise the quality of product and hence affect the patient's health, through the efficient and appropriate use of available resources. This includes facilities involved in the production of APIs and the use of excipients.

The inspectorate should have a clearly defined philosophy, approach and procedures for risk management, which should be periodically reviewed and updated (at least annually) and should address all types of inspections and other related activities (48).

The appropriate risk assessment tools should be used and the risk assessment for a site to be inspected should be documented using a standard format, for example, on a risk assessment worksheet (48).

In addition, application of risk-based auditing approaches and risk-based selection of methods is an important factor when a full scope audit is not possible (63).

GP3.5 Plan and conduct inspections based on assessment of risk

Proper planning of regulatory inspections is necessary for:

- efficiency;
- prioritization of high-risk locations, companies and products;
- compliance with local laws;
- prevention of serious problems;
- transparency and accountability;
- continuous improvement.

Planning for inspections of regulated facilities should be risk based. A quantitative system should be used for risk ratings, with clear descriptions for each rating.

The inspection workforce and the required duration of the inspection and its scope should be determined based on the risk rating for the site (48).

The scope of inspections should:

- be based on the assessment of risk, including previous inspection findings and corrective and preventive action, where applicable;
- cover the overarching aspects of regulatory audit based on the appropriate legislation;
- apply to manufacturing and distribution at the facility.

Documentation of pre-inspection risk assessment should be completed for each facility.

At the planning phase, risk ratings should normally be done only for sites that have been previously inspected. However, for sites that have not been previously inspected, a modified risk rating may be developed, considering that the absence of information about the facility's management and compliance with good practices is itself a risk factor.

Risk factors

Risk factors may vary according to the type of inspection. Factors that should be considered to inform risk assessment include (48):

- outcomes of inspections by other regulatory authorities or WHO;
- the number and types of complaints related to the facility or products;
- outcomes of previous inspections;
- previous cases of substandard and falsified incidents;
- the complexity of the site (for example, buildings, utilities);
- complexity of the product (for example, sterile, non-sterile);
- the type or nature of the product (for example, biologicals, oral solution with diethylene glycol, or product with nitrosamine impurities);
- significance of changes (for example, equipment, key personnel);
- results of product quality testing;
- risk to the patient;
- complex routes of synthesis (API);
- polymorphism (API);
- biopharmaceutical classification of the product;
- innovative or emerging technologies.

The criticality of the products may also be determined based on predefined criteria, considering situations such as:

- public procurement programmes for medical products;
- essential medicines, vaccines and medical devices considering the country's priority diseases;
- immunization programmes;
- lifesaving medical products.

In addition to consideration of the complexity of the establishment, critical nature of the products, and regulatory risk (compliance with good practices, recalls, analytical reports), the risk assessment should be conducted considering the type of establishment (for example, manufacturers, storage and distribution facilities, pharmacies). Where there is variation in criteria for the selection of establishments, risk assessment worksheets should be developed to match the type of facility to be inspected.

Weights should be assigned to the various risk factors, as their impact and importance may vary. The points assigned will be directly related to presence, complexity and importance of the various factors. Numerical ranges used to classify the level of risk should be clearly explained using qualitative descriptions.

Risk assessment tools

Various tools may be considered for risk assessment by the inspectorate, as no single tool or combination of tools may apply to every situation. Examples of tools for quality risk management, including risk assessment matrices, are provided in the *WHO guidelines on quality risk management* (48). Appendix 4 to the present annex includes tables that provide examples of:

- a risk assessment of a facility based on compliance with good manufacturing practice and inspection frequency, based on ICH Harmonised Guideline Q9 on quality risk management (62) and WHO TRS 996, Annex 4 (64) (Table 1);
- an inspection planning tool (Table 2);
- a decision matrix for frequency of inspections based on regulatory and intrinsic risks (Table 4).

Tools developed by the NRA, or adapted from other sources, should be pre-tested by users, and modified prior to implementation. The NRA should periodically review and update risk assessment tools, as needed to ensure their relevance and consistent application by officers.

Regulatory reliance for inspections

The inspection of a site that has not been inspected previously should be waived only where a recognition or reliance procedure exists for use of findings by other

regulatory authorities or organizations that meet the criteria for reliance for inspections. In addition, there should be supporting evidence of compliance with good practices of the facility that indicates an acceptably low risk to products and patients.

GP3.6 Conduct regulatory inspections throughout the supply chain

In practical terms, regulatory inspections should be conducted in facilities at different points in the supply chain to:

- evaluate facilities prior to licensure or authorization;
- monitor compliance with good practices (manufacturing, storage, distribution, disposal);
- follow up proposed action plans or take corrective and preventive action in relation to non-conformities identified in prior inspections;
- investigate quality issues or incidents, including tampering and product damage, taking into consideration complaints received from the public.

Where inspections are conducted in the informal market (unlicensed facilities), the NRA should:

- consider the presence and relative proximity of licensed facilities (public or private) in the area;
- evaluate the need for support of law enforcement authorities, such as the police and customs officials, to enforce regulatory actions.

Lack of access to licensed facilities due to distance, and lack of law enforcement, may encourage the emergence and continuation of operations of unlicensed facilities.

Conducting inspections and reporting

Inspections should cover the premises, equipment, personnel, activities, quality management system, the application of quality risk management, qualification and validation, and other related aspects.

An inspection report should be prepared and provided to the inspected entity within a specified period following the last day of the inspection (54).

In addition, the inspection report should align with the basic principles of documenting observations and findings, as described in the WHO model inspection report (64).

Inspection reports should contain findings and observations, with appropriate classification of departures from good practices based on risk, for example, “critical”, “major” or “minor” (65).

Standard operating procedures describing the inspection process, reporting, classification process, the application of qualitative and quantitative ratings of inspection findings, and applicable assessment tools (including software) should be developed and available for reference.

As far as possible, inspection findings and other forms of surveillance information should be integrated into the NRA's systems, for example, use of a unified dashboard, to support interdepartmental collaboration for regulatory activities.

GP3.7 Review corrective and preventive actions, and schedule routine inspections

Corrective and preventive actions addressing the observations listed as non-compliance in the inspection report, with reference to the respective national legislation and guidelines, should be submitted for review by the NRA within the defined period (54).

The NRA should consider the review of corrective and preventive actions in determining ongoing acceptability of the site and adjust risk ratings accordingly.

The timing of the subsequent inspection and frequency of routine inspections should be defined based on the risk rating and capacity of the NRA.

5.4 Risk-based surveillance

5.4.1 Purpose and framework

Market surveillance programmes aim to assess the quality of medical products provided to patients and generate the data that can help to formulate strategies and plans to ensure provision of good-quality products (45). Risk-based approaches enable NRAs to establish effective, affordable and sustainable market surveillance systems.

GP3.8 Develop a strategic framework for risk-based market surveillance

The NRA should plan and implement risk-based market surveillance in the context of a strategic framework that includes, at a minimum (66):

- assessment of local needs of the medical product sector;
- prioritization of objectives of the market surveillance programme;
- planning for implementation of risk-based market surveillance, including determination of sampling and testing priorities;
- implementation of risk-based sampling and testing;
- data analysis, investigation protocols and communications;
- development of protocols for escalation of regulatory actions and coordination in response to findings;

- monitoring and evaluation of operations for regulatory system strengthening.

NRAs should lead and supervise market surveillance activities, with engagement of the relevant stakeholders involved, including other government offices (for example, regional regulatory offices and public health programmes), law enforcement agencies, affiliated laboratories (for example, private laboratories and universities), manufacturers, importers or procurement agencies, distributors and health professionals. A national taskforce may be established with the NRA as the secretariat for coordination of market surveillance activities.

Medical product sector assessment

An assessment of the medical product sector may inform the development of market surveillance goals, objectives and activities, both prior to initiating a market surveillance programme and periodically during routine planning. The information collected may provide insight on potential high-risk products, regions or locations for sampling and testing activities, identify gaps in capacity, and assist in ensuring that market surveillance activities are relevant to local needs and priorities.

Prior to initiating the national market surveillance programme, or on review of the existing programme, the NRA should assess the medical product sector, including (66):

- local capacity for quality assurance and quality control activities and systems throughout the product life cycle (for example, market registration, procurement, storage and distribution practices, licensing, enforcement);
- applicable legal frameworks, including regulations;
- relevant sectors related to product quality (for example, manufacturing, importation, wholesalers, retail markets, online sellers);
- all relevant stakeholders and their role in market surveillance and control.

After the full assessment, the NRA should conduct periodic review of any changes in the regulations and the medical product landscape that may affect surveillance activities.

5.4.2 Priorities for risk-based market surveillance

The capacity and effectiveness of market surveillance of the NRA may vary among countries. It is therefore important for NRAs to consider priority activities for

market surveillance. Market surveillance objectives, activities and priorities may be influenced by various factors, including:

- the extent to which the quality of the medical products has been assessed prior to registration, for instance where the regulatory system of the country requires strengthening;
- compliance with good manufacturing practice and good storage and distribution practice during the manufacture, storage and distribution of medical products;
- the number and variety of products that are imported;
- the prevalence of informal markets, including unauthorized online sales;
- external (regional or international) notifications of substandard and falsified medical products that may affect the local market;
- access to dedicated human, financial and technical resources, for example an accredited quality testing laboratory with the appropriate scope of testing and adequate capacity.

GP3.9 Determine priorities for market surveillance using available market information and related information systems

Information sources

Different information sources may inform priorities for market surveillance, including product quality complaints from consumers or health workers, suspicious observations in the supply chain (for example, unexpected stock levels, unauthorized sellers), discrepancies during verification of unexpected adverse events reported for a given product, notices of substandard and falsified products from other NRAs or country networks, notices or alerts from regional or international organizations, findings from quality surveys, evaluation of online sellers and promotions, independent research, and reports from manufacturers, marketing authorization holders or distributors.

The NRA should utilize findings from the medical product sector assessment, along with information from local regulatory agencies, regional and international agencies or networks, to inform priority sampling and testing needs for market surveillance. Examples include:

- product quality complaints from consumers, including via media reports;
- case reports of quality defects, therapeutic failure, adverse drug reactions, adverse events following immunization, or adverse events involving medical devices from health professionals;

- queries, reports or notifications from customs or other national security agencies or authorized personnel;
- findings from searches of virtual environments for promotions or sales of unauthorized medical products;
- national quality surveys by the NRA or independent researchers, and inspection reports (routine or ad hoc) by the NRA (for example, with regard to contaminated cough syrups);
- inspection reports, quality audits or confirmation of compliance with good manufacturing practice issued by a reference NRA;
- signals from national, regional or international vigilance systems (local, regional, international);
- medical product quality notices or alerts from regional networks or international databases, for example, WHO GSMS rapid alert system;
- targeted product surveillance by other NRAs, regional regulatory authorities or networks, or international organizations;
- product recalls by manufacturers or marketing authorization holders;
- notices or alerts by international organizations that provide information about substandard and falsified medical products detected or seized globally, such as INTERPOL pharmaceutical crime operations,¹⁴ WHO medical product alerts,¹⁵ and the World Customs Organization;
- publications by nongovernmental organizations that monitor the quality and safety of medical products and may report irregularities;
- open-source intelligence for public health decision-making, for example, WHO's Epidemic Intelligence from Open Sources system,¹⁶ which may be a resource for reports on substandard and falsified products detected or related regulatory activities by other NRAs.

5.4.3 National vigilance and market surveillance

The functions of vigilance and market surveillance and control by NRAs often intersect, as the adverse outcomes that arise from the use of substandard, falsified or both medical products are often detected through reports of adverse drug

¹⁴ <https://www.interpol.int/en/Crimes/Illicit-goods/Pharmaceutical-crime-operations>.

¹⁵ <https://www.who.int/teams/regulation-prequalification/incidents-and-SF/full-list-of-who-medical-product-alerts>.

¹⁶ <https://www.who.int/initiatives/eios>.

reactions, adverse events following immunization, treatment failure (for example, due to antimicrobial resistance), and adverse events or incidents involving medical devices. Reviews of reports received, data mining or trend analyses enable the detection of clusters of adverse events that may alert the NRA to substandard, falsified or both medical products. This intersection requires clear mandates, roles and responsibilities of the different regulatory units of the NRA to avoid duplication of efforts and wastage of resources.

GP3.10 Utilize existing vigilance systems to detect substandard and falsified medical products

The NRA should clearly describe the relationship, workflow procedures and tools for reporting and information sharing between vigilance personnel or units and their counterparts in market surveillance and control, or post-market surveillance, where these are separated.

Policies, criteria and procedures should be developed and implemented to support the workflow, information exchange, joint activities, data analyses and risk assessment for effective interdepartmental collaboration.

Product quality complaints from consumers and reports from health professionals should be reported to the NRA via a national reporting system to encourage complete and reliable reporting. User feedback submitted to manufacturers of medical devices should be shared with the NRA where there is a threat to public health.

Reports received by the NRA should be verified and assessed to determine the next steps, including case investigation, risk assessment, sampling, testing and further regulatory action ([59](#)). Additional recommendations for reporting systems for market surveillance will be provided in section 5.5 on reporting systems.

GP3.11 Leverage notifications and alerts from other sources and stakeholders

When considering alerts from external sources (other NRAs, regional or international networks), the NRA should consider:

- information about the source (product manufacturer or supplier) and current compliance with good practices;
- the type of deviation, for example, substandard, falsified or unregistered product;
- product names (nonproprietary name, trade name) and active ingredients;
- the location where the product was found (geographical, border proximity, type of establishment or facility);

- common sources (manufacturers, suppliers), shared distribution channels or borders;
- the packaging or label language.

The NRA should develop policies, criteria, requirements and procedures for reliance based on product quality alerts and inspection reports or notifications from other NRAs, regional regulatory agencies, and regional or international organizations. The policies should include a risk assessment of the alerts, and subsequent steps for verification, investigation and action.

Sampling and testing priorities

Sampling and testing priorities that an NRA may consider include (66):

- monitoring newly marketed medical products, particularly innovator products;
- monitoring based on risks associated with complex manufacturing, dosage form, impurities, stability, safety, efficacy, demand or therapeutic indication;
- monitoring the quality of medical products at key ports of entry;
- coordination with ongoing initiatives for sampling and testing, for example, activities by national health programmes, regional initiatives by networks and surveillance activities by other stakeholders;
- medical products included in substandard and falsified alerts by other NRAs, or by NRAs with the same sources of production or distribution.

5.4.4 Planning for market surveillance: sampling and testing

The purpose of sampling and testing programmes in the context of active market surveillance may vary as these may be organized to achieve different objectives, for example, to confirm that products used by patients are of appropriate quality, to check the effectiveness of the regulatory system, or to address a suspicion that patients are not receiving quality medical products. At a programmatic level, objectives may include one or more of the following (45):

- to evaluate the quality of selected medical products available in formal or informal markets (where warranted) in selected areas, regions or countries, and at various levels of the supply chain, with the aim of assessing the exposure of patients to poor-quality products and preventing harm;
- to evaluate the quality of specific medical products used in national treatment programmes and, more broadly, to identify those that are not compliant with standards;

- to compare the quality of domestically produced and imported medicines, where applicable, based on previous regulatory or research findings;
- to identify vulnerabilities of the supply chain, including possible factors that may contribute to or have contributed to patient or user exposure to substandard medical products (for example, storage conditions of facilities, poorly regulated entry points, and compliance with good manufacturing practice);
- to test the quality of selected medical products for the identification of manufacturers that are not compliant with regulatory standards;
- to determine if any falsified or unlicensed medical products within a selected category have penetrated the market in selected areas, regions or countries.

Quality surveys inform the NRA about the quality of medical products available to patients and are important to all regulatory systems, irrespective of the maturity level or regulatory strength. These tools provide an indication of product quality and patterns and, when done correctly, they inform effective interventions, though their effectiveness and reliability depend on the survey design, sampling and analysis, and available resources ([45](#)). When planning for quality surveys, the NRA should consider the subsequent activities or actions that would be needed in response to the detection of poor-quality medical products.

GP3.12 Determine the scope and frequency of sampling and testing activities

National surveillance sampling and testing activities should be planned and executed annually, if possible, using a risk-based approach to determine sampling and testing priorities across different medical products in the public and private supply chains ([6](#), [66](#)).

Planning for sampling and testing should involve (a) establishing objectives for each activity prior to implementation; (b) defining the criteria for the implementation of the activities; and (c) describing the methods, information sources and risk-based considerations for quality surveys, including sampling and the subsequent testing of the samples.

Irrespective of the maturity level of the NRA, planning should include monitoring the regional and global landscape for emerging threats.

The NRA should also describe how the findings of market surveillance would be used to inform regulatory actions, revision of policies and development of strategies for regulatory strengthening.

Prior to implementing sampling and testing programmes, NRAs should determine:

- the formal markets or sectors to be surveyed or monitored;
- the sources of information for market intelligence (including advertisements, social media) and vigilance (local, regional, global systems);
- the criteria and procedures for retrieval and use of information or reports (see section 3.2);
- criteria to identify at-risk medical products, and procedures for risk assessment;
- resources required to initiate and sustain active surveillance (personnel, competence, data access);
- feasibility and sustainability of surveillance activities.

Planning for individual quality surveys

GP3.13 Clearly define objectives of quality surveys

For each quality survey, the NRA should clearly define the objectives of the survey, which should include the survey rationale, identification of medical products to be included in the survey, sites of sample collection, surveyed areas and tests to be conducted ([45](#)).

To ensure that quality surveys provide the necessary information, the NRA should develop appropriate and relevant questions to be answered or issues to be addressed when results become available. Examples include:

- What proportion of sampled products failed testing?
- What was the proportion of sampled products that failed quality testing across the different levels of the regulated supply chain?
- What are the critical deficiencies and risks (for example, treatment failure, potential for patient harm)?
- What is the registration status of the sampled products and the respective proportions?
- What are the supply chains or channels by which poor-quality medical products are distributed, and their respective market segments?

Additional examples of questions for consideration during planning are provided in WHO TRS 996 on the conduct of quality surveys ([45](#)). In addition, WHO provides guidance on testing of suspected falsified products in TRS 1010, Annex 5, which includes information on sampling and documentation that may inform planning of quality surveys ([60](#)).

5.4.5 Survey management

GP3.14 Identify partners and agree on survey management plan

The regulatory authority and the relevant partners should collaborate and agree on the survey plan before it is implemented. Key roles, responsibilities and tasks of various actors (for example, survey coordinator) should be identified, including persons responsible for monitoring, analysis and report preparation. The lines and means of communication should be agreed on, and meetings should be held with local coordinators and survey personnel (where possible) to explain and discuss the survey protocol ([45](#)).

Risk-based sampling

Sampling comprises the operations designed to select a portion of a medical product for a defined purpose. The number of medical products authorized for the market, which varies greatly among countries, makes sampling of all products difficult and often unfeasible. This is compounded by limited financial, technological and human resources, particularly in low- and middle-income countries. Therefore, risks related to products, locations or facilities should always be considered, irrespective of the overall purpose of the survey.

GP3.15 Apply a risk-based approach to select products, locations and facilities

The NRA should apply a risk-based approach for efficient use of resources, based on where the risks are greatest, considering ([61](#), [66](#)):

- risks associated with the medical products, their APIs and excipients;
- risks associated with the geographical location;
- risks associated with the facilities downstream in the supply chain, where medical products are stored or dispensed.

Examples of risk considerations are shown in [Table A9.7](#). These were adapted from the United States Pharmacopeia *Guidance for implementing risk-based post-marketing surveillance in low- and middle-income countries* ([66](#)).

Table A9.7

Risk considerations for selection of medical products, areas, sampling sites and methods

Aspect	Risk considerations
Medical products	• Newly introduced products on the market • Medical products with limited safety and efficacy data • Medical products with complex formulations or high-risk excipients • Medical products known to have issues involving bioequivalence, sterility, stability or impurities • Medical products to which antimicrobial resistance is increasing • Medical products in high demand or of public interest • Manufacturers or suppliers with previous quality issues, including non-compliance with standards, for example, good manufacturing practice or good storage and distribution practice • Products and product classes with validated case reports from vigilance systems where incidents were deemed related to product quality • Consumer or user complaints that have been validated • Product recalls issued by importers, distributors or manufacturers
Geographical areas	• Poor storage conditions • Distribution sites or locations for substandard and falsified products • High disease burden • Population size • Locations of informal markets • Complexity of the supply chain • Porous border zones
Collection sites or downstream facilities, and methods	• Public or private sector • Facilities with a history of substandard and falsified products • Centralized or decentralized supply chain • Sites that are difficult to access • Virtual markets • Informal outlets in unregulated sectors • Ability to maintain chain of custody to preserve integrity from collection site to testing laboratory • Appropriate handling, storage and transportation • Cold chain requirements

NRAs should assess available tools that support the development of sampling protocols and adopt or adapt the tools that are appropriate to its needs and local context.

The selection criteria for medical products to be sampled, target areas, collection sites and sampling methods should be clearly identified and applied during planning in collaboration with key stakeholders and based on NRA knowledge of the supply chain ([66](#), [67](#)).

The roles and responsibilities of various personnel, including stakeholders involved in sampling at different stages, should be clearly established, described and agreed upon ([67](#)). Stakeholders may include personnel at customs receiving imported medical products, managers responsible for storage and dispensing facilities, and field inspectors involved in sampling medical products.

Survey methodology

GP3.16 Develop and communicate a predefined survey protocol

All surveys should be conducted according to a predefined survey protocol to avoid inaccuracies, insufficient sample collection, incorrect sampling or testing that would produce inaccurate results ([45](#), [67](#)).

The sampling procedure should be appropriate to the purpose of sampling, to the type of controls intended to be applied to the samples, and to the material to be sampled.

The survey protocol should include the plan of survey activities, personnel responsible and estimated time frame ([45](#)).

The protocol should clearly describe the following:

- selection of areas to be sampled;
- selection of medical product, API or excipient;
- selection of sample collection sites;
- sampling design;
- sampling plan, specifying dosage units and quantities;
- sample collection approaches, methods and instructions for samplers;
- chain of custody, storage and handling of samples;
- criteria and protocol for laboratory testing;
- identification of testing laboratory;
- test methods and specifications;
- ethical considerations.

Sampling should be carried out by trained inspectors according to approved standard operating procedures ([66](#)).

Additional details and tools

When developing more procedures for the conduct of quality surveys by inspectorates, NRAs may utilize the relevant guidance in TRS 996, Annex 7, which includes an example of a survey plan (45). Similarly, the principles and procedures described in the *WHO guidelines for sampling of pharmaceutical products and related materials* (67), *WHO guidance for post-market surveillance and market surveillance of medical devices, including in vitro diagnostics* (7) and *WHO guidance on testing of “suspect” falsified medicines* (60) may support NRAs seeking to develop or revise these procedures.

In addition, WHO has developed toolkits for the following procedures to supplement the guidance (see Appendix 9). These may be accessed by NRA staff via the WHO GSMS e course, and are listed below:

- taking good photographs of suspected substandard and falsified medical products;
- identifying suspected substandard and falsified medical products;
- sample collector’s checklist;
- preparing and using tamper-proof (evidence) bags and team leader’s checklist;
- sample collector’s aide-memoire.

Table 1 in Appendix 5 presents information on content recommended for inclusion in quality survey protocols. It summarizes key recommendations and elements of the survey protocol provided in WHO TRS 996, Annex 7: *Guidelines on the conduct of surveys of the quality of medicines* (45).

5.4.6 Risk-based quality testing

The limited availability of human and financial resources in many countries, particularly the lack of a functional national quality control laboratory, makes it difficult for NRAs to test medical products when needed to ensure the quality of medical products throughout the supply chain. Even in countries with a functional national quality control laboratory, resources need to be used responsibly and efficiently to avoid wastage and ensure the sustainability of services.

GP3.17 Implement efficiencies in quality testing to reduce waste and unnecessary costs

A risk-based, tiered approach to testing products is recommended, particularly in limited-resource settings (45, 60, 68).

Depending on the objectives of each sampling and testing activity, implementing a tiered approach to testing can reduce the number of samples needed and the types of tests to be performed without affecting the overall quality

of market surveillance. For instance, if the objective of a quality survey is to assess the stability of medicines at different levels of the supply chain (that is, the storage conditions), then it may not be necessary to conduct full compendial testing at each level of the system (66). The three-level approach provides a framework for product testing through three successive, complementary and increasingly rigorous levels of analysis: (a) visual and physical inspection; (b) screening with quality-assured field-based methodologies; and (c) laboratory testing utilizing compendial or manufacturer methodologies ([Table A9.8](#)).

Table A9.8

Framework for tiered (three-level) approach for medicine quality testing

Level	Description	Key decision points
1: Visual and physical inspection	Initial check of registration status, expiration date, labelling, packaging, appearance, physical properties, use of traceability or authentication technologies	Fails (any aspect) – report to NRA for appropriate action – no further testing needed
		Passes – proceed to level 2 (screening) Suspicion of substandard or falsified product – proceed to level 2
2: Field-based screening	Use of field-based screening technologies, which may test for identity, content and other quality attributes	Fails product identification – report to NRA for further action – no further testing needed Passes product identification and other screening tests as applicable – no further testing may be needed
		Passes product identification but fails other screening test – proceed to level 3 (compendial) testing, starting with failed test Confirmation of results required based on protocol – proceed to level 3 testing

Table A9.8 *continued*

Level	Description	Key decision points
3: Compendial Testing	The use of pharmacopoeial methods or other validated methods approved by the NRA	Failure of any of the following is sufficient to stop, and report to the NRA. No further testing is required: • disintegration, identification, assay, dissolution • uniformity of dosage units, related substances • bacterial endotoxin, sterility • other tests required for the dosage form • confirmation of failed level 2 result using compendial method
		Passes all tests – stop and report to NRA – no evidence of poor quality

The framework uses each level in a stepwise manner to decide if a medicine may be considered of good quality, substandard, or falsified, and whether it should continue to the next level for further analysis, or if there is a need for corrective regulatory action. United States Pharmacopeia guidance on post-marketing surveillance provides flowcharts that illustrate the use of this approach for decision-making (66). For instance, the outcome of a screening test may serve as an initial indication of a substandard product issue or confirm a falsified product (no active ingredient). Further analytical techniques may be considered necessary if the NRA wants to quantify the active ingredient or further confirm the detection of a falsified medical product. However, where screening has confirmed that the product lacks the active ingredient or the levels are well below the label levels, the finding should be sufficient to trigger regulatory action, particularly in limited-resource settings or where quality testing is not readily accessible or is cost prohibitive. It is important, however, to include in the country's regulations that results obtained with screening methodologies are sufficient for the NRA to implement subsequent regulatory actions. Reports and publications about the use of this approach in countries from Africa, Asia, and Central and South America are included in the Medicines Quality Database.¹⁷

¹⁷ <https://www.usp.org/global-public-health/medicines-quality-database>.

Any tiered approach and criteria used for quality testing of medical products within market surveillance programmes should be clearly described in survey protocols and testing guidance (45, 66).

If a sample fails the visual inspection, the NRA should communicate with the manufacturer or marketing authorization holder to confirm its authenticity and trace its sources, to determine if it is falsified or substandard.

The NRA should clearly describe the procedures for products that fail visual inspection or screening, including confirmatory testing or immediate regulatory action to protect public health. Further confirmatory testing can be determined based on the risk assessment, conditions of discovery and survey objectives, in accordance with national regulations for subsequent regulatory and legal action (68).

The NRA should consider the characteristics of products to be tested, the purpose of testing, its local capacity, available resources and the test attributes to select the appropriate testing level due to the limited number of attributes tested through level 1 and level 2, which may only be detected by compendial methods in level 3.

For level 3 testing (laboratory), NRAs should (45, 66):

- implement a scheme for prioritizing compendial testing based on the type of product, risk associated with samples, costs of tests and technical complexity – this should start with tests that are less labour intensive or those that assess critical quality attributes;
- ensure access to an accredited quality testing laboratory (national quality control laboratory or contracted official medicines control laboratory) for testing according to registered specifications;
- ensure the use of pharmacopoeial methods or other validated methods that have been approved (for example, methods submitted in the product dossier).

Regional networks

Examples of collaborative risk-based surveillance may be found in the Caribbean, through its regional quality testing programme at the Caribbean Public Health Agency (69), and the European Union, through its sampling and testing of centrally authorized products programme¹⁸ and its scheme for post-marketing surveillance of mutual recognition and decentralized procedure products.¹⁹ The latter approach enables the sharing of surveillance responsibilities and minimizes

¹⁸ <https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/compliance-marketing-authorisation/sampling-testing>.

¹⁹ <https://www.edqm.eu/en/d/240851>.

redundant efforts by replacing individual national testing systems for centrally authorized products with a single, harmonized procedure across all European Union and European Economic Area Member States. By pooling expertise, it provides access to advanced technologies and analytical methods, ensuring optimal use of laboratory resources throughout the European Union and European Economic Area. It complements national surveillance programmes that focus on products authorized nationally or through mutual recognition and decentralized procedures. National authorities of the European Union may also test centrally authorized products as part of their own oversight responsibilities (70, 71).

The network also provides examples of planning for periodic surveillance activities through coordination and collaboration between the European Union network of official medicines control laboratories,²⁰ the European Medicines Agency,²¹ and the European Directorate for the Quality of Medicines and HealthCare. These approaches may be utilized by other NRAs working within a regional network, or by those using decentralized laboratories. The NRAs may maximize benefits and minimize use of resources by initially considering risk based on pre-marketing conditions and post-marketing information generated during implementation.

Preparation for quality testing

After medical products have been collected following the sampling protocol and in accordance with the objectives of the surveillance exercise, the samples need to be subjected to quality control to verify their compliance with the requirements established during registration. Compliance with these quality parameters is critical to ensure that the safety and efficacy profiles established for registration are maintained.

The quality control of sampled products consists basically of two processes:

- the visual inspection of the package and product insert, the physical characteristics, and the organoleptic properties (when applicable) of the dosage form;
- the assessment of the quality attributes of the dosage form via screening technologies or compendial testing.

Different tests enable the detection of different quality issues. Examples of issues detected in the context of medicines are shown in [Table A9.9](#).

²⁰ <https://www.edqm.eu/en/d/144225>.

²¹ <https://www.ema.europa.eu/en/compliance-marketing-authorisation-0/sampling-and-testing/sampling-testing-questions-answers>.

Table A9.9
Medicine quality issues detected by certain tests

Test (type)	Possible product quality issues	Comments
Visual inspection	Falsely or incorrectly labelled Poor appearance or odour Unregistered	Comparison with innovator or registered products in the country. The national medical product registration database is a good source of information
Identification	Incorrect or absent active ingredient or excipient	Techniques may vary depending on capacity and technology
Assay Uniformity of dosage unit	Quantity of active ingredient inconsistent with claim on label	See pharmacopoeia sections on uniformity of dosage units (QAS 15.635)
Disintegration Dissolution	Dosage form performance	Harmonized across pharmacopoeia – United States Pharmacopeia, European Pharmacopoeia
Related substances, impurities	Degradation or impurities	Product specific – refer to pharmacopoeial or other standards
Endotoxin Sterility Foreign particulate matter Viscosity pH	Toxicity or contamination of liquid and sterile formulation	Microbial testing may be needed

Adapted from United States Pharmacopeia *Guidance for implementing risk-based post-marketing surveillance in low- and middle-income countries* (66).

5.4.7 Visual and physical inspection

Visual and physical inspection stand as pivotal methods for the detection of substandard and falsified medical products due to their simplicity, speed and cost-effectiveness, while providing sufficient immediate information for supporting prompt regulatory action in some cases, reducing the need for laboratory testing. The visual and physical inspection of the samples can help identify important characteristics related to product quality in the examination of the

package (for example, registration status, expiry date, packaging information and integrity status), or the physical characteristics of the dosage form (for example, presentation, colour, texture or other organoleptic properties), and support other testing techniques (72). Product protection features can serve as additional measures to screen and authenticate suspicious packaging components. In addition, there may be overt (visible) or covert (hidden) marks embedded in the package of genuine products to enable authentication, which may help identify falsified products during visual inspection (72). Both features can be integrated for enhanced authentication.

Overt and covert technologies

Table A9.10 outlines the features of both overt and covert technologies that may assist NRAs in determining the appropriate technologies for product authentication. Additional information regarding both overt and covert technologies is presented in the report on the WHO *Member State Mechanism on substandard/spurious/falsely-labelled/falsified/counterfeit medical products*, Appendix 2 (72).

Table A9.10
Features and examples of overt and covert technologies

Overt technologies	
Features	Examples
• Visible security features discernible to the naked eye • May be costly to replicate • Patients and health care professionals can authenticate medical products, with appropriate education • Integration of overt technology with tamper-evident features is advisable, to prevent falsification • This technology is considered short lived because it can be easily copied	• Tamper-evident measures • Holograms • Optically variable devices • Colour-shifting security inks and films • Fugitive inks • Security graphics • Scratch-off technologies • Overt use of a covert technology

Table A9.10 *continued*

Covert technologies	
Features	Examples
• Allow NRAs, marketing authorization holders and other stakeholders to identify substandard and falsified medical products • Typically, patients are unaware of the presence of covert features and lack the means to verify them • Design should make features difficult to detect or replicate without specialized expertise • Access to their details should be restricted on a “need to know” basis to prevent duplication	• Invisible printing • Latent image or three-dimensional (3D) intaglio printing • Embedded image • Watermarks or filigrees • Digital watermarks • Hidden marks • Microtext or microprinting • Anti-copy or anti-scan designs • Safety fibrils or filaments • Laser coding • Marks in a die-cut profile • Substrates • Odour

GP3.18 Use visual and physical inspection, authentication technologies and databases to detect substandard and falsified medical products

Visual inspection should be done by the inspector or personnel assigned after the medicine samples are collected to detect substandard, falsified or unregistered products.

Regulatory personnel and enforcement officers involved in sample collection or inspection (for example, inspectors, customs officials or other assigned personnel) should undergo training in conducting visual inspections and the use of national databases and traceability systems, emphasizing the identification and documentation of irregularities or inconsistencies such as spelling errors, unusual batch numbering systems, atypical batch number printing and signs of repackaging.

Marketing authorization holders or manufacturers should be contacted to provide the NRA with reference samples of products in cases of suspected falsified products. Adequate quantities of samples or their components (for example, labels, primary and secondary packaging) should be available to facilitate meticulous visual comparison with genuine authorized products, aiming to achieve the intended goal of visual inspection.

The physical conditions and appearance of the product label, packaging and presentation should be assessed. In addition, the following aspects should be assessed based on dossier information ([45](#), [66](#)):

- trade name of the medicine;
- the scientific, generic or international nonproprietary name²² of the product;
- appearance of company trademarks (for example, logo, colours);
- manufacturer's name and full address;
- dosage form, route of administration, number of units per container;
- integrity of primary and secondary packaging;
- product colour, texture, presence of particulates;
- batch number;
- dates of manufacture and expiry.

When performing visual inspection for the detection of substandard or falsified products ([60](#)), additional characteristics should be noted, including:

- irregularities or inconsistencies, such as spelling mistakes;
- unusual batch or lot numbers;
- unusual printing of batch or lot number and shelf-life;
- unexpected or modified manufacturing or expiry dates;
- signs of e-packing, for example, to circumvent inspection activities;
- instructions in a language that does not match the area of their distribution;
- organoleptic characteristics;
- presence or absence of overt or covert marks implemented by the legitimate manufacturer to facilitate authentication.

Products that fail any aspect of visual inspection should be reported for follow-up by the NRA to determine appropriate action, for example, communication with the manufacturer for confirmation of falsification or authenticity of the product, investigation, and other appropriate regulatory actions.

Suspicious products should not be sent for analytical testing until proper follow-up has been completed. Consideration should be given to the need for samples of the product by law enforcement for criminal proceedings.

The quality attributes of medical products that pass visual inspection should be tested according to an established testing protocol in the proper setting, and by trained and authorized personnel using the appropriate tools, technology and equipment.

²² <https://www.who.int/teams/health-product-and-policy-standards/inn>.

The NRA should provide guidance to health professionals and the public on the use of national databases of authorized medical products to authenticate authorized products and thereby identify substandard and falsified medical products. Additional recommendations for reporting by health professionals and the public are provided in section 5.5 on reporting systems.

5.4.8 Screening (field testing) technologies

Screening technologies enable a more efficient and risk-based market surveillance process by increasing the amount of samples that can be rapidly assessed in field settings, thereby reducing the number of samples transported to and tested at quality control laboratories, thus facilitating increased coverage and a reduction in the overall cost of surveillance (73). These field technologies allow rapid testing of certain key quality attributes and are recommended when:

- the product has passed visual inspection, but additional information is needed to confirm the presence or absence of the API and its quantity, where applicable (66);
- the NRA has limited or no access to a laboratory with the required scope, resources or capacity (technical or financial) to perform the tests;
- the number of samples is too large to ensure prompt delivery of results if full analysis is performed at the laboratory.

The advantages of utilizing screening analytical techniques and the scope of the technologies used are significant (66, 68):

- facilitation of field analysis of several samples in a short period of time;
- facilitation of rapid detection of substantial quantities of substandard and falsified medical products, enabling a rapid response to prevent these products from reaching patients;
- operating with minimal use of financial and human resources;
- use of low-cost equipment that requires low maintenance;
- usage may be performed by adequately trained personnel;
- provision of reliable results;
- particularly beneficial for decentralized, non-urban or rural areas, due to the ability to conduct these techniques in the field.

Many of the available screening technologies rely on paper-based or spectroscopic methodologies, allowing mostly the identification of the API, and in some cases, semiquantitative assessment of its content.

The NRA may utilize a range of simple, rapid and cost-effective techniques to identify defects in critical quality attributes of products, such as identity, assay, impurities and disintegration for solid dosage forms. These different technologies used by screening devices may be implemented either in the field or by the national quality control laboratory (60). Prior to use, screening technologies should be validated for the intended purpose.

The selection of the technologies to be utilized depends on various factors, including:

- the anticipated advantages of the technology and its potential contributions to post-marketing surveillance activities;
- the expected benefits from the information and scope of each technology, considering its applicability, method complexity and performance characteristics;
- the availability of adequately trained local operators and, if needed, the associated training costs;
- operational costs, including periodic calibration, qualification and recurring expenses, as well as material costs for consumables, reference materials, libraries and maintenance;
- the purpose of product testing and specific goals to be attained, considering the survey objectives;
- for more advanced screening technologies or devices, the capacity of the national quality control laboratory, considering the availability of skilled personnel, testing requirements and equipment costs;
- the product specifications and validated analytical techniques submitted during market authorization.

The NRA should develop clear policies and procedures governing the use of screening technologies, including the conditions for the use of confirmatory testing, and the application of findings in regulatory decisions.

Appendix 1 of WHO TRS 1010, Annex 5, *WHO guidance on testing of “suspect” falsified medicines*, provides examples of analytical techniques that may be used for package identification, screening and testing of suspected falsified medical products (60).

GP3.19 Evaluate appropriate screening technologies

The NRA should consider the scope of the surveillance, the type of medical products it will be assessing and the areas where they will be sampled before deciding the use and type of screening technologies.

Many of the screening devices utilize paper chromatography and varied spectroscopy methods. The NRA should consider the various properties of

spectroscopic technologies and their advantages and disadvantages before use. Properties to be considered include ability to identify the active pharmaceutical ingredient and impurities, invasiveness of methods, utility in the field, training requirements and cost.

The NRA should evaluate screening technologies under consideration to determine the appropriate tools to support market surveillance. The *United States Pharmacopeia General Chapter <1850> Evaluation of screening technologies for assessing medicine quality* provides guidance that may assist NRAs in their evaluation (74). In addition, the NRA may utilize published technology reviews (75) or comparisons between screening technologies (73) to identify and select possible tools for assessing medicine quality.

5.4.9 Laboratory testing

Quality control testing at the laboratory may involve comprehensive analytical assays based on NRA requirements and approved methodologies or the methodologies outlined in official standards, such as pharmacopoeias accepted by the NRA. Typically, identity and assay testing are required and conducted, with impurity testing justifiable based on the information provided via the product dossier during market authorization. The dossier should be used as a comparative reference for information on the product specifications, analytical methods and standard certificate of analysis of the unexpired product. The selection of parameters to be tested should be based on the purpose of the test (for example, detection of impurities), the availability of the number of units required, the formulation or presentation and risk assessment (for example, previous incidents related to the product) (45).

GP3.20 Establish collaborative relationships with quality control laboratories

Where there is no quality control laboratory within the NRA, the NRA should collaborate with the respective national quality control laboratory or an alternative official medicines control laboratory. The national quality control laboratory or official medicines control laboratory should advise on the purpose of testing, determining sampling requirements, feasibility, laboratory capacity, standards, testing methods and any other technical aspects. Such information should be provided and fully documented by the NRA and the national quality control laboratory.

Collaboration between the national quality control laboratory and inspectors is essential for obtaining detailed information regarding the rationale behind sampling, including quantities required for testing and subsequent verification of results if that would be necessary. In case the numbers of product units required for certain tests would be unavailable, the national quality control laboratory should inform the inspector of the minimal amount of product forms

required. Such information should be fully documented and provided to the NRA and, when the national quality control laboratory or official medicines control laboratory requires information relevant to testing of the product, the NRA should have a procedure and channels to facilitate communication or information exchange.

NRAs are encouraged to review best practices for quality control laboratories and the minimum equipment requirements outlined in WHO TRS 1052, Annex 4, *WHO good practices for pharmaceutical quality control laboratories* (76). Additionally, WHO TRS 1010, Annex 5, *WHO guidance on testing of “suspect” falsified medicines*, includes the minimum requirements necessary for national quality control laboratories to conduct appropriate analytical procedures (60).

Requirements for product quality testing laboratories

GP3.21 Ensure quality testing laboratories are adequate

The institutions responsible for testing the quality of medical products should be clearly identified, including any duly authorized laboratory or network of laboratories operating in the country.

The NRA should ensure that the assigned laboratory follows good laboratory practices (76) that have been (or are working towards becoming) prequalified by WHO²³ or accredited based on relevant international standards (for example, ISO/IEC 17025) conferred by a recognized, national or international accreditation body, such as the International Laboratory Accreditation Cooperation.²⁴

The NRA should ensure that the following skills and capacities exist within the national quality control laboratories:

- sufficiently skilled laboratory personnel with critical thinking abilities, ongoing training, exposure to external institutions, and access to contemporary scientific expertise;
- maintenance in optimal condition of adequate technical equipment for conducting tests to assess product quality attributes and to confirm suspected falsified medical products;
- consideration of equipment costs and ancillary expenses such as consumables, reagents, standards, databases, libraries, installation, maintenance, and training, where relevant;
- implementation of standard operating procedures to uphold good practices and internal quality management systems, ensuring meticulous planning, execution, and documentation of laboratory activities for each testing request;

²³ <https://extranet.who.int/prequal/medicines/medicines-quality-control-laboratories-0>.

²⁴ <https://ilac.org/about-ilac/>.

- development of a suitable analytical testing programme to detect suspect substances, incorporating expected experimental outcomes, tested parameters, and available sampling units to devise a scientifically sound testing plan;
- adherence to general good practices for interpreting and reporting laboratory testing results.

The national quality control laboratory or official medicines control laboratory, or any other contracted quality testing laboratory, should provide written information on the available tests, the scope of testing (for example, medicines, vaccines), the minimum number of samples required, costs, and conditions for transporting the samples to the laboratory.

When the national quality control laboratory or official medicines control laboratory lacks the technical capacity to perform certain tests, the NRA should contract and assign an accredited laboratory with the required capacity to perform the tests.

To avoid false positive or false negative results and bias in analysis, specific points should be considered:

- a scientifically based justification for selecting the methods used in the testing plan, with reference to any limitations of these methods;
- separating sampling from testing;
- blinding staff performing each analysis on the testing plan to the results of other analyses as far as possible;
- the laboratory should ensure full traceability of samples and results;
- the laboratory should follow rigorous procedures to preserve the integrity of samples and documentation, ensuring a chain of custody that can withstand scrutiny in case of legal action;
- traceability of the measurements, with links to all physical material and to the original sample on which the test was done.

GP3.22 Ensure that testing is appropriate for the respective product

The NRA should require testing based on the registered specifications of the product.

The NRA should inform the national quality control laboratory or official medicines control laboratory about the relevant analytical procedures to test each product. These may be in house or the manufacturer's validated methods or the relevant compendial methods accepted during market registration.

All medical products should be tested, at a minimum for the identification of the API or diagnostic substance (where applicable) and content (assay).

The NRA should determine if additional specifications should be tested according to the objectives of the survey, the product form, or the active ingredient or substance. Consideration should also be given to the number of dosage units required to perform additional tests. These may include:

- disintegration and dissolution for oral dosage forms;
- fill volume, pH, and sterility tests for certain liquid dosage forms, including injections;
- purity and related substances that apply to many medicines and vaccines irrespective of the dosage form.

Note that the full extent of quality attributes that need to be assessed may involve additional tests depending on the medical products. Additional recommendations for testing protocols for quality surveys are provided in Appendix 5 of the present annex, and in WHO guidance on the conduct of quality surveys ([45](#)) and the testing of suspected falsified products ([60](#)).

5.4.10 Survey reports

GP3.23 Establish a database, procedures and tools for survey reports

The NRA should implement procedures for documenting and interpreting results from any of the testing approaches used, whether inconclusive findings, passing results, or non-compliance or out-of-specification findings. These procedures should include the steps to be followed to decide if the product is deemed substandard or falsified ([72](#)).

A detailed survey report should be prepared that includes all test results on the collected samples together with their interpretation. An example outline for the survey report content is provided in Appendix 3 of WHO TRS 996, Annex 7 ([45](#)).

The NRA should remain cognizant that satisfactory test results for a specific sample do not provide insight into the product's manufacturing process. Therefore, the NRA should implement policies for case handling, investigations and risk assessment following the detection of substandard and falsified medical products, considering risk factors and associated actions, activities and behaviours. Case handling and investigations will be addressed in section 5.6 on handling and investigation of complaints and reports.

Data obtained during collection and testing of samples should be summarized and appropriately organized in a database (using appropriate applications or software for epidemiological studies), linking each sample with the data gathered ([45](#)).

Suitable precautions should be taken to avoid errors, irrespective of the size of the dataset.

5.5 Reporting systems

Reporting suspected substandard and falsified medical products to NRAs is an essential step in increasing their detection in all settings, irrespective of system maturity. Product quality complaints from public and health professionals, reports from the industry, alerts from traceability systems, alerts from global or international organizations, and regional alerts from other NRAs related to substandard and falsified medical products enable the detection of local incidents that may require further investigation. For medical devices, user feedback is typically sent to the manufacturers, who then classify and determine reportability to the NRA (7). Regardless of the source, a functional reporting system, in tandem with knowledgeable stakeholders, is needed. This need is even greater where routine market surveillance of all medical products is not feasible or is partially implemented (77).

The objective of the system is to enable reporting from various stakeholders to the NRA, to increase the opportunities for early detection of suspected substandard and falsified medical products. All NRAs are responsible for establishing such a system for market surveillance and control, irrespective of maturity level (9, 77).

To complement other methods of detection, in 2013, WHO launched the WHO Global Surveillance and Monitoring System (GSMS)²⁵ for substandard and falsified medical products to work with Member States to improve the quality, quantity and analysis of data on substandard and falsified medical products, and to use the information to enhance prevention, detection and response to these products in order to protect public health. In general, the GSMS was established to:

- improve reporting of substandard and falsified medical products;
- assess more accurately the scope, scale and harm caused by substandard and falsified medical products;
- provide immediate coordination, resources and technical support in emergencies;
- issue medical product rapid alerts;
- gather and analyse a detailed body of validated evidence for Member States to enable evidence-based policy-making and investment;
- strengthen regional and national regulatory capacities to prevent, detect and respond to substandard and falsified medical products.

Additional details are provided in the WHO GSMS terms of reference (13).²⁶

²⁵ <https://www.who.int/who-global-surveillance-and-monitoring-system>.

²⁶ https://cdn.who.int/media/docs/default-source/substandard-and-falsified/gsms_terms-of-reference160517.pdf?sfvrsn=e2ff0ccf_3&ua=1.

The GSMS is designed for reporting by trained focal points in national (or regional) regulatory authorities, who submit reports of either validated or suspected medical products. In addition, WHO provides training for NRA focal points to understand the portal and use it to submit reports and search for similar cases reported by other NRAs.

However, the ability of NRAs to identify and report products requires effective in-country systems for health care professionals and other stakeholders to report suspected substandard and falsified products, and incidents arising from the use of those products, including poor efficacy. These systems may be dedicated to reporting of substandard and falsified medical products or product quality complaints, or they may be integrated into national vigilance systems ([78](#)).

GP3.24 Establish a reporting system or integrate reporting into existing vigilance systems

The NRA should ensure that there is a functional system to facilitate reporting of suspected substandard, falsified or unregistered medical products that is accessible to all stakeholders, with relevant guidelines for reporting.

The NRA should develop and implement policies and procedures to ensure that all complaints and other information concerning potentially defective products are carefully recorded and reviewed.

Manufacturers of medical devices (or their economic operators or representatives) should be required to submit an investigation report to the NRA under the following circumstances ([5](#), [6](#));

- detection of a serious public health threat;
- when use of a device has led to death or serious deterioration in the health of a user, patient, client or other person; or when there is failure, malfunction, improper or inadequate design, manufacturer, labelling or user error of the medical device that could have led to death or serious deterioration in the health of a user, patient or client, or other person.

All feedback from the use of medical devices received by manufacturers should be reported to the NRA as part of a periodic summary of post-market surveillance reports.

Manufacturers of medical devices, either directly or through its authorized representative, should be required to report to the NRA in a timely manner any field safety corrective action it is undertaking in the country.

Additional recommendations on reporting of user feedback to NRAs, including time to report, are provided in the WHO *Guidance for post-market surveillance and market surveillance of medical devices, including in vitro diagnostics* ([7](#)), and by the International Medical Device Regulators Forum ([79](#)).

5.5.1 Reporting channels and tools

There may be variations in the organization and channels through which suspected products may be reported, according to the local context, including the type of products. For example, in some countries, in addition to health care provider and patient or consumer reporting, regulators also receive reports from wholesalers and distributors who have been offered medical products that they consider to be suspicious (for example, unusual quantities, regularity or prices). Complaints about defective or ineffective products may indicate non-compliance with good practices, including good manufacturing practice, good storage and distribution practice, and quality assurance processes by responsible stakeholders. Therefore, ensuring timely detection of suspected products is a shared responsibility of all stakeholders, including manufacturers, wholesalers and distributors, and the vigilance programme.

As far as possible, the NRA should utilize existing national vigilance databases to facilitate reporting of substandard and falsified medical products by various stakeholders.

NRAs should implement procedures for information sharing about suspected quality issues for medical products between the divisions for vigilance and market surveillance and control, if separate, particularly cases or clusters of adverse drug reactions or adverse events following immunization that are suspected to be caused by quality issues.

Reporting forms or digital reporting interfaces (for example, mobile applications) for vigilance systems should encourage reporters to indicate if a suspected adverse drug reaction or adverse event following immunization may be related to a product quality issue where the product may be substandard or falsified.

The NRA should select various media and methods to sensitize the public about reporting channels and tools, considering the local context, for example:

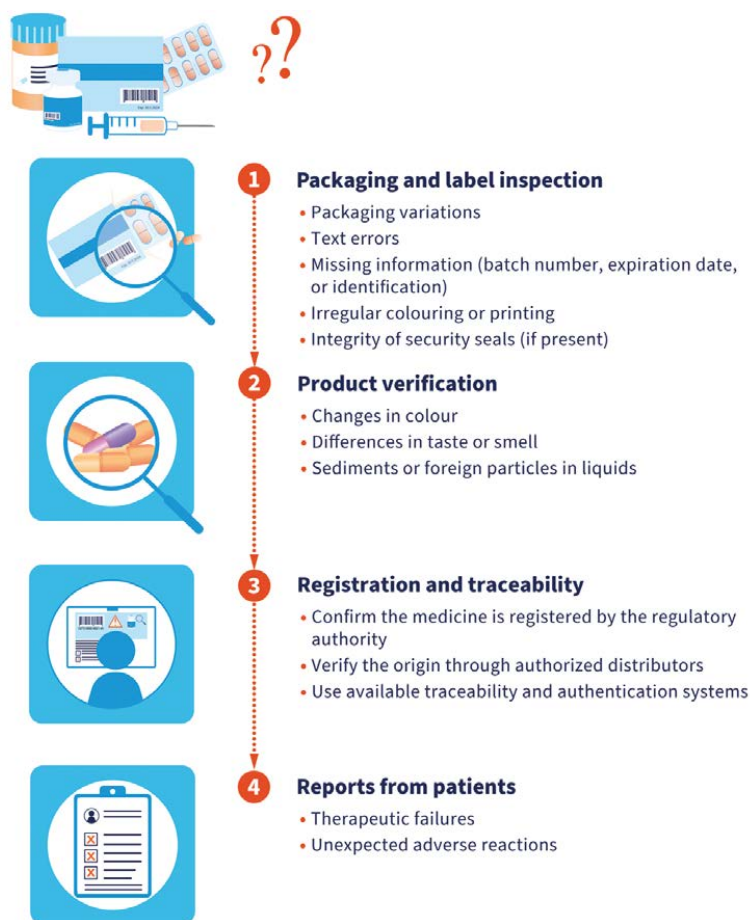
- posters or information leaflets;
- provision of information through traditional media, such as radio or television programmes, or social media;
- provision of information via health centres, hospitals, general practitioners, pharmacists and others in the health system;
- communication on substandard and falsified medical products and market control activities through websites, newsletters, social media and internet advertisement;
- engagement meetings with patients, consumers, community-based organizations, health care professional organizations and other health institutions (for example, hospitals);
- education about substandard and falsified medical products (for example, schools, communities, curriculum of health care professionals).

Although the content communicated may vary depending on the media and methods, it should include key messages that are regularly repeated in all communications, such as:

- how to identify suspected substandard and falsified products (Fig. A9.10);
- the importance of promptly reporting suspicious products and reporting channels.

Fig. A9.10

Infographic: how to identify substandard and falsified medical products



Source: Pan American Health Organization.²⁷

²⁷ <https://www.paho.org/en/documents/infographic-how-identify-substandard-and-falsified-medical-products>.

Awareness of reporting channels

In addition to the prevention of substandard and falsified medical products, approaches such as public education and awareness campaigns are key to the detection of substandard and falsified medical products through reporting by stakeholders ([78](#)). To achieve this, stakeholders should be aware of the various reporting channels.

GP3.25 Develop and launch campaigns to create awareness of substandard and falsified medical products

In addition to creating awareness of the dangers of substandard and falsified medical products, public awareness campaigns and guidance for potential reporters should include information on the identification and reporting of suspected substandard and falsified medical products, including reporting channels ([78](#), [80](#)).

Key target reporters should include ([78](#), [80](#)):

- patients, caregivers, and consumer or patient organizations;
- health care professionals (public and private sectors) and relevant associations;
- personnel of manufacturing companies, import agencies, and distributors of medical products;
- officers of national or regional vigilance centres;
- law enforcement officers, for example, customs officials, judicial authorities, police and border authorities.

Additional details on addressing lack of awareness are provided in section 4.1 on “Education and awareness”.

Addressing barriers to reporting

Barriers to reporting are multiple and complex, and may include ([81](#)):

- lack of time;
- non-supportive or weak legal framework, governance or policy;
- language or literacy limitations or differences;
- complexity of the reporting form or the reporting system;
- lack of knowledge, awareness or training, for example, uncertainties on what to report, or uncertainties on who should report;
- lack of trust;
- lack of motivation;

- perception of product or quality;
- guilt or shame;
- fear of consequences;
- lack of positive feedback to reporters.

[Table A9.11](#) lists barriers to reporting of substandard and falsified medical products to NRAs by external stakeholders, and barriers to reporting by NRAs to WHO GSMS.

Table A9.11

Barriers to reporting suspected substandard and falsified products

Barriers to reporting to NRAs identified by health care professionals and others (42–44)	Top five barriers to reporting to WHO GSMS identified by WHO Member States (82)
1. Lack of awareness about the existence, extent and risks of substandard and falsified medical products 2. Inability to distinguish substandard and falsified products (especially where errors and polypharmacy can make it difficult to pinpoint adverse events or unexpected inefficacy) 3. Lack of user-friendly reporting systems, particularly technical or logistical challenges with existing reporting systems 4. Fear of repercussions (reputational risks and personal safety), linked to whistleblowing 5. Lack of trust in the system	1. Unavailability of investigation mechanisms for suspected substandard and falsified medical products 2. Staff workload 3. Untrained national focal points 4. Inadequate or insufficient coordination among regulatory pillars 5. Immaturity of the system in terms of the scope and diversity of reportable incidents

Addressing ability, utility and fear

The following provides recommendations relating to barriers 2–4 identified by non-NRA reporters in [Table A9.11](#).

GP3.26 Implement approaches to encourage and facilitate reporting

5.5.2 General principles for reporting systems

To encourage reporting, the reporting process should be simple, effective, easy to use and accessible to most of the people. Specifically, reporting systems should encourage reports through:

- clear guidelines for reporting by health professionals, the public and other stakeholders;
- easily accessible reporting tools for health professionals, community health workers and the public;
- mechanisms to maintain confidentiality and protect whistleblowers;
- reporting through multiple modalities or media;
- inclusion of languages for locals and migrant populations.

The NRA should develop and provide clear and easily accessible guidelines for reporting, highlighting what to report, when to report, who should report, and how reports may be submitted.

- Reporting should be via a single entry point for suspected medical products without any restrictions on the type of products or issues to be reported.
- Information on what happens to the report after submission should be included.

The NRA should develop and provide guidance to health professionals and authorized dispensers (pharmacists and operators of licensed over-the-counter medicine outlets) on visual and physical inspection of medical products, specifically:

- packaging: to check for anomalies in packaging and indicators of authenticity (for example, holograms), for comparison with authentic products, where possible;
- labelling: to check accuracy of product information, such as the correct name, dosage form, batch number, manufacturing and expiry dates, and manufacturer's details;
- tampering signs, such as broken seals or glue residue;
- appearance or physical characteristics of the product, for example, colour, size, shape or markings;
- consistency, for example, crumbling tablets or cloudy solutions;
- smell and taste: note any unusual smells or tastes that are not characteristic of the medicine;
- supporting evidence: gather any supporting documents or evidence, such as photographs of the product, packaging and purchase receipts.

In addition, health professionals and licensed dispensers should be educated to monitor the medical product's performance, safety and efficacy, where applicable, particularly:

- observation of the efficacy and performance of the medical product (for example, therapeutic efficacy, diagnostic performance);
- identification of unexpected adverse reactions or adverse events;
- noting multiple reports or increased reports of expected adverse reactions (clusters) from the same source (for example, batch, supplier, dispensary).

To support the verification of the product, the NRA should provide guidance to health care professionals and licensed dispensers on:

- source verification, including:
 - verification of registration or licensure with the NRA, and other applicable certifications from recognized bodies;
 - availability of appropriate documentation;
 - verification of product authenticity via use of databases to check product identifiers or authentication features with the respective manufacturers;
- verification of regulatory status, including:
 - market authorization status and conditions;
 - notices, alerts and warnings issued by regulatory authorities.

Enhanced reporting tools and systems

The NRA should ensure that effective, user-friendly reporting tools are available and accessible to all health professionals, licensed medicine dispensers, the public and other stakeholders.

Appropriate technologies. The NRA should facilitate different means of reporting, such as a website, web applications (for example, online forms), email, in person, postal services, dedicated telephone hotlines, or social networks, customized to the target audience for maximizing the outreach. The selection of the reporting tools should consider the local context (for example, population density, population or community literacy, accessible communication channels), including available infrastructure.

User-friendly systems. The NRA should ensure that reporting systems are simple and easy to use (for example, avoiding complex forms and technical language). Reporting forms should enable the NRA to contact the reporter if further information is required, under strict and guaranteed confidentiality conditions.

Information requirements. The information collected should include, at a minimum, an identification and description of the problem, product name (international nonproprietary name and brand name, if any), batch, manufacturing date, expiry date, manufacturer, location or source where the product was acquired (for example, physical address, web address), and description of any adverse events or other observations linked to the use or application of the suspected substandard and falsified medical product, if any.

The form should facilitate submission of photographs of the medical product, including the product unit (for example, tablet), the product packaging and the label from different angles.

Whenever feasible, the use of standardized field structures, drop-down menus or other devices is recommended to improve data quality.

Time and capacity constraints. Ensure that reporting can be done quickly and efficiently (for example, avoiding overlong forms or delays in responding to phone calls).

Minimizing financial barriers. Ensure that reporting systems are free of charge for users (for example, toll-free hotlines, prepaid postal services).

Availability of reporting modalities. The NRA should conduct periodic checks to ensure that forms are available, functional (for example, on websites or web or mobile applications), and accurate in information (for example, contact email addresses, phone numbers).

Coverage. The NRA should ensure that the forms or reporting tools are accessible by health professionals, marketing authorization holders, dispensers and other users of medical products. This includes users operating in the private sector (for example, private clinics, retail pharmacies and licensed over-the-counter dispensers) and the voluntary sector (including nongovernmental organizations and religious organizations that provide health care).

The NRA should ensure that appropriate user and technical support is in place, including:

- up-to-date systems and prompt responses to technical issues;
- regular training of and support to all users (in public, private and voluntary sectors);
- maintenance of human resources and infrastructure to support reporting, including reliable electricity supply and internet and phone connectivity.

The NRA should provide clear and user-friendly instructions to health professionals, licensed dispensers and any other reporters on maintaining records

for submitted reports, and documentation of any follow-up on developments or updates from the authorities regarding the reported issue.

NRAs should provide a timely response and appropriate follow-up to reporters (44). There should be a system of positive and confidential feedback or acknowledgement of reports to encourage further reporting, for example, thanking the reporter and acknowledging receipt of each report.

The NRA should periodically provide feedback to the public and stakeholders on reports and decisions on substandard, falsified or both medical products via appropriate channels, for example, newsletter, social networks or fact sheets.

NRAs should share case reports on suspected substandard, falsified or both medical products promptly (or as soon as possible) with the World Health Organization via the WHO GSMS portal.²⁸

NRAs should also share information promptly, as appropriate, with other authorities and agencies, including:

- national law enforcement agencies (for example, the police, standards authorities, customs and excise);
- relevant international organizations;
- manufacturers and marketing authorization holders.

NRAs should develop and maintain systems for tracking reported cases, analysis of data and monitoring trends.

The NRA should create and support a transparent reporting culture via (42, 44, 78, 83):

- education and training of health professionals and other stakeholders on identifying and reporting substandard and falsified medical products;
- increasing public awareness about the importance of reporting suspected substandard and falsified medical products, including among community or patient advocacy groups.

The NRA should facilitate confidential reporting by:

- ensuring the confidentiality of the reporter and any sensitive information provided;
- creating a culture of zero tolerance towards corruption or intimidation.

²⁸ <https://www.who.int/who-global-surveillance-and-monitoring-system>.

5.5.3 Systematic recommendations

The reporting system should be further supported by (78, 82, 83):

- an adequate quality management system that ensures appropriate workflow, validation, document management, report transmission, digital intelligence, updating of reports, database management and management of alerts;
- a dedicated database;
- enough dedicated, trained, qualified staff;
- a communication strategy and system to provide information, updates and feedback to relevant stakeholders, including the media.

There should be adequate and formalized coordination with relevant stakeholders and departments, for example, inspection and vigilance. Ideally, complaint reporting should be complemented by media monitoring for any information related to potential substandard and falsified products.

Additional details and guidelines on reporting systems may be found in published standards of the ICH, WHO guidance documents on pharmacovigilance, and Pan American Health Organization documents, as listed below.

GP3.27 Monitor and periodically evaluate the performance of the reporting system for substandard and falsified medical products

The NRA should establish key performance indicators to monitor and evaluate the performance of the market surveillance and control reporting system, or the vigilance system (where applicable), in accordance with the national context. Key performance indicators for the reporting system may include such elements as the types of reporters utilizing the system, the number of reports received from various stakeholders, timeliness of reports reviewed against target timelines, seasonal trends in reporting, number or percentage of reports for which investigations were conducted, and the number or type of actions taken in response to investigations. Key performance indicators should be part of action plans and reports, as reporting on them will be critical for transparency and accountability.

Key performance indicators should be periodically evaluated to determine whether changes or improvements may be needed to the reporting system or procedures.

Guidance for reporting systems

Additional details on reporting systems are provided by the following WHO guidance documents:

- WHO: *Safety monitoring of medicinal products: reporting system for the public*;²⁹
- Pan American Health Organization: *Good pharmacovigilance practices for the Americas*;³⁰
- WHO *Global Surveillance and Monitoring System for substandard and falsified medical products*;³¹
- WHO fact sheets on substandard and falsified medical products.³²

Additional resources from regional regulatory networks include:

- Africa Medical Devices Forum: *Standard operating procedure for handling reports of substandard/falsified medical devices including in vitro diagnostics*;
- European Medicines Agency: *Good pharmacovigilance practices (GVP)*;³³
- South-East Asia Regulatory Network: *Action Point 10: Strategy to stimulate reporting*.³⁴

5.6 Handling and investigation of complaints and reports

GP3.28 Develop procedures for handling of complaints and incident reports

The NRA should develop procedures for handling, review and investigation of reports of suspected substandard and falsified medical products from consumers, health workers, marketing authorization holders, distributors, border control officials and other participants in the supply chain.

The NRA should facilitate the receipt and review of complaints or market reports that are associated with confirmed public health risks involving substandard and falsified medical products. This approach should also apply to alerts or reports by other NRAs or international reporting systems, and peer-reviewed publication of research for local monitoring and lessons that may improve the local reporting system and associated procedures.

The severity and seriousness of the reported harms or potential health hazards should be promptly assessed by the NRA at the time of receipt, followed by the investigation and root cause analysis. In the case of a substandard medical

²⁹ <https://www.who.int/publications/i/item/9789241503198>.

³⁰ <https://www.paho.org/sites/default/files/Series-Red-PARF---5-Eng.pdf>.

³¹ <https://www.who.int/publications/i/item/9789241513425>.

³² <https://www.who.int/news-room/fact-sheets/detail/substandard-and-falsified-medical-products>.

³³ <https://www.ema.europa.eu/en/human-regulatory-overview/post-authorisation/pharmacovigilance-post-authorisation/good-pharmacovigilance-practices-gvp>.

³⁴ <https://searn-network.org/action-point-10-strategy-to-stimulate-reporting>.

device, the manufacturer is responsible for undertaking the investigation, conducting the root cause analysis and submitting the report to the NRA (7).

When the NRA has limited capacity to conduct investigations, resources should be prioritized to investigate those complaints or incidents that indicate harm has occurred or those that have the greatest potential to cause harm.

The NRA should clearly identify the procedures and tools for the assessment of complaints received.

Investigation of complaints and incidents is an iterative process, and the conclusions and next steps should be regularly reviewed based on emerging information. The review procedure should enable the NRA to conduct periodic reviews of cases, update risk assessments, and develop relevant conclusions and regulatory responses based on new information or findings.

When investigating complaints and incident reports, the NRA should assess the quality of the information received and the relevance of the report, code information in a consistent manner using internationally recognized terminology and conduct relevant assessments. The process should include identification and removal or consolidation of duplicates (78, 80).

For complaints and reports that may involve a substandard or falsified product, the NRA should implement procedures and processes to report information on suspected or validated substandard and falsified medical products to the GSMS, regional networks and other concerned NRAs, according to established agreements for information sharing and exchange (13).

Investigations of suspected substandard and falsified medical products should include efforts to evaluate the validity of the information on the substandard and falsified medical products by follow-up activities, including (84):

- checking for similar issues or similar products in other information submitted to the NRA as well as shared data sources, such as rapid alerts and GSMS, and research;
- contacting the legal manufacturer to assess whether the product is genuine;
- reviewing the report for indicative information, such as packaging or labelling issues (spelling, grammar mistakes, missing or inconsistent expiry dates) or quality concerns (unusual or unexpected look, feel, smell).

According to the level of risk and the nature of the problem, the NRA should determine if further investigations, analyses or inspections are needed. NRAs are encouraged to utilize the WHO aide-memoire, which provides guiding steps to be taken by NRAs when handling incidents of substandard and falsified medical products (see Appendix 9) (84).

In the context of medical devices, manufacturers or marketing authorization holders should be required to undertake risk assessments of incidents involving

suspicious devices, determine if corrective actions are needed, implement them, and report any field safety corrective action implemented to the NRA (7).

GP3.29 Document and facilitate sharing of investigations of complaints and incident reports, as appropriate

The NRA should maintain all documentation of complaints, incident reports, investigations performed and supporting material. Where feasible, the information or critical elements thereof should be maintained in a searchable electronic repository.

Based on the severity of the incident and the quality of the information received, the complaint investigation may need to be further acted upon or shared with other stakeholders, where appropriate. The NRA should have procedures or other measures in place to ensure the effective communication of complaint investigation information to other stakeholders.

The finished product manufacturer and other relevant supply chain representatives (for example, distributor) should be promptly informed of the discovery of the suspected substandard or falsified product and involved to provide additional information, including confirmation of authenticity and product tracing.

The NRA's procedures or measures should ensure involvement and communication among stakeholders, including other organizations within the NRA, other governmental agencies involved in implementing follow-up action related to an investigation (including quality testing and tracing) and regulatory actions (for example, recall), as well as other organizations or partners for which there are agreements to share information or assistance in communicating with the public (such as regional partners or nongovernmental organizations).

5.6.1 Information sharing with external partners

International and regional networks or organizations involved in activities to detect substandard and falsified products are key partners in protecting public health. In addition, these may support NRAs and their respective national security agencies in the detection of and response to falsified medical products. This includes notification of substandard and falsified medical products that are reported to the WHO GSMS and then confirmed by the WHO technical unit responsible for substandard and falsified medical products. If needed, WHO will issue a WHO Medical Product Alert that would be publicized (85). Timely information sharing is thus a critical hallmark of a mature regulatory system and essential for ensuring coordinated global efforts for access to medical products of reliable quality.

In addition, NRAs should inform their regional networks, where applicable, and other NRAs that may be impacted by substandard and falsified

medical products detected and confirmed in local markets (49, 77). This may be facilitated by the establishment of agreements and maintenance of guidelines involving NRAs and national single points of contact (59). The South-East Asia Regulatory Network's Strategy to Facilitate Information Sharing provides a regional example, which may apply to information sharing for regulatory reliance and market surveillance and control activities.³⁵

5.6.2 Incident risk assessment

Risk assessment is a systematic process of organizing information to support a risk decision to be made within a risk management process. It involves the evaluation of the hazards involved and the potential severity of each risk associated with exposure (48).

GP3.30 Assess risks using a standardized approach and tools

In the context of exposure to substandard and falsified medical products, the NRA should apply this process to assess reports of quality defects, non-compliance with practices that may lead to substandard products, local or regional alerts, incidents of confirmed substandard and falsified medical products, and the respective facilities, where applicable, to inform appropriate regulatory actions.

The NRA should collaborate with the manufacturer to gain insight into the risk assessment that was done for the respective product. However, the NRA may conduct its own risk assessment depending on the nature of the feedback from users, and the risk to public health (7).

Reports of investigations received from manufacturers should be reviewed for scientific rigour, evidence of documented procedures, timeliness and rationale. In addition, the NRA should seek clarification as needed, and determine acceptability of reports, suggested corrective or preventive actions, suggested actions for users, and risk management updates (7).

The NRA should establish a team of suitably qualified or trained personnel capable of performing initial risk-based assessments of quality defects promptly, in accordance with the risk posed by the quality issue.

The NRA should establish a procedure and develop or adapt a tool to perform risk assessment in order to achieve standardized and reliable results during the evaluation of alerts, notifications and reports related to substandard and falsified medical products, and for unregistered or unlicensed medical products (86).

Where there is a need to develop a risk assessment tool, the NRA may customize existing risk analysis and management tools or develop its own

³⁵ <https://searn-network.org/action-point-3-strategy-to-facilitate-information-sharing>.

procedure or tool. Irrespective of the approach used, the NRA should consider the following in developing or revising its risk assessment tools (86):

- support required for analysis of risks considering the reporting quality, types of risks (for example, severe, minor), and expected case load (quantity);
- the availability of the necessary data or information from valid and reliable sources.

The tool should provide the necessary information with sufficient detail to support analysis and provide recommendations to safeguard public health.

In developing an effective risk assessment tool, the NRA should ensure the following steps are included (86):

- risk factor identification: to identify the applicable risk factors;
- risk importance: to establish the importance level or weight of each risk factor, for example, high, medium or low importance;
- risk categorization: taking into consideration criteria related to severity of the case and possible impact on the patient's or population's health and safety, for example, high, medium or low risk;
- risk prioritization: predetermined groups assigned based on the risk importance and the risk categorization.

WHO has developed a tool for prioritization of risks (87) in the context of substandard and falsified medical products, and questions to consider when assessing risk (88) for use by NRAs (see Appendix 9).

Risk factors

GP3.31 Identify relevant risk factors to be considered

To avoid inconsistencies in data interpretation and risk assessment, the NRA should clearly identify the main elements involved in the risk assessment ("risk factors") relevant to substandard and falsified medical products that may impact health outcomes. At the subsequent stage, the NRA should assign appropriate weights to each factor, ensuring that the final categorization accurately reflects the likelihood of harm arising from the incident.

The NRA should consider various factors when assessing a risk for categorization. These should include:

- the nature and severity of the defect or non-compliance: for instance, a defective package may affect the quality of a product, or may indicate an absence of product registration or facility licensure;

- the potential clinical consequences of the defect: the NRA must evaluate the intended use along with severity of the defect to estimate the health and safety implications for patients, including acute and long-term risk;
- the intrinsic risks associated with the product, its potency or therapeutic index, patient population, route of administration, and site of action.

The NRA may consider additional factors that may not have direct impact on risk categorization but may impact the action plan. These include:

- the probable site of deviation;
- the exposed population;
- the reported frequency of occurrence;
- the potential or confirmed impact on the patient;
- market turnover and expiration of the product;
- extent of use (for example, in national or international public health programmes, mass vaccination campaigns);
- detectability of non-compliance;
- the relevance of the investigated case to the community (for example, falsified childhood vaccine);
- the history of the manufacturer and the product involved;
- the frequency of alerts, notifications and reports about the specific type of problem;
- the relationships with microbiological risk or toxicity of the product;
- the existence of analytical reports or other scientific evidence of non-compliance.

For examples of quality defects in products, risk factors, and details about the risk assessment matrix, readers may refer to the report of the WHO Member State Mechanism *Recommendations for health authorities on criteria for risk assessment and prioritization of cases of unregistered/unlicensed, substandard, and falsified medical products* (86). [Table A9.12](#) identifies examples of risk factors and considerations that NRAs may use in developing a risk assessment tool.

Table A9.12
Examples of risk factors and implications

Factor	Examples of implications
Main factors to be taken into consideration	
Potential clinical consequences	Therapeutic failure, adverse reactions including hospitalization or impairment, withdrawal effects, reduced quality of life, antimicrobial resistance, or death
Nature and severity of defect	- Minor defect: for example, poor adhesion of labels, poorly cut blister packs, incomplete packs - Major defect: for example, microbial contamination of non-sterile products, degradation prior to expiry date - Critical defect: for example, microbial contamination of a sterile injectable or ophthalmic product, or the presence of impurities with serious medical consequences
Therapeutic indication	The intended use, which may be: - critical or lifesaving or emergency treatment product - long-term management of a chronic disease - used for acute treatment - used for prevention of outbreaks or widespread disease
Route of administration and site of action	Substandard parenteral products with systemic effects pose higher risk than non-parenteral products with local effects Defective devices for parenteral products pose higher risk of therapeutic failure and contamination than transdermal or topical delivery systems
Patients' vulnerability	Use by vulnerable patient groups where likelihood of harm is higher, for example, neonates, paediatric patients, older adults, pregnant women, and immunocompromised patients
Main factors to be taken into consideration	
- The level of supply chain where the suspected product was detected and its distribution to different geographical areas. - Market turnover and expiration date: this information might help NRAs to decide what action should be taken, especially for product recall, based on the probability of having the non-compliant medical products still on the market and the possibility of collecting samples for official analysis. - The availability of the product and alternatives: this information must be considered before the decision to withdraw a medically critical product from the market or ban a company, considering the possibility that a shortage or lack of a specific product might cause more damage than the presence of a marginally substandard product.	

Source: WHO Member State Mechanism (86).

The identification and review of risk factors should be used to inform an importance ranking or categorization, whereby risks of greater potential to result in damage to public health are prioritized ([86](#)).

Risk importance and categorization

GP3.32 Categorize risk factors and establish importance

The NRA should assign weights to the risk factors identified based on the factor's importance to public health, specifically access to and use of quality, efficacious and safe medical products.

The NRA may apply or adapt the risk categories developed by WHO to classify risks in assessing the associated scope, scale and harm presented by substandard and falsified medical products (Appendix 9) ([87](#)):

- **Critical risks:** those that may compromise public health or the integrity of the supply chain:
 - public health
 - supply chain (access)
 - geographical
 - environmental.
- **Associated risks:** those that do not directly impact public health but may impact systemic factors, including efficient care delivery and patient access to quality, safe and efficacious medical products:
 - public interest
 - political
 - economic
 - legal.
- **Historical risks:** reports from other NRAs, regional or global databases, or international security agencies may indicate a history of a problematic medical product or facility:
 - product history
 - supply chain elements.

Appendix 9 provides risk-level criteria that may be adapted to the local context. This tool will be finalized for the upcoming publication of the WHO GSMS toolkit, but it may be adapted by NRAs for initial use until publication. It should be used in conjunction with the WHO “Questions to consider when assessing risk” ([88](#)).

The following should be considered high-risk cases (86):

- Falsified products, whether confirmed or suspected, where immediate investigation is needed to mitigate the risk.
- Unregistered or unlicensed products: given that their origin is unknown, the products have not been evaluated by the NRA, and agents are acting without authorization from the NRA or regional regulatory authority. Incident evaluation should be conducted in such cases, as information from other NRAs may confirm appropriate registration elsewhere, which could minimize their potential risk.

For substandard products, the nature and severity of the defect should be evaluated to categorize the risk appropriately.

The NRA should collaborate with internal stakeholders (for example, regulators responsible for vigilance, market authorization), health care professionals, law enforcement officers, and other stakeholders in the supply chain to acquire information needed to assess risk.

Risk assessment matrix

GP3.33 Combine risk categories and weights in a risk assessment matrix

The NRA should combine risk categorization based on risk factors with the importance or weight assigned to each risk factor, as some factors are more relevant than others.

The NRA should establish qualitative descriptions and quantitative values to be assigned to importance levels and risk categories, considering national and regional circumstances.

In developing an effective risk assessment tool, the NRA may consider the following steps:

1. Identify the risk factors to be used for the relevant quality defects or non-compliance.
2. Establish importance level or weight of each risk factor, which may be a priority rating, for example: 3 – high priority, 2 – medium priority, or 1 – low priority.
3. Define the different categories of each risk factor, considering the severity of the case and potential impact on patients' health and safety. Three risk categories may be applied: 1 – low risk, 2 – medium risk, or 3 – high risk.
4. Multiply the importance level of each risk factor by its risk category and sum the subtotals for each risk factor. The result should fall into one of the predetermined priority groups, serving as an index for

prioritization against other assessed cases. The higher the index, the greater the severity and importance of the case for public health.

The NRA should adapt or periodically revise risk matrices according to local context, for example, considering additional levels for importance, likelihood or risk categorization.

Once the risk matrix is developed or revised, the NRA should implement procedures to validate the tool within a specified period or via a number of notifications to confirm that the results are relevant before full implementation (86).

Examples

Examples of a probability versus impact matrix (Table 5) and a risk assessment matrix (Table 6) are provided in Appendix 4. The latter highlights how impact and probability may be classified into different levels, with increasing weights as either or both increase (48). WHO TRS 981, Annex 2, includes additional tables depicting examples of consequences based on probability and impact, and examples of common risk management tools (48). Additional information on risk management principles, tools and trainings may be found in the ICH guideline for quality risk management (Q9) (62).

GP3.34 Establish risk mitigation procedures and appropriate regulatory actions

The NRA should establish risk mitigation procedures and regulatory actions according to the findings of the risk assessment. The NRA should consider the following possibilities according to actual or potential risk (7):

- no action, if the case is not confirmed;
- additional product or supply chain surveillance, according to the case;
- issuance of safety alert to users;
- issuance of requirements for the manufacturer to make changes to the product, manufacturing process or product information provided;
- enforcement of corrective or preventive regulatory actions (for example, recalls, safe disposal, market withdrawal);
- submission of data to a database for ongoing monitoring and evaluation of cases and trends.

The NRA should delineate the scenarios necessitating further investigations, including laboratory testing based on incident risk assessment. It is imperative that urgent cases are immediately classified as such, with an accelerated schedule for actions (for preventing avoidable morbidity or mortality).

Structured guidance for incident management and risk assessment should be developed to align with the decision-making process regarding the need for immediate laboratory testing or other investigations to confirm the presence of substandard and falsified medical products. This guidance should be comprehensive, featuring a systematic and transparent arrangement of steps for evidence-based decision-making (84).

6. Response

In order to protect the public from substandard and falsified medical products, NRAs will need to develop and implement regulatory responses to mitigate the risks, including law enforcement where appropriate. After the detection, confirmation and risk assessment of substandard and falsified medical products, the NRA should implement appropriate regulatory action according to the situation, which may include:

- risk communication: issuance of notifications, alerts or warnings;
- requiring the implementation of appropriate corrective and preventive actions or field safety corrective actions by manufacturers, including revision of product information or labelling;
- suspension or withdrawal of registration or licensure of local manufacturers, authorized representatives, importers or distributors;
- recall of product batches;
- withdrawal of the product from the list of marketed medical products;
- quarantine and safe disposal of substandard, falsified or both medical products;
- criminal prosecution of persons and organizations involved in and responsible for the manufacture, storage, distribution and sales of substandard and falsified medicines by unlicensed entities.

It is critical that there are relevant regulatory frameworks and adequate dedicated resources to support the regulatory and judicial response to substandard, falsified or both medical products. These responses should include collaboration with law enforcement, key stakeholders (such as manufacturers, marketing authorization holders, distributors, health professionals, academics, civil society), other NRAs, and regional and international networks (6).

6.1 Alerts and recalls

Sharing of alert and recall information with the public and other stakeholders is essential to the response to substandard and falsified products, particularly where there may be widespread distribution or in circumstances where the distributors are unable to identify persons who received the substandard and falsified products. In addition, alerts and recalls issued by or impacting neighbouring countries, or alerts and recalls by reference NRAs where the same product is involved, may inform priorities for the detection of the respective substandard or falsified product in the market and subsequent action by NRAs. Similarly, information from global systems (for example, WHO GSMS, INTERPOL) enables the identification of other affected countries or regions where substandard and falsified products have been detected and recalled.

6.1.1 Risk communication

Risk communication is a vital component of an effective regulatory response to the detection of substandard and falsified medical products in local or international supply chains. Its primary role is to ensure timely, accurate and transparent dissemination of information to all stakeholders, including health care providers, patients, supply chain actors, and international partners, thereby minimizing harm and restoring trust (89). Appropriate risk communication enables the target population to make informed decisions about the recommended measures for avoidance of harm and risk mitigation.

When substandard and falsified medical products are identified, swift communication enables prompt containment actions, such as recalls, alerts and public advisories. In addition, timely, effective risk communication empowers health care professionals to recognize and manage risks and helps patients avoid treatment failure and preventable harm. It also supports coordination among NRAs, international organizations and manufacturers, fostering a collaborative response. Moreover, clear messaging helps prevent misinformation, reduces concerns, suspicions and panic, and encourages compliance with regulatory measures. In the long term, robust risk communication strengthens public confidence in the health system, enhances vigilance across the supply chain, and promotes a culture of safety and accountability.

Levels of communication

The communication risks related to substandard and falsified medical products detected during market surveillance and control activities may be categorized into three levels:

- internal communication within the NRA for promoting regulatory effectiveness, and cooperation among all relevant departments;

- communication among relevant stakeholders (for example, the public, distributors, law enforcement, health professionals, industry) for engagement, increased awareness, mutual understanding, risk mitigation, cooperation and the development of trust and confidence;
- communication with other countries, regional networks, and international organizations for information sharing and coordinated global efforts.

GP4.1 Develop a risk communication strategy

The NRA should develop and implement a risk communication strategy that should include (89):

- how the risk management team will provide regular feedback on the risk assessment and the respective format;
- clearly defined roles and responsibilities for communications;
- the scope and content of the risk communication (for example, awareness of reporting, avoidance of substandard and falsified medical products);
- the guidelines and procedures for escalation of risk communication;
- the method, media and format for the information to be presented to stakeholders, including the public.

Strategic planning for risk communication should be done in advance and undergo periodic revision. It should be sensitive to stakeholders' needs, encourage participation, be responsive to context, and integrate feedback from affected groups (90).

The NRA should develop and implement a systematic approach to risk communication for substandard and falsified medical products for rapid dissemination of information, and the identification of preventive actions and mitigation measures (89). This should include policies for escalation of risk communication according to the urgency and level of risk, predefined triggers, communication pathways and documentation requirements.

Stakeholders

NRAs should engage with various stakeholders when disseminating information. Building relationships with the media is crucial to ensure fast and widespread dissemination of trustworthy messages and to avoid misinformation. A trained focal point should be designated to communicate with the media (91).

NRAs should adopt different communication approaches tailored to the targeted audience when disseminating messages about the same incident. Both

internal and external stakeholders who may be directly or indirectly impacted by substandard and falsified medical products should be identified, as well as the types of communication to be used, and the nature and content of the communication to be issued.

Social media and organizational websites are useful tools in risk communication to engage the public, facilitate communication with stakeholders, create awareness, monitor and respond to feedback (for example, rumours, public reactions and concerns that arise), and to facilitate local-level responses. Official social media of the NRA, the NRA's websites and traditional media should be included in the communication strategy along with other forms of communication to ensure accuracy of information and consistency of messages disseminated (90).

Procedures and personnel

The NRA should implement policies to identify the types of risk communication that would be appropriate for various types of incidents, for instance (13):

- communications that require an urgent national health alert to the public;
- communications involving market recalls;
- notifications providing warnings or restrictions regarding the use of a product;
- alerts about theft or diversion of authorized medical products, which must be reported to the NRA by the marketing authorization holder, manufacturer or importer;
- warnings regarding misleading advertisements or unauthorized online sales of medical products through websites or social media;
- recommendations for health care professionals outlining appropriate actions when a health alert or product recall is issued;
- notices to regional networks and international agencies, for example, WHO GSMS or INTERPOL.

The NRA should establish transparent procedures for communicating confirmed incidents of substandard and falsified medical products, utilizing clear messaging appropriate to the nature of information disclosed (77).

Where capacity allows, the NRA should establish a dedicated team for designing and delivering necessary messages under their purview. This team should be prepared to provide additional clarification in case of ambiguity, maintain detailed records of incidents, and stay updated. They should also be responsible for planning the communication agenda.

GP4.2 Develop messages appropriate to the target audience and risk level

When disseminating messages related to any incident involving substandard and falsified medical products, the NRA should (41):

- identify the target audience who will be directly or indirectly impacted by the message, assess their level of understanding, identify differences in language and literacy levels, particularly in multilingual settings, understand their possible fears and concerns, and clarify what is expected from them;
- ensure that the message content is clear and comprehensive, preferably accompanied by illustrative materials and a method for addressing further inquiries;
- select the method of communication based on the targeted audience's familiarity with it, considering the speed and reach of message dissemination.

As far as possible, the NRA should validate the messages to ensure clarity and comprehension by pre-testing with the target audience.

The NRA should clearly define the levels of risk communication required for response to substandard and falsified medical products, for example, notifications, warnings or alerts.

Risk communication messages involving substandard and falsified medical products should (90):

- avoid explaining risk in technical terms;
- be issued from different information sources using consistent content;
- promote specific actions persons can take realistically to protect their health.

For communication to the public, the NRA should aim to “alert but not alarm” by providing information about the product and its discovery, clear guidance on how to handle the product, whether to stop using it, and actions to be taken, specifically when to seek advice from health professionals, or whether to refer to the NRA (90).

It is essential for NRAs not only to disseminate information but also to provide guidance to health professionals, procurement agencies, wholesalers and importers on how to address substandard and falsified medical products and ensure the availability of genuine, authentic products.

For health professionals, the information provided should be clear and concise and should include information about the product and its discovery, along with instructions on identification of the substandard or falsified medical

product and subsequent actions to be taken, specifically reporting of cases to the NRA, and handling of the product (90).

The NRA should provide guidance to manufacturers describing the minimum content required for risk communication to the public in the context of substandard and falsified medical products, for example, formats for field safety notices or product recalls. Manufacturers of medical products should be required to provide a draft of the planned alert or field safety notice to the NRA for review prior to issuance (7). In the absence of alerts or field safety notices from the manufacturer, the NRA should implement a standard notice to be used.

The NRA should establish communication strategies for disseminating relevant national, regional and international alerts, and for facilitating information exchange and inquiries with other NRAs and international organizations, as needed. The flow of information should be clearly mapped and should include the urgency or risk level, type of information, responsible entities, modes of communication and timelines.

6.1.2 Managing alerts and recalls

The NRA should develop a recall strategy that aligns with the risk level of the product and the distribution scope.

Identification and assessment

GP4.3 Identify incidents that require alerts and recalls

The NRA should establish policies to identify thresholds for response, specifically the types of incidents involving substandard and falsified medical products and the criteria for alerts, with or without recall. Where local regulations specify that manufacturers conduct product recalls, the NRA should ensure that the manufacturers have implemented relevant policies describing the criteria and conditions for product recalls and any other relevant field safety corrective actions.

Information to be assessed before issuing an alert include:

- public health threat or risk related to the incident;
- distribution of the related product within and outside the country;
- potentially related warnings or alerts (for example, by searching GSMS or other systems);
- continuity or resolution of the risk;
- the appropriateness of measures taken by relevant stakeholders.

The criteria or thresholds for responses should be informed by the findings of the risk assessment, and the respective policies and guidance for their application should be clearly documented.

GP4.4 Establish procedures for alerts and recalls

The NRA should issue alerts to inform health care providers and the public about product recalls, where applicable, ensuring transparency and public safety, while avoiding undue panic or hoarding.

In addition to determining the need for notifications and alerts associated with recalls, and identifying appropriate communication strategies, the NRA should have policies and procedures in place for recall that identify:

- the criteria for determining the level of response needed, for example, notification, alert, recall or product withdrawal;
- the criteria for determining the appropriate level of the supply chain for the recall, for example, distributor, wholesaler, or point of sale or user level;
- guidance for manufacturers, importers, exporters and distributors on how to handle a recall, including formats for notices;
- requirements for coordination between the NRA and manufacturers, distributors, or marketing authorization holders for recalled products;
- mechanisms for feedback on recall effectiveness and corrective actions taken;
- practices for safe storage and disposal of recalled products;
- documentation of regulatory decision, including related regulatory sanctions or legal prosecution.

When alerts are warranted, including those related to locally recalled products, the NRA should determine whether communications should include the public, health professionals, other countries, regional networks and international organizations (for example, rapid alert notifications and WHO GSMS) ([10](#)). The NRA should establish expedited alert-sharing mechanisms with neighbouring NRAs and regional and international networks, according to information-sharing agreements for strengthened regional surveillance and coordinated responses.

Alerts and recall notifications should be:

- shared with the target audience in a timely manner to minimize impacts from the substandard or falsified product;
- consistent with good risk communication principles, with consideration of the local context and the target audience.

Recall notifications to the public, including health professionals, should include such information as how to identify the recalled substandard or falsified product (for example, images, lot numbers), information on where the product

was distributed (if known), the health hazard associated with the substandard or falsified product, and instructions on what consumers or health professionals should do if they have identified a substandard or falsified product (see section 5.5 above on reporting systems).

The NRA should develop standardized formats or templates for the recording and communication of notices, alerts and field safety notices. Information on content and format of field safety notices may be found at WHO's "Field safety notice" page,³⁶ with reference to additional guidance from ISO 13485:2016.³⁷

The NRA should maintain documentation of recalls and any associated alerts or notifications. Where possible, recall information should be maintained in a centralized web-based location accessible to and searchable by the public.

WHO has published various guidance for risk communication and community engagement, particularly for emergencies, which may assist NRAs in developing policies and procedures for promptly responding to substandard and falsified products (92).

6.2 Storage and disposal

Improper disposal of substandard and falsified medical products may be harmful to human health, animal health and the environment, particularly when it leads to contamination of soil, water supplies or local sources used by nearby communities or wildlife. There should be consideration of health and environmental risks, such as (93):

- contamination of water, air and soil through leachate from poorly located landfills;
- destruction of bacteria needed for treatment of sewage or aquatic life by disposal of antibiotics, antineoplastics and disinfectants into waterways;
- release of toxic pollutants into the air via burning of pharmaceuticals at low temperatures or in open containers;
- diversion of products or packaging for resale, reuse or falsification, particularly in the absence of suitable disposal sites, adequate security or qualified personnel to supervise disposal.

The NRA may encounter substandard and falsified medical products under various circumstances, warranting regulatory action where removal of the product from the official or informal market is needed, including the following:

³⁶ <https://extranet.who.int/prequal/vitro-diagnostics/field-safety-notice>.

³⁷ <https://www.iso.org/standard/59752.html>.

- **seizure by the NRA:** when the NRA seizes falsified or unlicensed products following investigations triggered by complaints, reports from law enforcement agencies or whistleblowers, or alerts from regional networks or international organizations (for example, WHO, INTERPOL);
- **seizure by other stakeholders:** when other stakeholders, such as customs or legal regulatory bodies, seize products that are later confirmed as substandard or falsified;
- **product recall:** issuance of recall letters by manufacturers or marketing authorization holders for batches of products that do not comply with regulatory standards.

The NRA should be empowered by legal frameworks to regulate the safe disposal of unwanted medical products, including substandard, falsified or both medical products. Given that the responsibilities for regulation of waste management and environmental protection are usually outside the remit of NRAs, the respective regulatory activities should be conducted in coordination with key stakeholders, including law enforcement, to ensure that substandard, falsified or both medical products are handled according to public health, legal and environmental requirements. This structured approach promotes responsible management, mitigates risks to public health, and prevents the reintroduction of substandard and falsified medical products into the supply chain.

The NRA's responsibilities in the context of regulating the disposal of medical products may be grouped into four main areas, according to local context:

- **policy and legislative development:** establishment of laws, regulations, standards and policies governing medical product waste, including waste classification, storage, transport and disposal, spill response, and extended producer responsibility;
- **oversight and enforcement:** approval of disposal, ensuring compliance with international conventions and standards, monitoring and audit of facilities and processes, and enforcement of penalties (where applicable);
- **coordination and communication:** participation in advisory committees, coordination with environmental authorities and law enforcement, and engagement with stakeholders for emergency preparedness;
- **public education:** campaigns targeted to the public, education for health workers about the safe disposal of substandard and falsified products, support for take-back programmes.

For the purpose of this document, this section will highlight recommended actions for the NRA in the oversight of the disposal of substandard and falsified medical products. Additional guidance on the management of health care waste is provided in the WHO publications *Safe management of wastes from health-care activities* (94) and *Safe management of pharmaceutical waste from health care facilities: global best practices* (95).

GP4.5 Ensure the implementation of a comprehensive waste management system for substandard and falsified medical products

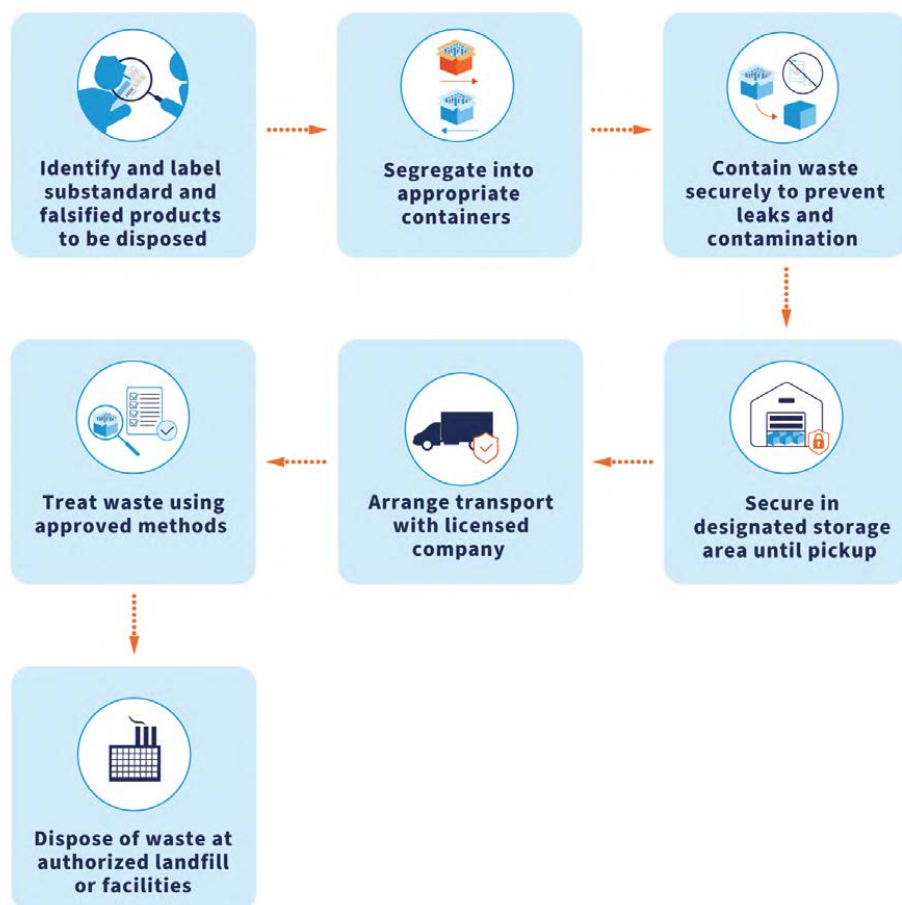
The NRA should ensure that comprehensive waste management systems are in place that include the disposal of substandard and falsified medical products by the licensed stakeholders (for example, manufacturers, distributors, health facilities) according to their defined roles and responsibilities. The medical product waste management system should be integrated into an environmental management system for efficient documentation, implementation and auditing, and should include (92):

- identification of facilities for storage, treatment and disposal of medical product waste;
- clear identification of responsible personnel and contracted companies;
- policies for waste generation and minimization;
- policies and procedures for identification, classification and separation of medical product waste;
- policies and procedures for handling, packaging, labelling and storage;
- policies and procedures for reverse logistics and transportation between facilities;
- methods of treatment of waste and disposal of residues;
- risk assessments considering occupational, public and environmental health and safety;
- information management systems for documentation, tracking, and analysis of products, storage locations, facilities, and processes;
- stakeholder and community awareness and education;
- research and development of improved technologies and practices.

The removal, storage, transport and disposal of falsified medical products should remain the responsibility of the NRA and the responsible law enforcement agency following seizure, according to defined roles and responsibilities.

[Fig. A9.11](#) provides an overview of processes for the disposal of medical product waste.

Fig. A9.11

Overview of medical product waste disposal processes

Source: Adapted from <https://ecomeddissposal.com/regulated-medical-waste/>.

GP4.6 Establish procedures for storage and safe disposal of substandard and falsified medical products

6.2.1 Storage

Proper storage of substandard and falsified products is essential to preserve product integrity (particularly in cases where confirmatory testing or retesting may be required), to support any further legal proceedings, and to prevent re-entry into the market through theft or diversion. The NRA should establish well defined standards and procedures for the safe storage of substandard and falsified medical products that should be clearly communicated to NRA

personnel and licensed establishments, such as manufacturing facilities, importers, distributors, health facilities and waste management companies. The NRA or responsible stakeholder should maintain dedicated warehouses for the temporary storage of substandard and falsified medical products. In the absence of such facilities, appropriate storage locations should be identified, including arrangements with providers for security and transportation. The storage and transportation conditions and procedures should comply with national legal provisions and standards (good practices) to ensure that products are maintained under suitable conditions consistent with the products' specifications.

Substandard products that have been recalled by the manufacturer or returned to the manufacturer, health professional, health facility, or dispensary should be stored away from other products under appropriate conditions, sorted and classified by risk level, and appropriately labelled in secured locations.

Similarly, substandard, falsified or unlicensed medical products that have been seized by the NRA should be sorted, labelled and stored in a secure facility pending disposal. Where storage at the site of discovery would be insufficient or inappropriate (for example, due to poor security or falsified products), the products should be transferred for storage at an approved warehouse with sufficient capacity and security.

Comprehensive technical reports should be prepared and maintained describing the substandard and falsified products that have been recalled, returned or seized, including generic name or INN, brand name, form or presentation, strength, pack size, batch number, unique identifiers for traceability, quantity and the reason for storage. Records of the delivery and dispatch of the products should be maintained at the facility to enable tracking. The NRA should inspect the respective storage facility to confirm the location, appropriate sorting and storage conditions, and security of the products. The inspection report should confirm the products stored, their specifications, quantities, locations within the facility and the conditions of storage.

Substandard and falsified medical products should be disposed of within a reasonable time to avoid accumulation at storage facilities. The responsible entity and the management of the storage facility should determine the limits of storage periods according to product specifications, capacity and workflow.

6.2.2 Disposal

The NRA should develop and implement standards and guidelines in collaboration with the national environmental protection authorities for appropriate treatment and safe disposal of substandard and falsified medical products to avoid adverse public health and environmental impact, such as leakage into soil or water and the emission of gases.

For the disposal of substandard and falsified medical devices, the NRA should ensure that the appropriate procedures for disposal are implemented, particularly where the device contains active substances or reagents (for example, mercury).

Each NRA should establish a dedicated unit for oversight of the disposal of medical products, with documented guidelines and procedures that are publicly accessible. These procedures and guidelines of the unit should describe:

- **licensure of facilities:** the criteria and process required for approval of the medical product waste disposal facilities by the NRA or by the relevant environmental authority;
- **steps for requests for disposal:** the process that licensed institutions must follow to submit disposal requests to the NRA;
- **required documents:** a list of all necessary documents for disposal requests;
- **time frames:** specific time frames for each step of the process, including the time frame for applications, the deadline for document submission, the time frame for review and approval, and scheduling of supervision of the disposal.

Additionally, the procedures should outline the steps that the NRA will take throughout the disposal review process, including decision-making, to ensure transparency and efficiency.

The responsible entities should be required to follow procedures for the safe disposal of medical products, including substandard and falsified products. These recommended procedures are described below.

Decision to initiate disposal

The licensee, warehouse, pharmacy, hospital or other responsible organization should document the decision for disposal of substandard and falsified medical products, which should be followed by a request to the NRA for approval of the disposal. This should include the following documents ([95](#)):

- **request letter:** a letter issued and signed by the company, addressed to the NRA, requesting approval for the disposal of the products listed, and contact information of the requester;
- **product list:** a detailed list of the products to be disposed of, including the reasons for disposal;
- **product information:** name (including the INN or generic name, brand name), batch numbers, expiration dates, quantities, weights and volumes;

- **risk level:** the risk classification of the products, denoting if hazardous, along with appropriate cautions;
- **plan:** the complete details of the intended disposal site, equipment, procedures, estimated density (for example, 0.2 tonnes per cubic metre), the methods of treatment and disposal, and all relevant documents verifying the facility's licensure and suitability for handling the disposal;
- **notification letter:** identification of the responsible personnel for document issuance and receipt, and persons who would be attending the disposal process;
- **written pledge:** confirmation that the products are not subject to legal prosecution.

Where the NRA is responsible for the decision to dispose of the substandard and falsified medical products, the decision-making process, personnel involved, rationale for disposal, and conclusions should be documented.

Approval process

The NRA should review the submitted documents and communicate the decision to the requester within a determined time frame regarding:

- **Approval.** If all documents meet the NRA's criteria, where applicable, the requester should be advised to schedule the disposal within a specified time. The NRA should clearly communicate conditions under which requests may be cancelled, for example, failure to dispose of products within the specified period.
- **Additional documents required.** If documents are missing, the requester should be notified about the additional documents required and advised to submit these within a specific period. Following submission, review of the documents, and approval, the requester should be advised of the approval.
- **Additional reports.** If additional reports are needed from other authorities or institutions (for example, compliance reports or environmental approvals), the requester should be informed, and the relevant administration should be contacted to submit the report within a specified period (for example, five working days).
- **Non-approval.** If the request is denied, a description of the reasons for the denial should be provided with information on the deficiencies to be addressed. The requester may be permitted to resubmit after corrective actions are taken.

Forming teams for oversight of disposal

The NRA should identify auditors to participate as technical members of the committee formed to supervise the disposal process. The composition of the oversight committee should include experts and officers selected according to the type of products, methods of treatment and disposal, and facilities. The committee may include:

- representatives of the manufacturer, sponsor or distributor, where applicable;
- a responsible pharmacist or auditor for the NRA;
- a representative of the environmental health authority;
- a technical expert in waste management;
- a law enforcement or judicial representative.

Sorting

Auditors or inspectors should ensure that the products listed for disposal are appropriately sorted and match the approved list of products for disposal. After verification, the items should be sealed by the auditor prior to transportation.

Disposal

After sorting, disposal should follow as soon as possible, supervised by assigned personnel to verify the integrity of the seals before removal, and the correct methods of treatment and disposal. If any discrepancies in safety seals, product type, quantity or disposal method are observed, the disposal process should be halted, and a report issued. The applicant should submit a corrective and preventive action report and request to proceed with justification. After the NRA has completed its review of the corrective and preventive action report and request from the applicant, supervised disposal may resume.

The completed disposal process should be documented, and reports should be sent by the disposal facility to the NRA, environmental protection authorities, legal authorities, and other regulatory authorities, as per established roles, responsibilities, and procedures for reporting. Where applicable, the NRA should submit any required reports to international or regional networks according to agreements or conventions, for example, the International Narcotics Control Board ([95](#)).

GP4.7 Ensure that appropriate methods are used for the treatment and disposal of substandard and falsified medical products

The NRA should ensure that appropriate methods are used for the safe disposal of substandard and falsified medical products according to the nature and

characteristics of the products to avoid harm, prevent the dissemination of confidential information (for example, medical devices that capture user data), and minimize the environmental impact. Examples of treatment and disposal options for safe disposal of pharmaceuticals to minimize risk to public health and the environment are provided in Appendix 6, Table 1 ([95](#)).

For medical devices, WHO provides guidance on the decommissioning of medical devices in its Medical Devices Technical Series ([96](#)).

NRAs should adapt or adopt these guidelines according to needs and local context to ensure effective and safe disposal practices.

6.3 Transparent legal processes

The principle of transparency has been established as a good regulatory practice in section 3: “National regulatory systems”. The following are additional recommendations specific to the response to substandard and falsified medical products.

GP4.8 Establish and communicate appropriate sanctions based on legal provisions and responsibilities

The legal provisions, regulations and guidelines should clearly define the roles, responsibilities and requirements of the NRA, manufacturers and other stakeholders in addressing issues related to substandard and falsified medical products ([9](#), [34](#), [77](#)).

The legal processes and levels of enforcement or sanctions relevant to regulatory actions should be clearly described and communicated to stakeholders via regulations and guidelines.

The relevant legal processes should describe the various tiers of regulatory responses for substandard and falsified medical products detected in official or regulated markets and informal markets, the discovery of unregistered medical products, and the discovery of unethical promotion of medical products, including misinformation.

When guidelines, the legal procedures and related sanctions are revised, stakeholders should be consulted prior to finalization and implementation.

Supplemental resource

The MEDISAFE reference manual on legislation on substandard and falsified medical products provides guidance on aspects that should be considered and integrated into legal frameworks to enable the regulatory response to substandard and falsified medical products, including administrative penalties, sanctions, remedial measures and criminal offences ([20](#)).

GP4.9 Clearly define standards and procedures for regulatory actions

NRAs should establish different enforcement actions appropriate to the detected violation, findings of the investigations and legal judgements. In cases where there is evidence of fraudulent intent or gross negligence, NRAs should refer to local criminal laws to determine the appropriate and proportional sanctions, penalties and prosecutions for violations of the applicable legislation ([97](#)).

Inspectors should have the judicial authority and the appropriate support from law enforcement authorities to seize or quarantine substandard and falsified medical products as a precautionary measure until the investigation concludes, if necessary.

It is critical that NRAs establish procedures for a system of escalation of enforcement actions in cases involving substandard or falsified medical products. These enforcement actions should cover measures targeting the product, responsible personnel, equipment and institutions involved. Actions against substandard and falsified medical products may include seizure, quarantine, suspension, recall or withdrawal of the product, temporary closure of the manufacturing or distribution site (pending corrective and preventive action), and withdrawal of the manufacturing or distribution licence ([98](#)).

The NRA should develop and communicate the criteria, timelines and processes for decision-making and regulatory actions, including suspension or revocation of market authorization.

The NRA should establish comprehensive procedures for appealing decisions, which may involve administrative processes for submission and review. There should be a well defined pathway for receiving, reviewing and investigating complaints related to the NRA's actions or those of their personnel, ensuring transparency, fairness and confidentiality. The resultant investigation and any corrective actions taken should be analysed and used to improve the system and performance of personnel, where applicable.

The NRA should also clearly define procedures for addressing unregistered or unlicensed medical products. While these products are not evaluated or approved by NRAs for local marketing, they may be illegally diverted but registered and marketed in other countries. Actions taken against these products should be based on thorough investigations and should comply with national regulations ([9](#), [14](#)).

GP4.10 Clearly communicate the enforcement of regulatory actions

The NRA should develop and implement policies and procedures for the communication of regulatory actions taken in response to confirmed substandard and falsified medical products, including target audience, content for notifications and channels. Communications should accurately reflect the roles and responsibilities of manufacturers, marketing authorization holders, health professionals and other stakeholders, including cooperation with the NRA.

As far as possible, information on regulatory actions or law enforcement against substandard and falsified medical products should be published via broadcast media and the website of the NRA, for example, media or press releases, print and electronic media, and posts on social media.

When notifying the responsible entity (for example, manufacturer, marketing authorization holder, importer or distributor) of the suspension or revocation of market authorization or of other regulatory penalties, the NRA should describe the reasons for the decision and the available procedures for appeal.

The level of the regulatory action should be appropriate to the offence, informed by the incident's risk level, and based on a thorough, unbiased investigation. It is recommended that NRAs engage in formal or informal communication with stakeholders during the investigation and before implementing regulatory actions.

The NRA should include findings of market surveillance and control activities and the respective regulatory actions in its public databases of authorized medical products and licensed establishments (9). The databases should include the following:

- information about the inspected facility, date of inspections, conclusions of inspections or investigations, and applicable regulatory actions, sanctions or penalties;
- information on products involved in market surveillance, including recalls and suspensions, describing relevant findings and regulatory decisions.

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Appendix 1. List of main good practices

National regulatory systems

GP1.1 Ensure that the regulatory framework clearly defines or describes the NRA's mandate

GP1.2 Ensure that the legal framework clearly describes the scope of medical products and facilities to be regulated

GP1.3 Ensure that legal provisions facilitate activities to prevent substandard and falsified medical products

GP1.4 Ensure that legal provisions enable the detection of substandard and falsified medical products

GP1.5 Ensure that legal provisions empower the NRA to respond to detected substandard and falsified medical products

GP1.6 Ensure that legal provisions empower the NRA to effectively manage, monitor and evaluate the market surveillance and control function

GP1.7 Implement good regulatory practices

GP1.8 Practice good governance

GP1.9 In the absence of a dedicated system for substandard and falsified medical products, utilize existing systems for vigilance

GP1.10 Utilize evidence in decision-making

GP1.11 Determine the use of regulatory reliance for market surveillance and control

GP1.12 Ensure transparency in legal processes

GP1.13 Develop an NRA competency model based on organizational requirements

GP1.14 Determine required staff competence and recruitment criteria, and assign staff according to functions

GP1.15 Implement appropriate staff training and development

GP1.16 Map key stakeholder groups

GP1.17 Engage stakeholders according to priority

GP1.18 Implement approaches for regulatory system strengthening

GP1.19 Utilize the WHO Global Benchmarking Tool for self-assessment and the development of national action plans

GP1.20 Develop a national action plan for substandard and falsified medical products

Table *continued*

Prevention
GP2.1 Support the development of competence of health professionals
GP2.2 Develop and implement appropriate education and awareness campaigns
GP2.3 Monitor and evaluate the impact of education and awareness campaigns
GP2.4 Ensure access to current critical product information and specifications
GP2.5 Ensure access to evidence of manufacturing quality
GP2.6 Apply a risk-based approach and efficiencies for preregistration inspection of manufacturing sites
GP2.7 Publish a database or lists of authorized medical products
GP2.8 Collaborate and communicate with customs and border agencies
GP2.9 Implement standards and procedures for importer authorization with relevant guidelines
GP2.10 Identify and monitor entry points for medical products
GP2.11 Establish appropriate clearance procedures for imported products
GP2.12 Assign inspectors according to risk, turnover and qualifications
GP2.13 Establish traceability systems, databases and standards for authorized medical products
GP2.14 Establish and communicate standards for storage and distribution of medical products
GP2.15 Conduct inspections according to established standards and guidelines
GP2.16 Ensure quality systems are implemented
GP2.17 Identify and separate substandard and falsified medical products
GP2.18 Collaborate with stakeholders to enforce standards for ethical product promotion
GP2.19 Establish systems to regulate and monitor channels for promotion, marketing and sales of medical products
Detection
GP3.1 Collaborate with border control agencies for detection of substandard and falsified medical products
GP3.2 Structure an inspection programme as part of the NRA's quality management system

Table *continued*

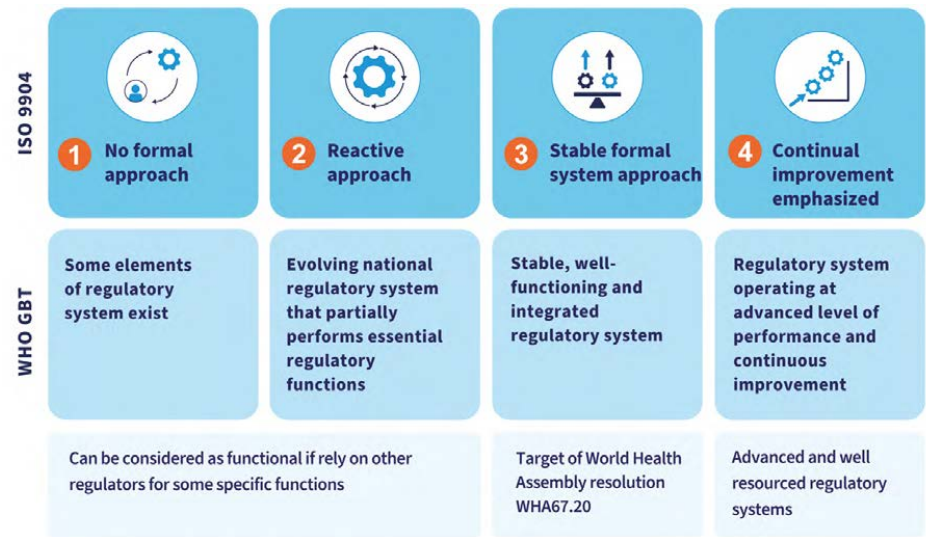
Detection <i>continued</i>
GP3.3 Establish and document quality risk management processes
GP3.4 Apply quality risk management approaches to inspection strategy
GP3.5 Plan and conduct inspections based on assessment of risk
GP3.6 Conduct regulatory inspections throughout the supply chain
GP3.7 Review corrective and preventive actions, and schedule routine inspections
GP3.8 Develop a strategic framework for risk-based market surveillance
GP3.9 Determine priorities for market surveillance using available market information and related information systems
GP3.10 Utilize existing vigilance systems to detect substandard and falsified medical products
GP3.11 Leverage notifications and alerts from other sources and stakeholders
GP3.12 Determine the scope and frequency of sampling and testing activities
GP3.13 Clearly define objectives of quality surveys
GP3.14 Identify partners and agree on survey management plan
GP3.15 Apply a risk-based approach to select products, locations and facilities
GP3.16 Develop and communicate a predefined survey protocol
GP3.17 Implement efficiencies to quality testing to reduce waste and unnecessary costs
GP3.18 Use visual and physical inspection, authentication technologies and databases to detect substandard and falsified medical products
GP3.19 Evaluate and select appropriate screening technologies
GP3.20 Establish collaborative relationships with quality control laboratories
GP3.21 Ensure that quality testing laboratories are adequate
GP3.22 Ensure that testing is appropriate for the respective product
GP3.23 Establish a database, procedures and tools for survey reports
GP3.24 Establish a reporting system or integrate reporting into existing vigilance systems
GP3.25 Develop and launch campaigns to create awareness of substandard and falsified medical products

Table *continued*

Detection <i>continued</i>
GP3.26 Implement approaches to encourage and facilitate reporting
GP3.27 Monitor and periodically evaluate the performance of the reporting system for substandard and falsified medical products
GP3.28 Develop procedures for handling of complaints and incident reports
GP3.29 Document and facilitate sharing of investigations of complaints and incident reports, as appropriate
GP3.30 Assess risks using a standardized approach and tools
GP3.31 Identify relevant risk factors to be considered
GP3.32 Categorize risk factors and establish importance
GP3.33 Combine risk categories and weights in a risk assessment matrix
GP3.34 Establish risk mitigation procedures and appropriate regulatory actions
Response
GP4.1 Develop a risk communication strategy
GP4.2 Develop messages appropriate to the target audience and risk level
GP4.3 Identify incidents that require alerts and recalls
GP4.4 Establish procedures for alerts and recalls
GP4.5 Ensure the implementation of a comprehensive waste management system for substandard and falsified medical products
GP4.6 Establish procedures for storage and safe disposal of substandard and falsified medical products
GP4.7 Ensure that appropriate methods are used for the treatment and disposal of substandard and falsified medical products
GP4.8 Establish and communicate appropriate sanctions based on legal provisions and responsibilities
GP4.9 Clearly define standards and procedures for regulatory actions
GP4.10 Clearly communicate the enforcement of regulatory actions

Appendix 2. WHO Global Benchmarking Tool maturity levels and indicators for market surveillance and control function

Fig. 1
WHO Global Benchmarking Tool performance maturity levels



Source: Broojerdi AK, Sillo HB, Dehaghi ROA, Ward M, Refaat M, Parry J. The World Health Organization Global Benchmarking Tool: an instrument to strengthen medical products regulation and promote universal health coverage. *Front Med.* 2020;7:457.

Table 1
Indicators and sub-indicators for market surveillance and control

Indicator	Sub-indicators	Maturity level
04 – Market surveillance and control		
Objective: To ensure compliance of the products placed on the market with preset criteria for quality, safety and efficacy		
Activity themes: (i) control of import activities, (ii) prevention, detection, and response to substandard and falsified medical products, (iii) market surveillance for monitoring quality of medical products throughout the supply chain, and (iv) control of promotional marketing and advertising activities		

Table 1 *continued*

Indicator	Sub-indicators	Maturity level
MC01 Legal provisions, regulations and guidelines required to define regulatory framework of market surveillance and control activities	MC01.01: Legal provisions and regulations are in place with respect to import activities including permanent regulatory intervention at designated entry and exit ports where medical products are being moved.	1
	MC01.02: Legal provisions and regulations authorize market surveillance and control activities which include product sampling from different points of the supply chain.	1
	MC01.03: Legal provisions and regulations address the role of NRA in dealing with substandard and falsified medical products.	1
	MC01.04: Legal provisions and regulations exist for the control of promotion, marketing and advertising of medical products to avoid communication of false or misleading information.	2
	MC01.05: Legal provisions and regulations exist for placement of a product's unique identification number on its outer packaging.	4
	MC01.06: Guidelines exist for importers that specify the format and content of the relevant applications and procedures to receive the necessary authorizations or permissions.	2
	MC01.07: Guidelines exist on the recall, storage and disposal of substandard and falsified medical products.	2
MC02 Arrangement for effective organization and good governance	MC02.01: There is a defined structure, with clear responsibilities, to conduct market surveillance and control activities.	2
	MC02.02: Documented procedures or mechanisms are implemented to ensure the involvement and communication among all stakeholders relevant to market surveillance and control activities.	3

Table 1 *continued*

Indicator	Sub-indicators	Maturity level
MC03 Human resources to perform market surveillance and control activities	MC03.01: Sufficient competent staff (i.e., education, training, skills and experience) are assigned to perform market surveillance and control activities.	3
	MC03.02: Duties, functions, and responsibilities of the staff in charge of market surveillance and control activities are established and updated in the respective job descriptions.	3
	MC03.03: Training plan developed, implemented and updated at least once a year for staff in charge of market surveillance and control activities.	3
	MC03.04: The NRA generates and maintains records of staff training activities and training effectiveness verification.	3
MC04 Procedures established and implemented to perform market surveillance and control	MC04.01: Documented and implemented procedures exist to grant the necessary authorizations or permissions for import activities.	2
	MC04.02: Documented and implemented procedures exist for regulation of promotion and advertisement of medical products.	3
	MC04.03: Documented and implemented procedures for active monitoring of the promotion and advertisement of medical products.	4
	MC04.04: Documented and implemented procedures exist for risk-based sampling of medical products from different points of the supply chain.	4
	MC04.05: Documented and implemented procedures exist to enable the public to report suspected substandard and falsified medical products.	3
	MC04.06: Documented and implemented procedures exist in the NRA to review any complaints or market reports received.	2

Table 1 *continued*

Indicator	Sub-indicators	Maturity level
MC04 Procedures established and implemented to perform market surveillance and control <i>continued</i>	MC04.07: Documented and implemented procedures and mechanisms exist to prevent, detect and respond to substandard and falsified medical products.	3
	MC04.08: Documented and implemented procedures exist to ensure safe storage and disposal of detected substandard and falsified medical products.	2
MC05 Mechanism in place to monitor regulatory performance and output	MC05.01: Database exists of approved and refused promotional and advertising materials along with the supporting documentation.	4
	MC05.02: Database for product batches that have undergone surveillance along with their relevant testing results and regulatory actions is established and periodically reviewed.	4
	MC05.03: Performance indicators for market surveillance and control activities are established and implemented.	4
MC06 Mechanism exists to promote transparency, accountability and communication	MC06.01: Market surveillance and control activities are appropriately communicated within the NRA.	3
	MC06.02: Findings and regulatory decisions of market surveillance and control activities are appropriately communicated to all national stakeholders including the general public.	3
	MC06.03: Findings and regulatory decisions of market surveillance and control activities of common interest are appropriately communicated and shared with other countries and regional and international organizations.	3

Source: WHO Global Benchmarking Tool (GBT) for evaluation of national regulatory systems of medical products: revision VI. Geneva: World Health Organization; 2021.

Appendix 3. Sample activities and competencies of inspectors

Inspection of various types of facilities throughout the supply chain is a key regulatory function in promoting and protecting public health from harmful products. Inspectors should be suitably qualified and trained to facilitate the conduct of various types of inspections for compliance with good practices to ensure that products are of good quality, safe and efficacious before and after marketing in the country.

Table 1
Sample activities and competencies of inspectors

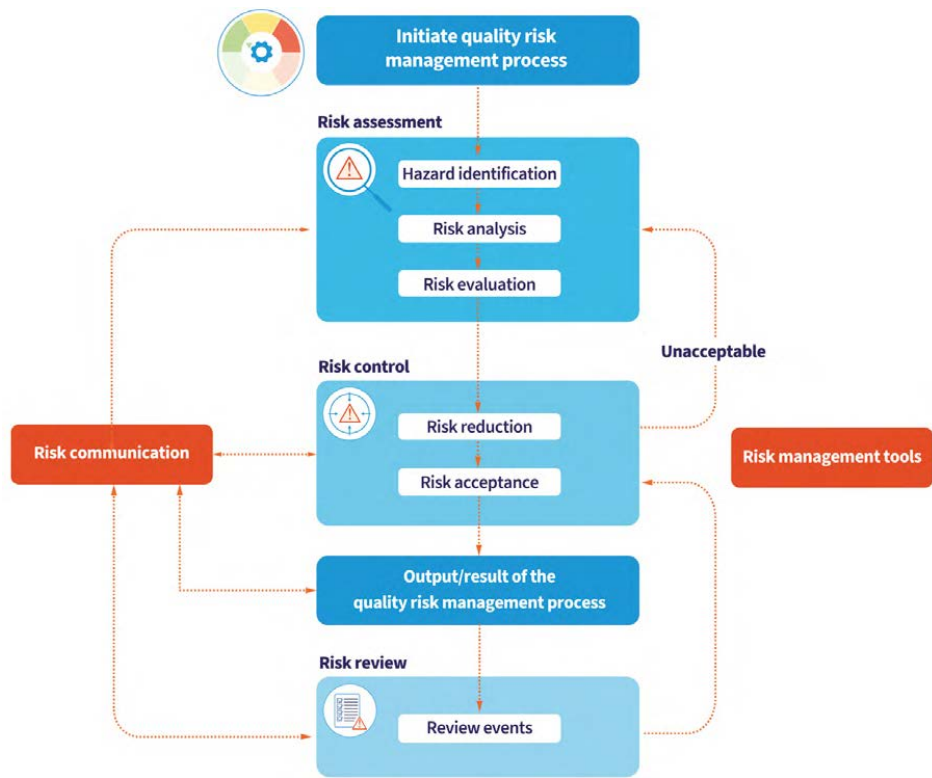
Role-specific activities	Examples of activities	Required competencies, knowledge and understanding of:
Performing review of applications for licences	Review of applications for manufacturing licences, wholesale and retail outlets	Regulatory requirements for applications for manufacturing licences, wholesale dealers' licences and retail outlets, along with their variations
Performing inspections for compliance with good manufacturing practice requirements	Evaluation of inspection reports from competent authorities Inspection of manufacturing facilities for compliance	Regulatory requirements, procedures for good practices Requirements of the quality management systems of manufacturers
Performing investigative inspections	Conduct of inspections in response to incident reports or complaints where investigation is needed Evaluation of submitted documentation relevant to needed investigations, e.g. manufacturing activities, clinical studies, operations at wholesale or retail outlets	Manufacturing environments, types of products, intended use and the required infrastructure and conditions for manufacture The process of investigations and implementation of corrective and preventive action by the manufacturers Procedures for good clinical practice, good laboratory practice or good manufacturing practice audits as part of an inspection or on their own

Table 1 *continued*

Role-specific activities	Examples of activities	Required competencies, knowledge and understanding of:
Performing inspections for compliance with good clinical practice	Evaluation of inspection reports from competent authorities Inspection of clinical sites in compliance with good clinical practice requirements	Biomedical literature, including clinical trial reports Scope of market authorization for investigative medical products Regulatory requirements for good clinical practice
Performing inspections for compliance with good storage and distribution practice	Inspection of premises and facilities that distribute, dispense or keep custody of pharmaceutical and other medical products for compliance with good storage and distribution practice requirements	Good storage and distribution practice regulatory requirements Regulatory requirements for product information and labelling

Appendix 4. Quality risk management process overview and examples

Fig. 1
Quality risk management process overview



Sources:

WHO guidelines on quality risk management. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: forty-seventh report (Annex 2). Geneva: World Health Organization; 2013 (WHO Technical Report Series, No. 981).

Quality risk management. ICH harmonized guideline Q9 (R1). Geneva: International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use; 2023.

Table 1
Example of a risk category assessment of a site depending on level of compliance and inspection frequency

Risk category of the site 	GMP compliance rating and related inspection frequency (in months)			
	Acceptable			Unacceptable
	Good	Satisfactory	Basic	
Critical (C)	24	18	12	Determine on a case-by-case basis
High (H)	30	20	15	Determine on a case-by-case basis
Medium (M)	36	24	18	Determine on a case-by-case basis
Low (L)	48	36	24	Determine on a case-by-case basis

GMP = good manufacturing practice.

Source: Guidance on good manufacturing practices: inspection report. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fiftieth report (Annex 4). Geneva: World Health Organization; 2016 (WHO Technical Report Series, No. 996).

Table 2
Example of a risk-based inspection plan

Complexity of the establishment					
Risk factors	Weight	1 point	2 points	3 points	Comments
Number of staff	2	< 50	Between 51 and 149	> 150	
Coverage of distribution	1	National	N/A	International	(e.g. national or international)
Presence of outsourcing activities	2	No production or quality control activities are carried out by third parties	N/A	Third-party companies are responsible for production or quality control activities	Inclusion of activities performed by third party, especially in products and quality control activities
Type of products	4	Other products	N/A	Cold chain products, biological products, sterile products, cytotoxic medicines, hormones, antibiotics	Assessment of the types of products manufactured, stored or distributed (e.g. veterinary products, food, cold chain products, hormonal products, cytotoxic, penicillins)
Types of equipment	5	Dedicated equipment	N/A	Multipurpose equipment	Assessment based on the nature of the equipment, if they are dedicated per product, partially dedicated or if they are multipurpose

Table 2 *continued*

Complexity of the establishment					
Risk factors	Weight	1 point	2 points	3 points	Comments
Diversity of production processes	4	Low diversity (< 3)	Medium diversity (4 or 5)	High diversity (> 6)	Evaluation of the number of different production processes on the site (e.g. non sterile solids, non sterile semi solids, non sterile liquids, hormones, cytotoxic, sterile products)
Number of products in portfolio	2	< 20	Between 21 and 49	> 50	
Process and equipment used	3	Production or manufacturing in enclosed equipment and the transfer of raw materials or intermediates through pipelines	Production restricted to the manufacturing of semi-solid products, carried out in an open system	Manufacturing that involves the dispersion of powders into the environment, either through the design of the equipment or the nature of the raw material or intermediate transfer operations	Assessment based on the type of equipment (e.g. closed systems, opens systems, manual feeding, manual transfer)

The intrinsic risk of the establishment will be the combination of the establishment's complexity and the criticality of the products, as shown in Table 3.

Table 3
Example of qualitative rating by complexity and criticality

	Complexity		Criticality	
	1	2	3	
1	low	low	medium	
2	low	medium	high	
3	medium	high	high	

Table 4 provides an example of frequency of inspection based on regulatory and intrinsic risks.

Table 4
Example of frequency of inspection based on regulatory and intrinsic risks

Regulatory risk	Intrinsic risk		
	Low	Medium	High
Low	Every 3 years	Every 3 years	Every 2 years
Medium	Every 3 years	Every 2 years	Annually
High	Every 2 years	Annually	Annually

In addition to the factors, the assignment of weights to the factors is recommended, as factors may vary in their impact on or importance to decision-making. The total points assigned will be directly related to its complexity. For this strategy, numerical ranges should be defined, along with qualitative descriptions. In this example, the ranges could be: 23 to 37 points (low complexity), 38 to 53 (medium complexity) and 54 to 69 (high complexity).

Table 5 provides an example of a probability versus impact matrix, and Table 6 gives an example of a risk assessment matrix.

Table 5
Example of a probability versus impact matrix



Impact					
Probability	Negligible	Marginal	Moderate	Critical	Catastrophic
Almost certain	5	10	15	20	25
Likely	4	8	12	16	20
Possible	3	6	9	12	15
Unlikely	2	4	6	8	10
Rare	1	2	3	4	5

Source: WHO guidelines on quality risk management. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: forty-seventh report (Annex 2). Geneva: World Health Organization; 2013 (WHO Technical Report Series, No. 981).

Table 6
Example of a risk assessment matrix

Risk factor	Importance level	Categories	Rate	Rate per factor
		Low risk	1	
		Medium risk	2	
		High risk	3	
Worked example: Medicine (tablet) used as the main treatment for breast cancer was tested, and the official laboratory detected an assay 40% below the specification				
Severity of defect / non-compliance	2	High risk: assay below specification	3	2 x 3 = 6
Potential clinical consequences	3	High risk: medicine used in breast cancer	3	3 x 3 = 9
Patient population	2	High risk: immunocompromised patients	3	2 x 3 = 6
Route of administration	1	Medium risk: internal use, oral	2	1 x 2 = 2
Total			23	

Source: Recommendations for health authorities on criteria for risk assessment and prioritization of cases of unregistered/unlicensed, substandard and falsified medical products. In: Seventh meeting of the Member State mechanism on Substandard and Falsified Medical Products. Geneva: World Health Organization; 2018.

Appendix 5. Elements of quality survey protocols

Table 1

Content and key considerations for quality survey protocols

Aspect	Key considerations	Examples of variables or elements
Selection of areas to be sampled	Collect from various geographical locations (rural, urban) unless targeted sampling is intended. Use appropriate mapping tools, risk profiles (heat maps), and digital platforms, where available. Justification of selection of sampled areas: areas with high risk of substandard and falsified medical products should be prioritized.	Population density Disease incidence or prevalence Risk of poor-quality products (e.g. near borders, previous sites) Income level of population in target area Complexity and security of distribution systems Presence of outlets selling mostly unregistered or illegal medical products
Selection of medical products	For optimal use of available resources, focus on medical products that pose greatest risk to patients. Conduct risk assessments of individual products to estimate risks posed. Include instructions for sample collection with consideration of dosage forms and strengths, multiple brands, storage conditions.	Categorization of medical products (e.g. API content, therapeutic group, formulation) Risks posed or public health impact: - probability of occurrence of a quality problem, considering manufacturing complexity, stability, compliance with good manufacturing practice, distribution chain complexity - Exposure of patients and seriousness of health impairment: extent of exposed population, vulnerability of target population, dosage form complexity, therapeutic properties

Table 1 *continued*

Aspect	Key considerations	Examples of variables or elements
Selection of sample collection sites	Classify types of outlets according to the relevant legislation or national situation. Sampling should usually be performed in both the public and private sectors as well as in the informal market. Explain the selection of types of sites for sample collection in light of achievement of the survey objectives. Include a map of sample collection sites. Consider the impact and implications of conditions of different supply levels.	Types of sample collection sites: • public (government), formal (licensed or registered for profit, or nongovernmental organizations), informal (unlicensed) • level in supply chain: - level 1 (points of entry) - level 2 (e.g. wholesalers, distributors, pharmacies, health facilities) - level 3 (informal outlets) - level 4 (virtual outlets)
Sampling design	The sampling design should be informed by the objectives of the survey, the risks and consequences of inherent decision errors and biases, and available resources. For efficiency, focus on outlets where the risk of poor-quality medical products is high. Include rules to guide the selection of the sample design, e.g. external validity, limitations, inclusion and exclusion criteria, substitution of sites.	Sampling designs: • convenience sampling • simple random sampling • stratified random sampling • lot quality assurance sampling • sentinel site monitoring <i>Note:</i> See WHO TRS 996 for descriptions of the various sampling designs, their advantages, limitations and applications

Table 1 *continued*

Aspect	Key considerations	Examples of variables or elements
Sampling plans	Ensure that the number of dosage units to be collected per sample is informed by the survey objectives, surveyed products, tests to be conducted, testing methods to be used and available resources. Collect sufficient amounts of samples for planned tests, investigation or confirmatory testing of out-of-specification samples, and in case of dispute.	Sampling plans for each area, region or country Sampling plans should specify: • individual sites for sample collection • medical products to be sampled (by API, dosage form, strength, pack size) • minimum number of dosage units • number of samples per medicine or medical product • total number of samples to be collected in the relevant area, region or country
Sample collection	Describe the rationale for the sampling approach to be used (overt or mystery shopper) and who should collect samples. This will depend on the survey objectives, the regulatory status of the target medical products and what is known about the knowledge and attitude of the sellers.	Overt sampling: if staff are anxious to avoid poor-quality products, and they are informed about objectives, overt sampling with feedback may support collection of more data and risk factors Mystery shopper approach: useful where seller knows or is concerned that stock is poor quality or illegal and may offer better-quality products <i>Note:</i> See WHO TRS 996 for strategies for use of covert or mystery shopper approach to sampling

Table 1 *continued*

Aspect	Key considerations	Examples of variables or elements
Sample collectors	The local coordinator should provide training for collectors to familiarize them with the project, survey protocol, sampling plan and instructions for sampling and data collection. Ensure that sample collectors have access to the tools and wear needed. Consider the cost of collected samples when determining number to be collected, and risk of product shortage after sampling. Replacement may be needed.	Instructions to sample collectors should include: • the minimum number of dosage units per sample, number batches • the target products, dosage forms, strengths, pack sizes • retention of samples in original primary and secondary packaging with labels and inserts • collection of samples with at least six months before expiry • use of sampling tools and protective wear • use of sample collection form • use of unique sample codes • batch certificates of analysis (where overt sampling is used) • documentation of storage conditions of the site (temperature, humidity, light, other observations) • maintenance of controlled conditions in line with product label requirements • Timing for sample collection and deadline for sending to laboratory

Table 1 *continued*

Aspect	Key considerations	Examples of variables or elements
Storage and transportation of samples	Follow the proper chain of custody for samples until they reach the laboratory, and ensure proper storage before and after they are received. Samples should be: • kept in original packaging • collected and kept under controlled conditions in line with the product label requirements • kept protected from light, excessive moisture or dryness • packaged and transported to avoid breakage and contamination • secured against theft.	Instructions for storage and handling of samples, including: • conditions for temperature, humidity, light • documentation of conditions using sample collection forms • copies of manufacturer batch certificates of analysis, if available

Table 1 *continued*

Aspect	Key considerations	Examples of variables or elements
Testing	Describe risk-based testing criteria, including the level of testing to be applied (see risk-based testing below). Visually inspect all samples to check product integrity, and for authentication. Clearly justify the choice of the testing laboratory or laboratories in the survey protocol, reports and publications. Ensure appropriate agreements are in place in advance, where laboratory testing needs to be outsourced. Include the following in the survey protocol: • name and address of laboratory or laboratories • contact person in the laboratory • medical products assigned for testing to the particular laboratory • Timing for testing and deadlines for sample submissions.	Testing laboratory criteria: • compliance with WHO good practices for pharmaceutical quality control laboratories, preferably WHO prequalified laboratory or one with equivalent standards • capacity and competence to conduct required tests • sufficient capacity to test required number of samples within specified period within budget Elements of agreements for laboratory services: • protocol: products and numbers of samples to be tested, tests to be conducted, specifications according to testing protocol • responsibilities of laboratory during survey • confidentiality declaration • acceptance of possible audit, access to records and retained samples

Table 1 *continued*

Aspect	Key considerations	Examples of variables or elements
	Establish agreement on laboratory testing protocols with quality control laboratory. Laboratory testing of all collected samples should be done according to the testing protocol, and according to a pharmacopoeial or analogous monograph.	Testing protocol should include tests to be conducted, such as: • identity verification • appearance • selected pharmacopoeial or analogous testing • special or specific tests, e.g. excipients • assay for APIs • dissolution, disintegration, uniformity of dosage units, dispersion (solid dosage forms) • pH value, volume in containers, extractable volume (liquid dosage forms) • sterility and bacterial endotoxin tests (parenteral products)

Table 1 *continued*

Aspect	Key considerations	Examples of variables or elements
	Select test methods and specifications according to: the survey objectives, and the product specifications registered with the NRA by the manufacturer. Use the same test method and specifications when samples from different manufacturers are collected that contain the same APIs in the same dosage form to enable comparison, unless different specifications and methods were approved by the NRA. Once the tests to be performed and the methods and specifications to be used have been selected, the testing protocol should be finalized. For each of the target products, the protocol should contain the list of tests to be conducted, reference to methods to be used and specifications to be employed.	Testing protocol should state: • pharmacopoeial or analogous testing methods • product specifications • reference monographs or validated method of the laboratory • justification of use of different specifications or methods for testing the same APIs in the same dosage form

Table 1 *continued*

Aspect	Key considerations	Examples of variables or elements
	Ensure that the procedures for receipt and testing of samples by the testing laboratory are available.	Procedure for receipt of samples by the testing laboratory should include: • sample inspection for conformity with information in sample collection form • storage according to required conditions, including compliance with cold chain requirements • testing in accordance with testing protocol and in compliance with good laboratory practices • preparation of complete analytical reports and certificates of analysis • retention of documents received with samples, records of testing of samples, and samples according to requirements

Source: Annex 7: Guidelines on the conduct of surveys of the quality of medicines. Geneva: World Health Organization; 2016 (WHO Technical Report Series, No. 996).

Note: See WHO TRS 996, Annex 7, for additional details and guidance, including an example of a sample collection form, content of analytical test report, and an outline of a survey report.

Appendix 6. Pharmaceutical disposal options

Table 1

Treatment and disposal options for pharmaceuticals

Type of option	Options
Optimal	• High-temperature incineration with flue gas cleaning that complies with the Stockholm Convention • Coprocessing in a high-temperature process (cement kiln) • Waste immobilization followed by disposal in an engineered landfill
Interim	• Safe burial in an engineered landfill • Waste immobilization followed by disposal in controlled landfill • High-temperature incineration or dual-chamber medium-temperature incineration with auxiliary burners with limited or no flue gas treatment
Immediate short-term (emergency situations)	• Encapsulation in uncontrolled landfill <i>Please note that if optimal or interim options are not available at the time, the temporary storage of waste should be considered before resorting to an immediate short-term response</i>
Not recommended	• Low- or medium-temperature incineration without auxiliary burners • Burial in uncontrolled or controlled (non-engineered) landfill

Technologies not applicable to the treatment of pharmaceutical waste: Low-heat thermal-based treatment technologies such as autoclaving, microwaving or frictional heat treatment are not suitable for the treatment of pharmaceutical waste.

Source: Safe management of pharmaceutical waste from health care facilities: global best practices. Geneva: World Health Organization; 2025 (<https://www.who.int/publications/i/item/9789240106710>).

Appendix 7. Market surveillance and control of medical products for public health emergencies

Introduction

Recent public health emergencies, such as the severe acute respiratory syndrome (SARS) outbreak and the COVID-19 pandemic, have highlighted the need for regulatory system preparedness and responsiveness to ensure access to quality medical products to manage diseases and prevent disability. However, efforts to prevent, diagnose and treat these diseases and to prevent disability are undermined and worsened with the entry and use of substandard and falsified medical products. As a result, national and regional regulatory systems play a critical role during public health emergencies. This section is included as a supplement to the recommendations provided in the main document to provide a quick reference guide for national regulatory authorities (NRAs) by highlighting additional recommendations for market surveillance and control towards the prevention of, detection of, and response to substandard and falsified medical products in preparation for and in response to public health emergencies.

Several of the recommendations have been extracted from the *WHO Guidelines on regulatory preparedness for the oversight of pandemic or other emergency use vaccines in importing countries* ([1](#), [2](#)). NRAs are encouraged to utilize this source for further details and recommendations, including required documentation, labelling, and pathways for product review and vigilance.

Regulatory systems

The roles and responsibilities of national and regional regulatory authorities remain the same during public health emergencies as during non-emergency periods. However, there is a need for rapid responses to emergencies for the application of regulatory functions to support the national response to a public health emergency where time and resources are limited. To ensure global health security and protect public health, NRAs will need to identify appropriate strategies and engage stakeholders to reduce the time required between the emergence of a pandemic pathogen and the availability of safe, efficacious medical products for diagnosis, prevention and treatment ([1](#), [2](#)). Plans for the NRA's response to public health emergencies should be guided by clearly defined objectives, with the aim of minimizing the impact of the emergency. Fig. 1 presents best practices and efficiencies arising from lessons from the COVID-19 pandemic, as identified by regulators in the Americas ([3](#)).

Fig. 1
Best practices and efficiencies learned from COVID-19 pandemic



Source: Adapted from Pan American Health Organization (3).

Supplemental to the recommendations provided for regulatory systems, NRAs should:

- develop written criteria to for emergency response, including circumstances, contingency planning and procedures for the conduct of regulatory activities during an emergency;
- be empowered by legal frameworks to permit expedited, risk-based assessment of medical products for public health emergencies for emergency use, a pandemic preparedness plan that includes emergency use authorizations or equivalent, technical procedures that use reliance and recognition of trusted authorities for emergency use authorizations, and a system to monitor the emergency use authorization products on the market;
- participate in national emergency response teams, including involvement in emergency planning;
- support the development and implementation of a national pandemic and emergency preparedness plan that includes strategies to facilitate timely availability, distribution and use of medical products for public health emergencies while ensuring quality, safety and efficacy;
- regularly review and test the national pandemic and emergency plans to ensure they are up to date;
- develop and test a risk-based emergency management plan for the NRA's response to public health emergencies prior to the occurrence of an event;
- collaborate with public health programmes (for example, immunization), marketing authorization holders and other stakeholders to develop strategies for enhanced market surveillance and vigilance to monitor the safety and quality of medical products for public health emergencies;
- establish communications and information-sharing systems for all stakeholders, including a public communications plan;
- be empowered by legal provisions to permit the application of regulatory reliance on the decisions and work of reference NRAs or WHO for key regulatory functions, for example, emergency use authorization, regulatory inspections or lot release;
- identify essential medical products needed to manage the public health emergency, including medicines, vaccines and medical devices.

Prevention

Supplemental to the recommendations for the prevention of substandard and falsified medical products, NRAs should:

- establish and apply criteria for emergency use authorizations or import approvals of medical products for public health emergencies where evidence of quality, safety and efficacy is required;
- develop prioritized evaluation and approval processes using flexible pathways or mechanisms, including emergency use authorizations, or compassionate use authorizations that do not compromise requirements of quality, safety and efficacy;
- enable flexibility of physical documentation requirements using information technologies and electronic communication systems to expedite submissions, review procedures, and approvals;
- develop pathways for the assessment or review of the quality of medical products for the diagnosis, prevention and treatment of public health emergencies (including donations), prior to market authorization, registration or import approval;
- develop guidelines describing conditions, criteria and requirements for special authorizations (for example, emergency use authorization, conditional authorization) of medical products during public health emergencies, including donations;
- where needed, update regulatory frameworks to ensure timely access to quality medical products during emergencies;
- apply a risk-based approach to the determination of emergency use authorizations;
- communicate new or revised guidelines for emergency authorization procedures for essential medical products for public health emergencies to sponsors;
- communicate any changes in evaluation and approval processes for non-emergency products to the respective sponsors to avoid backlogs and shortages;
- develop policies and procedures to avoid stock-outs of essential medical products for public health emergencies, such as modification of conditions of sale and points of sale;
- according to regulatory needs, utilize regulatory reliance to expedite market authorizations, according to national frameworks, and risk assessments of medical products;

- utilize established systems for collaborative assessment or prequalification of medical products for emergencies, for example, WHO emergency use listing;
- establish criteria for the selection of reference authorities for regulatory reliance, considering the reference authority's system maturity level, the range of essential medical products, and access to tools and reports for verification of sameness;
- collaborate with NRAs of donating countries to acquire essential product information to facilitate reviews for emergency authorization of donated medical products;
- during the interpandemic phase, establish liaisons with reference NRAs;
- develop criteria and conditions for maintenance of the emergency authorization, for example, submission of periodic safety reports, post-authorization changes;
- where implemented, extend reliance to post-authorization changes as far as possible, including major manufacturing process changes and shelf-life extensions;
- review and utilize approved post-authorization changes (for example, shelf-life, manufacturing processes) to update the database of authorized medical products for public health emergencies;
- collaborate with customs and border control agencies to establish conditions, priority entry points and import approval procedures for medical products for public health emergencies;
- establish a system and procedures for the importation of medical products for public health emergencies, including a procedure for approval of importation within five working days from the date of receipt of the application;
- where applicable (for example, non-vaccine-producing countries), establish procedures and requirements for lot release of emergency use vaccines by adopting recognition mechanisms.

Detection

Supplemental to the recommendations provided for the detection of substandard and falsified medical products, NRAs should:

- include medical products granted emergency use authorization in publicly available databases for authorized medical products to facilitate detection by stakeholders;

- apply a risk-based approach to market surveillance and control of medical products for public health emergencies and to inspections of facilities;
- develop and implement regulatory efficiencies to utilize decisions to inform licensing of establishments, such as recognition or reliance on inspections by trusted authorities;
- collaborate with marketing authorization holders or manufacturers for continuous monitoring and risk management of the medical products during marketing and use;
- maintain records of the distribution of lots of medical products for public health emergencies;
- where feasible, utilizing a risk-based approach, establish criteria, requirements and procedures for remote or virtual inspections of facilities;
- apply procedures and tools to enable prompt detection of substandard and falsified medical products, for example, visual and physical inspection, authentication procedures;
- establish procedures to increase monitoring of medical product promotion and informal markets, particularly social media and online vendors or platforms;
- adapt reporting tools to facilitate rapid communication and investigation of suspected substandard and falsified medical products for public health emergencies;
- communicate promptly with the respective marketing authorization holder when one of their products is identified as a suspected substandard or falsified medical product for authentication;
- closely monitor information produced by reference NRAs, regional regulatory networks, neighbouring NRAs, the WHO Global Surveillance and Monitoring System, and international security agencies (for example, INTERPOL) for warnings or alerts about medical products for public health emergencies;
- utilize reliance in vigilance and market surveillance by adapting warnings and precautions published by reference NRAs, regional networks, neighbouring NRAs, and the WHO GSMS where the same product is authorized for local use;
- empower customs officials, health workers, inventory managers and distributors to identify and report suspected substandard and falsified medical products for public health emergencies through education and training in visual and physical inspections.

Response

Supplemental to the recommendations for the NRA's response to substandard and falsified medical products, NRAs should:

- apply the principles, policies and procedures recommended during public health emergencies; these include transparency in legal processes, risk communication, and storage and disposal of substandard and falsified medical products, which continue to be applied during public health emergencies;
- implement systems for timely regulatory action with respect to emerging safety or quality issues identified post-authorization that may require reassessment of risk management and risk communication to the public and other stakeholders;
- periodically assess the effectiveness of regulatory systems and supporting legal frameworks in regulation of medical products for pandemics and other public health emergencies, to inform regulatory strengthening;
- regularly assess the effectiveness of response mechanisms in ensuring that the objectives of national pandemic plans are achieved, including transparency, avoidance of misinformation, assurance of quality medical products, and building trust in national systems.

References

1. Guidelines on regulatory preparedness for the oversight of pandemic or other emergency use vaccines in importing countries. In: WHO Expert Committee on Biological Standardization: seventy-eighth report (Annex 2). Geneva: World Health Organization; 2024 (<https://iris.who.int/handle/10665/376596>). Licence: CC BY-NC-SA 3.0 IGO.
2. Reliance for emergency use authorization of medicines and other health technologies in a pandemic (for example, COVID-19). Washington, DC: Pan American Health Organization; 2020 (<https://iris.paho.org/server/api/core/bitstreams/1ff5e19a-cf7b-471d-b1e0-efa7fd62e450/content>).
3. Regulatory system strengthening in the Americas: lessons learned from the national regulatory authorities of regional reference. Washington, DC: Pan American Health Organization; 2021 (<https://iris.paho.org/handle/10665.2/53793>). Licence: CC BY-NC-SA 3.0 IGO.

Additional guidance

- Chapter 9: Regulatory emergency response to the COVID-19 pandemic in the Americas. In: Regulatory system strengthening in the Americas: lessons learned from the national regulatory authorities of regional reference. Washington, DC: Pan American Health Organization; 2021 (<https://iris.paho.org/handle/10665.2/53793>).

- Guidelines for medicine donations, revised 2010. Geneva: World Health Organization; 2011 (<https://iris.who.int/handle/10665/44647>).
- Medical device donations: considerations for solicitation and provision. WHO Medical Device Technical Series. Geneva: World Health Organization; 2011 (<https://iris.who.int/handle/10665/378274>). Licence: CC BY-NC-SA 3.0 IGO.

Appendix 8. International agencies involved in prevention of, detection of, and response to substandard and falsified medical products

Agency	Description	Website
International Criminal Police Organization (INTERPOL)	An intergovernmental organization that supports police to make the world a safer place, including enabling sharing of and access to data on crimes and criminals. INTERPOL facilitates cooperation among police to seize falsified medical products and shut down illicit manufacturers, creates awareness of risks associated with buying medical products from unregulated websites, and facilitates training of law enforcement agencies and NRA personnel in illicit health products and pharmaceutical crime. Programme: Illicit Goods and Global Health Programme – Public Health and Pharmaceutical Crime	https://www.interpol.int/en/Crimes/Illicit-goods/Pharmaceutical-crime-operations
International Narcotics Control Board	The International Narcotics Control Board is the independent and quasi-judicial monitoring body for the implementation of the United Nations international drug control conventions. Publication: Precursors and chemicals frequently used in the illicit manufacture of narcotic drugs and psychotropic substances Guidelines: (https://www.incb.org/incb/en/publications/guidelines-for-use-by-competent-national-authorities.html) Guidelines for governments on preventing the illegal sale of internationally controlled substances through the internet Guidelines to prevent and investigate the diversion of materials and equipment essential for illicit drug manufacture	https://www.incb.org/incb/en/index.html

Table *continued*

Agency	Description	Website
United Nations Office on Drugs and Crime	The United Nations Office on Drugs and Crime strengthens the capacity of Member States to address the illicit manufacture and trafficking of falsified medical products, including provisions of guidelines and training in good legislative practices that provide the legal foundations to combat falsified products. Publication: Guide to good legislative practices on combating crime related to falsified medical products Programme: United Nations Office on Drugs and Crime Container Control Programme for training and awareness of customs officers for specific commodities	Falsified medical products: https://www.unodc.org/westandcentralafrica/en/falsified-medical-products.html
World Customs Organization	An independent intergovernmental body that enhances the effectiveness of customs administrations and their efficiency. The World Customs Organization raises awareness of the threats of falsified and illegal medicines to public safety and participates in Operation Pangea.	https://www.wcoomd.org/en.aspx
World Organisation for Animal Health	The World Organisation for Animal Health improves animal health globally by monitoring and disseminating knowledge about animal diseases and how to use scientific data to limit the adverse effects of animal diseases on society. Substandard and falsified veterinary products pose a threat to animals, and potentially to humans when they contribute to antimicrobial resistance.	https://www.woah.org/en/home/

Appendix 9. WHO tools for collection, handling and assessment of incidents

Document name	Languages	Short description
Aide-mémoire	EN – FR – SP – RU	Provides step by step guidance on how to handle incidents of substandard and falsified medical products, from establishing facts to risk communication.
Good photographs	EN – FR – SP – RU	Details what should and should not be done in order to capture the maximum amount of information when photographing samples of medical products.
Identification guide	EN – FR – SP – RU	Details “red flags” and other suspect elements to identify substandard and falsified medical products.
Prioritization and risk parameters	EN	Supplementary material to the WHO e-course on substandard and falsified medical products and must be used in conjunction with the “List of questions to consider when assessing risk” document. It lists suggestions that may be used to assess the level of different risk types, and prioritize resources to deploy appropriate mitigation and control measures.
Questions to consider when assessing risk	EN	Supplementary material to the WHO e-course on substandard and falsified medical products and must be used in conjunction with the “Prioritization and risk parameters” document. It proposes questions to consider when assessing different types of risk associated with an incident of substandard and falsified medical products.

Identifying suspect substandard and falsified medical products



Some substandard and falsified medical products are almost visually identical to the genuine product and very difficult to detect. The following signs should raise your suspicion

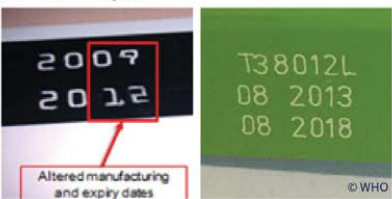
Please note that this guide is a non-exhaustive list

Outer packaging	• Packaging is not in good condition, and/or • Manufacturers details are not clearly stated, and/or • Incorrect language, with grammatical and spelling errors, and/or • Batch numbers and expiry/manufacturing dates appear altered
Inner packaging	• Batch numbers, manufacturing/expiry dates on inner packaging (e.g. blister pack) are different from outer packaging, and/or • Patient information leaflet is in the wrong language
Supply source	• Any suspicion is raise on the source or price of a product and authenticity of accompanying documents, and/or • Any suspicion is raised on quantities available, e.g. products that are usually in short supply are suddenly available regularly or in large quantities
Other factors	• Product does not look, smell, taste and feel correct (i.e. texture), and/or • Packaging components are empty or separated, and/or • Product was not properly stored

Substandard example



Falsified example



if in doubt, take good photographs of the suspect products and contact your national medicines regulatory authority for advice

Outer packaging and inner packaging

- Is the packaging in good condition? The container should appropriately protect the medical product inside (e.g. properly sealed, airtight).
- Are the manufacturer's details clearly stated and in correct language? The manufacturer's details (name, logo, hologram, full address, registration number, etc.) should be correct and in the appropriate language for the market or country in which the product is distributed.
- Are there any spelling or grammatical errors? There should be no spelling or grammatical errors, particularly for the trade (brand) name and active ingredients.
- Are the batch or lot numbers and manufacturing and expiry dates altered? They should be clearly indicated, should not be possible to erase, and should be easily readable, and there should be no irregularity in the embossing or imprinting.
- Is the dosage form or product strength clearly indicated on the label? The appropriate strength and dosage form for the product should be clearly indicated and be the same on all parts of the packaging.
- Is the information the same on the inner and outer packaging? This information should be the same on all parts of the packaging (with no signs of alteration and discrepancies).
- Is there a patient information leaflet and is it in the correct language? The information on the patient information leaflet should be clearly indicated and should match the information on other parts of the packaging. There should be no irregularity in printing quality or colour, shape, texture, and size of paper (e.g. ink should not be smudged, paper is not too rough).

Source of supply

- Is there any suspicion on the source, price, quantities available, or regular availability of products that are usually in short supply, or the authenticity of accompanying documents? Those engaged in the manufacture, distribution and supply of substandard and falsified medical products have shown they respond quickly to demand, thoroughly understanding the market and acting rapidly to exploit opportunities. Most commonly, substandard and falsified medical

products enter the legal supply chain at distribution level through hospitals, clinics, pharmacies and wholesalers, who have obtained medical products from unknown sources and intermediaries without checking their credentials or conducting any due diligence. For example, products whose price appears unusually low, or which are available in unusually large quantities, should raise suspicions, and further checks should be conducted.

Other factors

- Did you notice that the medical product looks, tastes, smells or feels different? Any irregularity in the uniformity of appearance (colour, shape, texture, size, clarity), flavour and smell should raise suspicion (for example, the product may be discoloured or degraded).
- Are there any empty or separated packaging components (bottle caps, spoons, bottles, flat packs, capsules)? Signs of empty or separated packaging components may indicate instances of smuggling or tampering.
- Was the product properly stored? Storage conditions (temperature, humidity, etc.) should be stated on the label and maintained. Signs of degradation may include leakage or discoloration.



World Health
Organization

Substandard and Falsified Medical Products Incident Management AideMemoire

1 REPORT RECEIVED

National regulatory authority receives a report of a substandard or falsified medical product. Focal point for substandard and falsified medical products must be informed.

2 ESTABLISH FACTS

First and foremost, it is key to establish the following facts:

- Contact the source of the information
- Assess the reliability of the source (*see overleaf*)
- Establish the facts
- Assess the credibility of the information received (*see overleaf*)
- Obtain samples of the product and photographs showing batch number and expiry dates

NATIONAL FOCAL POINTS FOR SF MEDICAL PRODUCTS SHOULD IMMEDIATELY SEARCH THE WHO DATABASE TO CHECK IF THE PRODUCT HAS PREVIOUSLY BEEN REPORTED

3 IDENTIFY & ASSESS RISKS

The risk to the population must be identified and assessed as soon as possible to determine appropriate response and resources deployed. These questions can help guide you:

- Is the suspected SF product available in hospitals, clinics, pharmacies, health centres?
- Is the product available in illegal markets, or through internet sites or smartphone applications?
- Have any adverse reactions been reported / Check with National Pharmacovigilance Centre?
- What quantities of suspected SF medical products have been discovered?
- Is there evidence that the product is in recent circulation?
- Is the product in wide circulation within your country, or neighbouring countries?
- Is this product in strong demand, or in short supply?
- Consult with the stated manufacturer of the reported product (*see overleaf*)

4 IMMEDIATE PUBLIC HEALTH PROTECTION

Once risks are identified, protect public health must be protected by removing the source of the risk that is represented by the SF product. This may include the following immediate actions:

- Quarantine or seize any suspected medical product dependent on risk
- Ensure the product is stored securely and in compliance with storage conditions
- Ensure appropriate treatment is available to affected patients

5 FIELD SCREENING AND LAB ANALYSIS

The product composition may reveal additional risks. Only laboratory analysis can confirm this – and help identify any possible remedies:

- If product is suspected of causing serious ADR's send samples directly to the laboratory
- Screen product with hand held screening equipment or other field testing equipment if available
- Secure up to 100 samples for testing, if not as many samples from the same source as the reported product as possible and store in controlled conditions
- Request sample from genuine manufacturer for comparison purposes
- Arrange for testing as a priority dependent on risk to public health

6 MANAGING THE INCIDENT

In order to efficiently manage an incident of SF medical products, it is necessary to:

- Establish a team of relevant regulatory specialists, appoint a lead person and invite relevant external stakeholders/experts
- Keep strict records of all meetings and all decisions that are made
- Focus on protection of public health, mitigating the risk posed by the product and investigating the origin
- Consider a recall of the medical product and associated communications and media messages
- Verify stocks/availability of genuine (quality assured) replacement product
- Consider an alert or public notice

REPORT THE INCIDENT TO THE WHO GLOBAL SURVEILLANCE AND MONITORING SYSTEM FOR SUBSTANDARD AND FALSIFIED MEDICAL PRODUCTS

Email: rapidalert@who.int ; Portal: <https://extranet.who.int/ssffcrapidalert>

Remember: These incidents attract a lot of attention – unnecessary delays are difficult to explain, public health and the reputation of your organization may be at risk.

National focal points are strongly advised to conduct a search of the WHO database when dealing with a suspected substandard or falsified medical product, at the earliest opportunity.

Irrespective of whether you receive a match with other products in the database, you should report the suspected or confirmed medical product to WHO as soon as possible. Other Member States may be seeing the same product in circulation and your report will assist them.

Remember, search results can match with your product on a separate continent or in another region. This information will help you assess risk and manage and respond to your incident more efficiently and effectively and, in serious cases, save lives.

Assessing the reliability of a source

Anonymous information should be treated with caution.

- Is the source a whistleblower or a current or ex-employee of a company that they are providing information about?
- What is the motivation for supplying the information?
- Is the source easily contactable?
- If contact details are supplied, are they accurate (dialling codes, telephone numbers, email addresses, physical addresses)?
- Has information been received from the same source previously? If so, was the information accurate?
- Is the source willing to be contacted or met, or supply further information?

Assessing the credibility of the information

- Has any similar information been received from different sources?
- Are there any other sources that can corroborate the information provided?
- Are there any obvious inaccuracies in the information?

Questions to manufacturers

1. Did you manufacture this product? If yes, does the product and packaging look genuine? Photographs and samples should be provided if available.
2. Are the manufacturing, batch, and expiry dates authentic? If the batch number is genuine, where and when was it distributed?
3. Have you had falsified or substandard versions of this batch reported previously? If so, when and where?
4. Have you received any complaints about this batch? If so, from whom, where, and when?
5. Have you received any reports of unexpected adverse reactions relating to this product or batch? If so, when, where, in what numbers, and with what severity?
6. Is there any other information we should be aware of?

Taking good photographs of samples of suspected substandard and falsified medical products

Good photographs of a suspect sample are critical to identify substandard and falsified medical products. A typical set of photographs includes one of each side of the sample, clearly displaying key information. When multiple photographs of multiple products or batch numbers are taken, it is essential to ensure that each photograph is appropriately associated with the correct product or batch number when sending the photographs. Examples are shown at the back.

Set the scene to ensure that clear photographs can be taken

To ensure the key information will be clear in the photographs, set the scene with the following tips:

- use camera with at least 10 megapixels (most smartphones have this);
- use a plain background and sufficient *indirect* lighting;
- avoid reflections (e.g. disable flash);
- prevent blurry photos (e.g. auto focus, stable support, use zoom if necessary);
- include a graduated ruler next to the sample in frame (or another scale);
- level camera close to and directly above the sample (no oblique views).

photograph *all sides* of the packaging to capture key information

Depending on the packing and shape of each product, more than one photograph may be necessary. Take photographs of all sides of the packaging (front, back, sides) to capture this information:

- product name;
- active pharmaceutical ingredients;
- dosage strength and formulation;
- batch number;
- expiry and manufacturing dates;
- name and address of manufacturer;
- detailed views of key features, including any authentication marks (e.g. hologram, logo, barcode).

Check that all information is captured exhaustively

To ensure the captured information is exhaustive, please also photograph, if available:

- registration number;
- all the different layers of packaging and the patient information leaflet;
- marketing authorization holder name;
- import or export licence numbers.

Examples



Capture maximum information

For the sake of simplicity, different elements are captured in a single frame below; it is better to take multiple photographs with higher definition to capture all details.

- 1 Product name, manufacturer name, registration number, and dosage clearly displayed

The example cannot show back of the box

- 2 Batch number and expiry-manufacturing dates in focus

- 3 Plain, non reflective background



- 4 Batch number and dates displayed on blister to compare with carton

- 5 Additional photograph needed for detail side view

- 6 Dosage form and primary packaging clearly displayed

- 7 Graduated ruler for scale

Ensure information is understandable

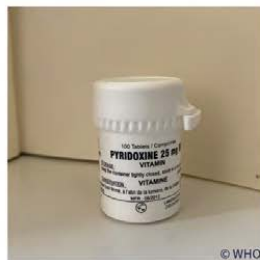
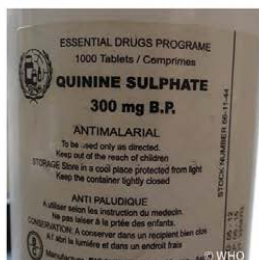
Importance of focus. The photograph below captures some of the key information (including batch number) but it is blurry, because the focus is on the foreground object.



Clear labelling. The photograph underneath is not associated with a product or manufacturer name. The information is useless because some information is not visible.



The importance of scale. The photographs below show how a product's size can be misleading if it is taken out of context or without indication of scale.



Prioritization and risk parameters

General remarks

Purpose and use of this document

This document is supplementary material to the e-course on substandard and falsified medical products and must be used in conjunction with the “*List of questions to consider when assessing risk*” document. The parameters listed below are suggestions only that may be used to:

- assess the level of different risk types;
- prioritize resources to deploy appropriate mitigation and control measures.

Users are required to adapt this support document to different contexts, taking into account their response capacity, socioeconomic context, reliance opportunities, etc.

Please note that:

- this document is not an official WHO guideline;
- this document does not cover medical devices, including IVDs. The WHO technical unit dealing with substandard and falsified medical products is always available for questions, clarifications, comments, etc., using the email rapidalert@who.int.

Risk assessment and management of incidents of substandard and falsified medical products

The goal of risk assessment is to identify and prioritize actions that will mitigate the threat.³⁸ Risk assessment is a systematic process for gathering, assessing and documenting information to assign a level of risk. It provides the basis for taking action to manage and reduce the negative consequences of substandard and falsified medical products.

The risk management cycle includes the identification and prioritization of mitigation measures, taking into account the likelihood of success, the feasibility of implementation and unintended consequences. Resources are never unlimited, and this applies to any organization. Resource management determines the quality of the response to an incident involving substandard and falsified medical products.

Multistakeholder engagement is always encouraged in order to establish all necessary facts. You may have limited confidence in the information available

³⁸ Threat is the potential harm that can occur. Risk is the likelihood that the harm will be realized.

initially: in which case, you will have to seek further information and repeat the risk assessment as appropriate. Risk assessment is a continuous (but not always sequential!) process and each new piece of information will help refine your assessment. Risk management also requires continuous monitoring and evaluation as the incident unfolds.

Handling incidents of substandard and falsified medical products in a professional, consistent, timely and proportionate manner requires sound risk assessment and will reflect well upon the organizations involved.

Prioritization and risk parameters by type

Critical risks

Type	Level	Criteria
Public health	High	- Fatalities or serious adverse reactions, or - Immediate or likely threat of fatalities or serious adverse reactions, or - Multiple numbers of persons affected > 500
	Medium	- Severe adverse reactions, or - Threat of severe adverse reactions, or - Multiple numbers of persons affected > 100
	Low	- Threat of severe adverse reactions, and - Low numbers of persons affected < 100, or - Isolated incident
Supply chain	High	- Product has entered the regulated or licensed supply chain, and - Reached hospitals, clinics, pharmacies or patients, or - Recall is likely to lead to a significant shortage
	Medium	Product is available only through unlicensed and unregulated outlets
	Low	Product not thought to be available to the public
Geographical	High	Product available on an interregional or global basis, including unregulated websites
	Medium	Product restricted to one region
	Low	Product restricted to small area
Environmental	High	Serious or immediate risk to the environment
	Medium	Potential risks to the environment
	Low	No environmental risks identified

Table *continued*

Type	Level	Criteria
Public interest	High	• Widespread and escalating international media interest, or • Patient group or other major stakeholder interest, or • Risk of serious reputational damage to public authorities or organization
	Medium	• Regional or national media interest, or • Subject of complaints to or from stakeholders
	Low	• No significant public interest
Political	High	• Subject of significant national government interest
	Medium	• Affects or is likely to affect, question or change policy
	Low	• Non-contentious issue
Economic	High	• Major financial gain or serious loss > US\$ 1 million
	Medium	• Financial gain or loss < US\$ 1 million
	Low	• No significant gain or loss
Legal	High	• Exposes your organization to litigation or other legal challenge
	Medium	• Exposes weakness in regulatory or legislative regime
	Low	• No perceived legal risks

Historical risks

Type	Level	Criteria
Product history	High	• Same batch or lot of product previously notified
	Medium	• Same product previously notified
	Low	• No previous reports
Supply chain elements	High	• Same importer, exporter, broker, wholesaler, distributor or retailer
	Medium	• Similar method of transportation, concealment or payment
	Low	• No previous reports

Questions to consider when assessing risk

General remarks

Purpose and use of this document

This document is supplementary material to the e-course on substandard and falsified medical products and must be used in conjunction with the “*Prioritization and risk parameters*” document. It proposes questions to consider when assessing different types of risk associated with an incident of substandard and falsified medical products.

The list is neither exhaustive nor prescriptive and should be adapted to each specific situation, by asking different questions or framing the questions differently, factoring in socioeconomic context, capacity to implement mitigation measures, timeliness of response, and other factors.

You do not need to answer all questions immediately. Critical questions should be identified and prioritized for immediate response. Numerous questions inform multiple risk types.

Please note that:

- this document is not an official WHO guideline;
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Handling incidents of substandard and falsified medical products in a professional, consistent, timely and proportionate manner requires sound risk assessment and will reflect well upon the organizations involved.

Critical risks

Public health risks

- Have there been any reports of health events associated with the reported or suspected substandard or falsified medical product, e.g. suspected adverse drug reaction? What is the severity?
- How many people or patients have been affected or may be affected in the future?
- What is the profile of the population exposed to the risk (children, vulnerable populations, people difficult to reach, etc.)?
- Were all the adverse drug reactions reported from the same location, point of care or health care professional?
- Is it likely that the adverse drug reaction was caused by method of administration or use (e.g. injection device, self-administration, self-medication) instead of the therapeutic itself?

Supply chain risks

- Is the reported or suspected substandard or falsified product in the regulated or in the illicit, unregulated supply chain? Was it duly registered and imported?
- At what point in the supply chain was the incident discovered (patient, pharmacy, hospital, wholesaler, manufacturer)?
- How many different batch numbers (and unique combinations of variable data) are involved?
- What volumes of product have been reported? How many doses or treatment regimens?
- Who supplied the reported or suspected substandard or falsified product, and who else may have received it (public or government procurement agency, nongovernmental organization or aid organization, private sector)?

- What other products have been procured through the same supply route?
- How many production sites of the reported or suspected substandard or falsified product exist? What is their production capacity? Could the issue affect other products produced in the same sites?

Geographical risks

- Where was the incident discovered (e.g. border area, major transport hub, in transit)? Where else is the reported or suspected substandard or falsified product likely to be available?
- Is the reported or suspected substandard or falsified product available in a wide range (various locations)? In which quantities?
- Could the reported or suspected substandard or falsified product have been smuggled or stolen in transit?
- Are there packaging elements that suggest that the reported or suspected substandard or falsified product may be widely available geographically (e.g. multiple languages, or multiple registration numbers)?

Environmental risks

- Are there factors associated with the environment, health status, behaviours, social or cultural practices, health infrastructure and legal and policy frameworks that increase a population's vulnerability to the reported or suspected substandard or falsified product?
- Would the services of a specialized laboratory be required? Is there any risk of environmental contamination if the reported or suspected substandard or falsified product was disposed of inappropriately?

Associated risks

Public interest

- Would a recall cause panic, distrust or other significant reaction from the public (alert versus alarm)?
- Is the media likely to report the incident? Is the subject material likely to generate media interest (e.g. involves children, high-profile organizations, a relatable story, multiple sources)?
- Has the incident been reported by multiple independent sources (e.g. residents, news media, health care workers)?

- How likely is the situation to escape control in terms of public communication? Is there an effective system for communication between incident managers and other stakeholders?
- Who supplied the reported or suspected substandard or falsified product (public or government procurement agency, nongovernmental organization or aid organization, etc.)?

Political

- Who reported or discovered the product in the first place? Could there be ulterior motives?
- Who has been involved and informed of the incident at this stage (e.g. are local authorities taking action, if so what are the local political implications to be taken in account)?
- Is the reported or suspected substandard or falsified product manufactured, distributed or procured by organizations or individuals who may have conflicts of interest?

Economic

- What is the volume or size of the seizure or discovery of the reported or suspected substandard or falsified product?
- Is the reported or suspected substandard or falsified product “expensive”? Will replacement by a quality and safe version have significant economic impact (and on whom)?
- Is the population who uses the product exposed to risk of financial hardship when procuring the product (economic vulnerability)?
- Would the response, mitigation or control measures have a significant cost (e.g. recall, laboratory analyses, replacement therapies)?
- Are there many pharmaceutical production or distribution sites that are affected?

Legal

- Is there a legal framework that empowers or enables the implementation of mitigation, response or control measures, especially if immediate or urgent action is required?
- Is my organization at risk of litigation in relation to the reported or suspected substandard or falsified product? Could any such legal process hinder the implementation of mitigation measures?

- Does my organization have any legal obligations as to the quality and safety of the reported or suspected substandard or falsified product? Were there any quality management processes that could have been overlooked?
- Who is legally responsible for the manufacture, distribution, quality of, and authorization of the reported or suspected substandard or falsified product?

Historical risks

Previous history of product

- Is it likely that the same adverse drug reactions may have occurred elsewhere previously and not been signalled?
- Is there a previous similar record (e.g. finished product, active ingredient, therapeutic indication) in your own database, the WHO GSMS database, the database of another similar organization, the media, or open source intelligence?

Previous history of supply chain element or stakeholder

- Has a similar incident been reported previously (e.g. with a similar presentation, affecting a similar population and geographical area, over the same time period)?
- Is there a previous similar record (e.g. manufacturer's name, importer's name) in your own database, the WHO GSMS database, the database of another organization, the media, or open source intelligence?
- Does the incident present a recurring pattern in terms of supply chain distribution or modus operandi?
- Have there been any associated or preceding events (e.g. product recalls, similar events in neighbouring countries)?

General questions to assess response capacity and impact

Which mitigation or control measures to prioritize

- Can the treatment be continued with an alternative product?
- What type of product is affected? For example, is the product on an essential medicines list? Is it a lifesaving product? Is this product marketed, distributed or available abroad, either formally or informally? Can help be obtained from colleagues in other organizations?

- Could there be any form of contamination in different batches or lots (e.g. pathogenic microorganisms, another active ingredient)? Is the issue at finished product or API level?
- Is specialized, lengthy or costly laboratory testing required? Can WHO assist?

Consequences of response measures

- Can the same number of doses be easily replaced or alternative treatment be provided? What would be the impact of a recall?
- What is the availability and acceptability of effective preventive measures and of treatment?
- Are there any access issues related to the reported or suspected substandard or falsified product (e.g. existing or projected shortage, likelihood of sharing medications between patients)?
- What is the market turnover and the shelf-life of the reported or suspected substandard or falsified product?

